ABCD ADHD menarche analysis: Output for longitudinal supplementary/sensitivity analyses

Model 2 longitudinal ADHD present output

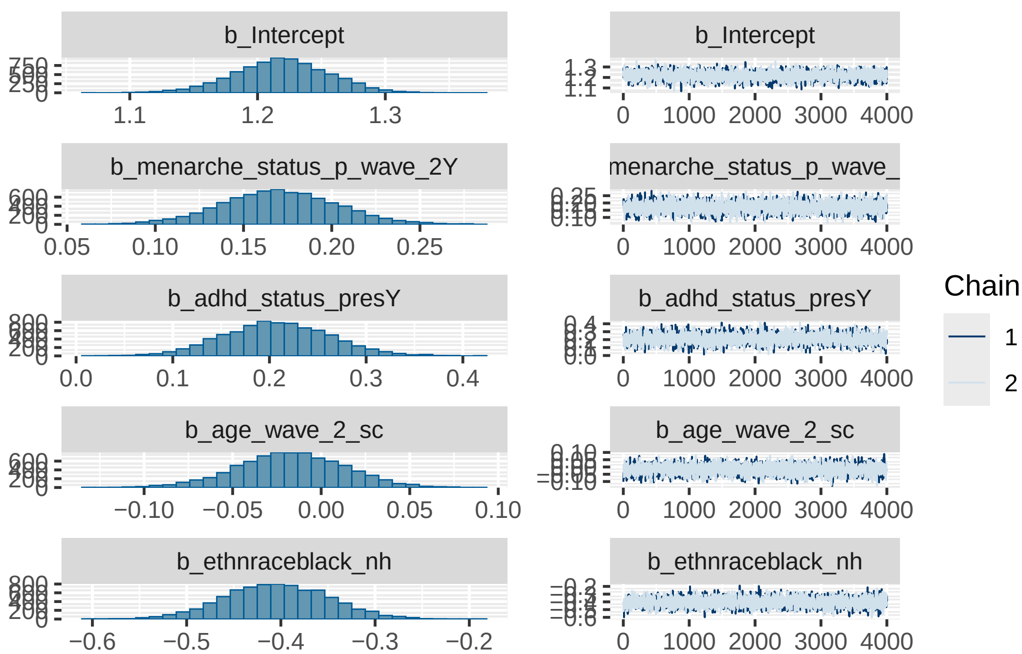
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8266999 0.004585641 0.8172962 0.8353203


 Family: poisson
  Links: mu = log
Formula: internalising_wave_3 ~ menarche_status_p_wave_2 * adhd_status_pres + age_wave_2_sc +
   Data: analytic_df_wide (Number of observations: 4496)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3915)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.844     0.015    0.815    0.873 1.000     1907     3757

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.109     0.033    0.052    0.181 1.004      994     1511

Regression Coefficients:
                                            Estimate Est.Error l-95% CI
Intercept                                      1.219     0.037    1.145
menarche_status_p_wave_2Y                      0.170     0.033    0.106
adhd_status_presY                              0.208     0.055    0.102
age_wave_2_sc                                 -0.017     0.030   -0.078
ethnraceblack_nh                              -0.404     0.055   -0.512
ethnracehispanic                              -0.103     0.048   -0.198
```

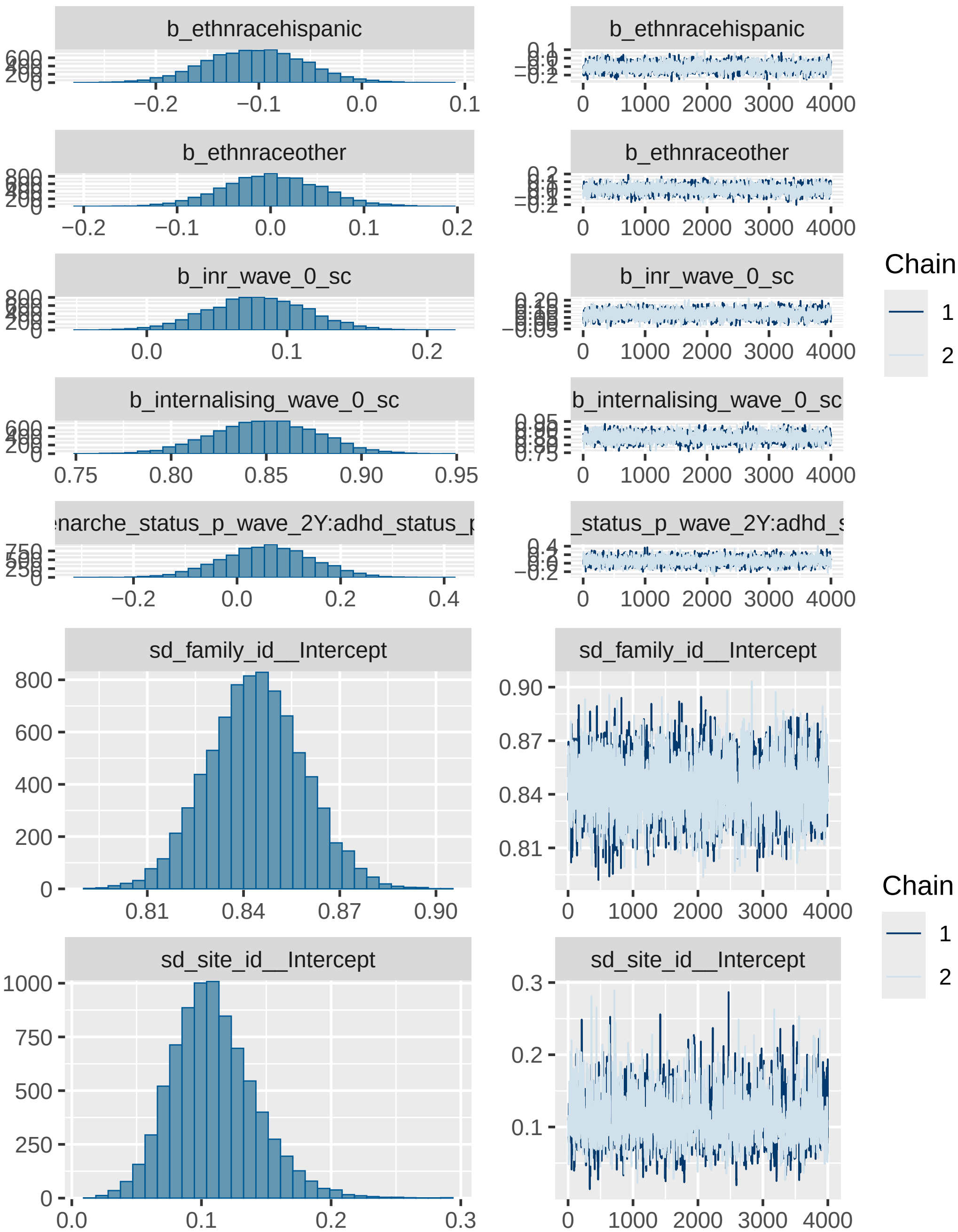
```
ethnraceother                                    0.001     0.051   -0.098
inr_wave_0_sc                                    0.079     0.035    0.010
internalising_wave_0_sc                          0.850     0.027    0.797
menarche_status_p_wave_2Y:adhd_status_presY      0.058     0.087   -0.113
                                              u-95% CI  Rhat Bulk_ESS Tail_ESS
Intercept                                        1.290 1.000     2055     3567
menarche_status_p_wave_2Y                        0.236 1.001     1993     3577
adhd_status_presY                                0.318 1.000     1975     3949
age_wave_2_sc                                    0.041 1.000     2401     4049
ethnraceblack_nh                                -0.296 1.000     2056     3413
ethnracehispanic                                -0.009 1.000     1693     3494
ethnraceother                                    0.102 1.001     1711     2988
inr_wave_0_sc                                    0.148 1.000     1899     3569
internalising_wave_0_sc                          0.902 1.001     1499     2913
menarche_status_p_wave_2Y:adhd_status_presY      0.227 1.001     1589     3268

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

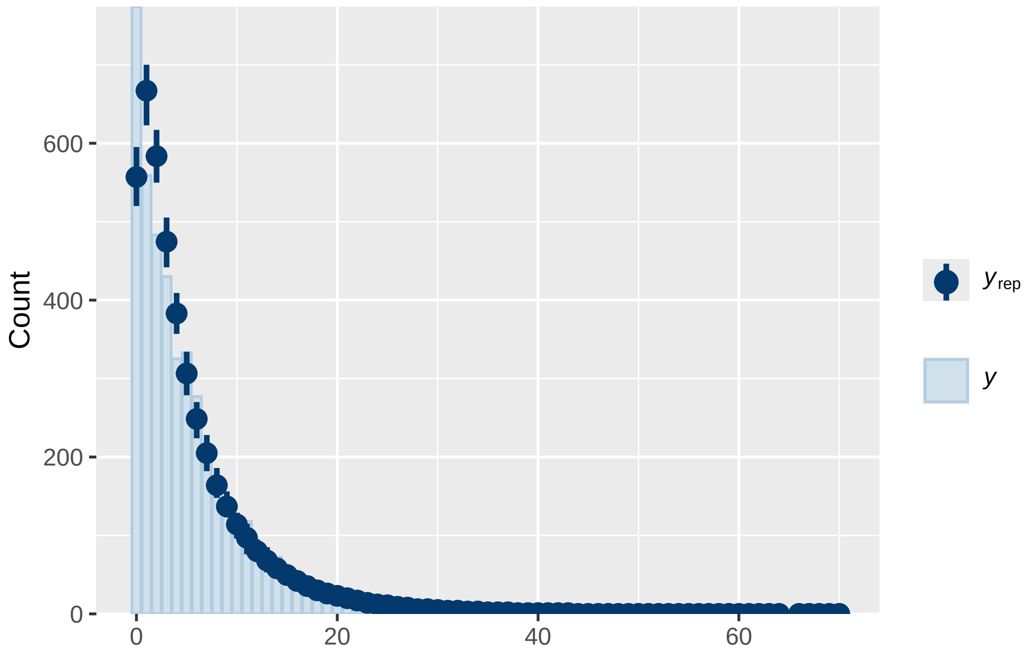
Model 2 longitudinal ADHD present diagnostics

```
                prior     class                                        coef
         normal(0, 1)         b
         normal(0, 1)         b                           adhd_status_presY
         normal(0, 1)         b                               age_wave_2_sc
         normal(0, 1)         b                            ethnraceblack_nh
         normal(0, 1)         b                            ethnracehispanic
         normal(0, 1)         b                               ethnraceother
         normal(0, 1)         b                               inr_wave_0_sc
         normal(0, 1)         b                     internalising_wave_0_sc
         normal(0, 1)         b                   menarche_status_p_wave_2Y
         normal(0, 1)         b menarche_status_p_wave_2Y:adhd_status_presY
 student_t(3, 1.4, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                                   Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                                   Intercept
     group resp dpar nlpar lb ub tag       source
                                             user
                                     (vectorized)
```

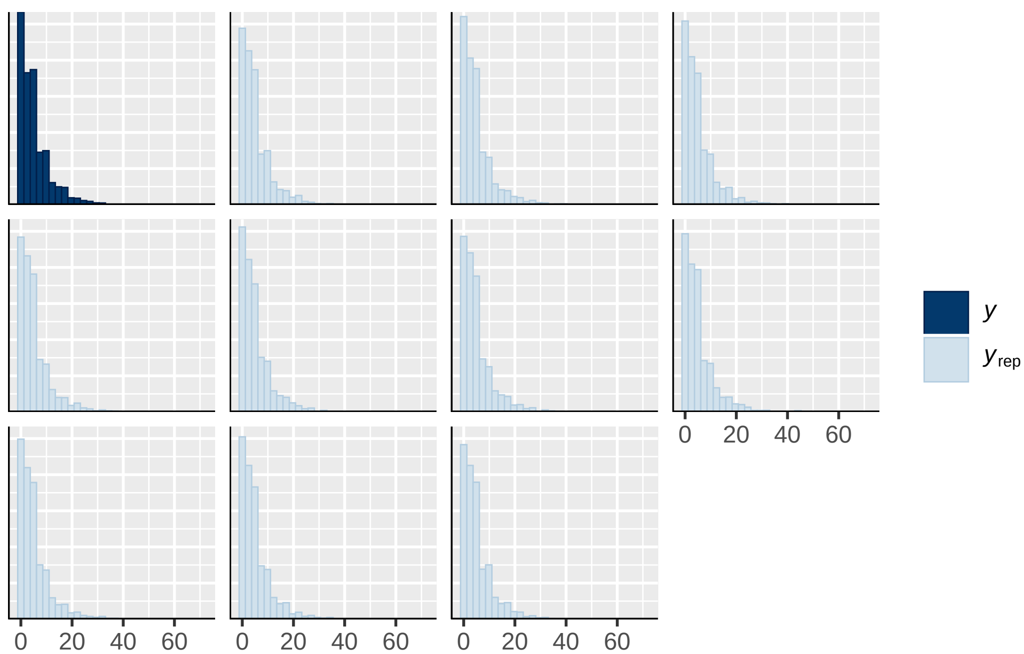
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		(vectorized)
		default
	0	default
family_id	0	(vectorized)
family_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)



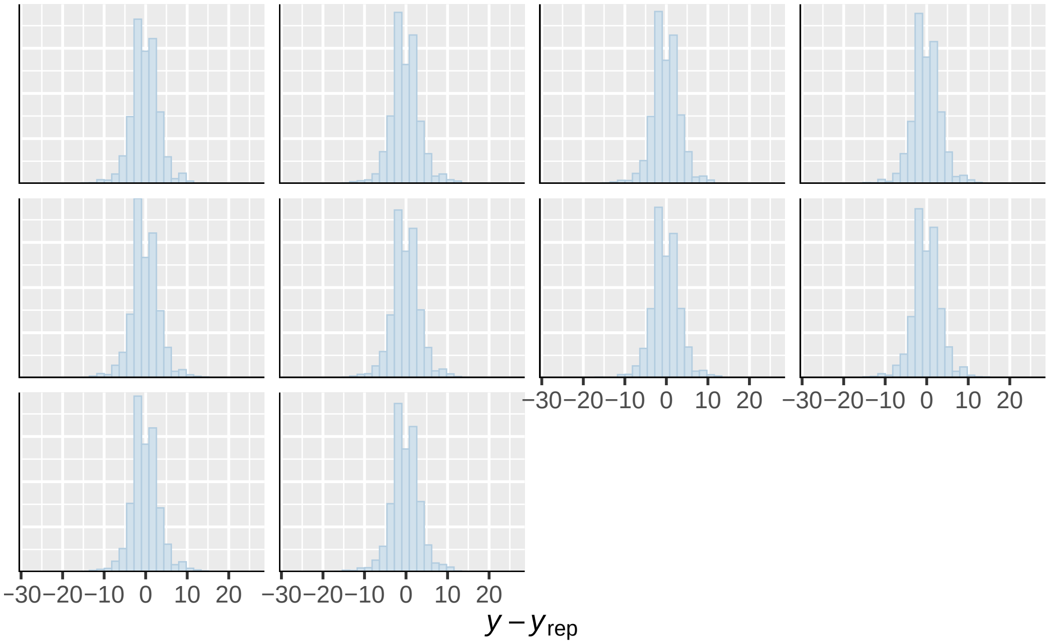

b_ethnracehispanic
−0.2
−0.1
0.0
0.1
b_ethnracehispanic
0
1000
2000
3000
4000
b_ethnraceother
−0.2
−0.1
0.0
0.1
0.2
b_ethnraceother
b_inr_wave_0_sc
0.0
0.1
0.2
b_inr_wave_0_sc
b_internalising_wave_0_sc
0.75
0.80
0.85
0.90
0.95
b_internalising_wave_0_sc
narche_status_p_wave_2Y:adhd_status_p
−0.2
0.0
0.2
0.4
_status_p_wave_2Y:adhd_s
Chain
1
2
sd_family_id__Intercept
800
600
400
200
0
0.81
0.84
0.87
0.90
sd_family_id__Intercept
0.90
0.87
0.84
0.81
0
1000
2000
3000
4000
Chain
1
2
sd_site_id__Intercept
1000
750
500
250
0
0.0
0.1
0.2
0.3
sd_site_id__Intercept
0.3
0.2
0.1
0
1000
2000
3000
4000



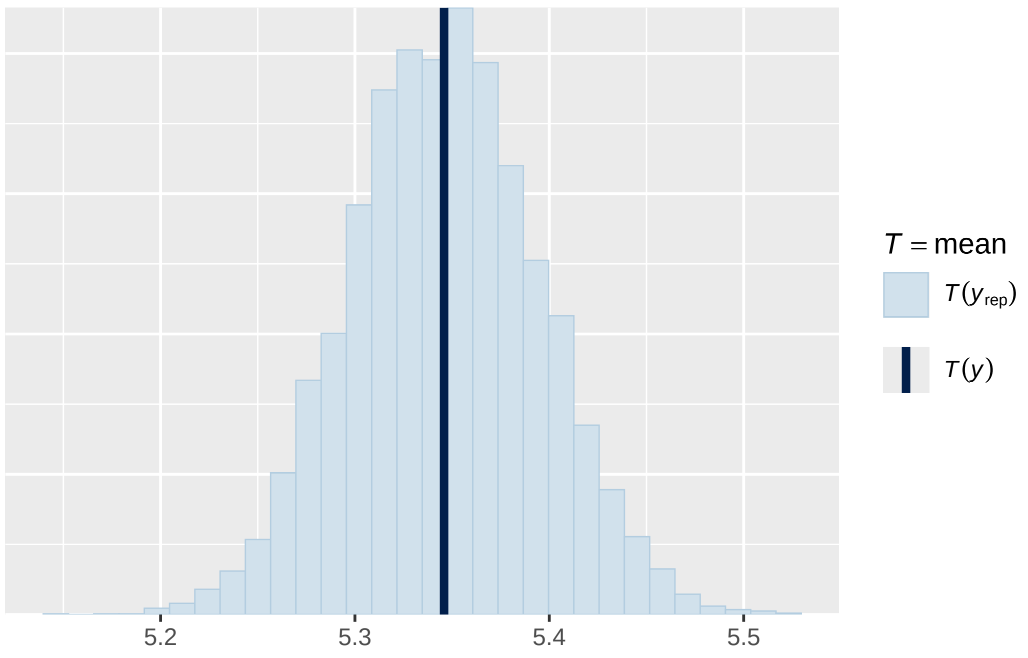

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



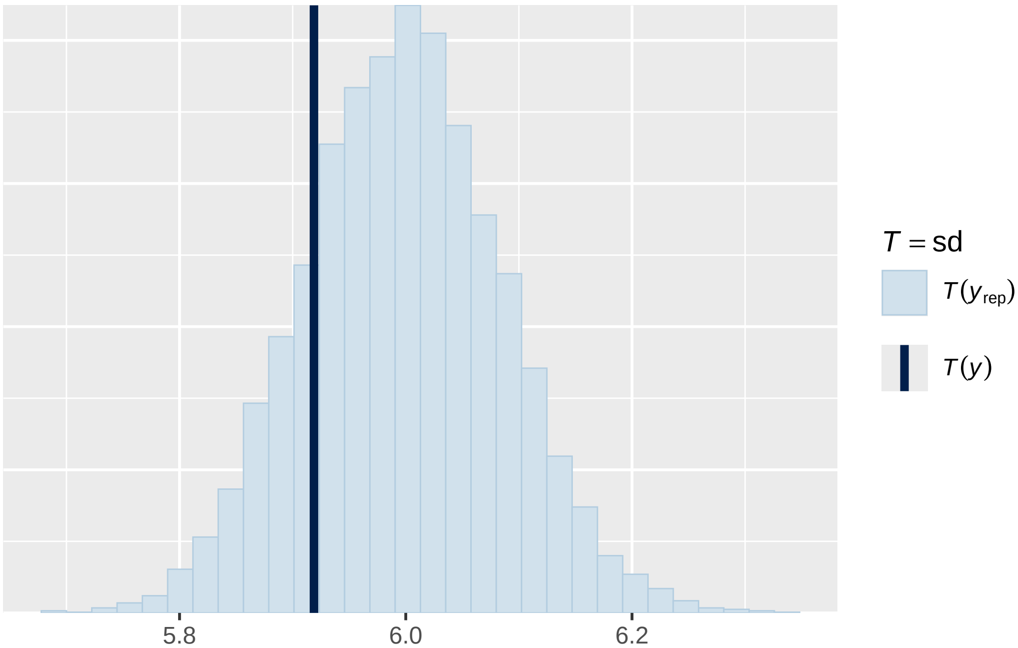

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



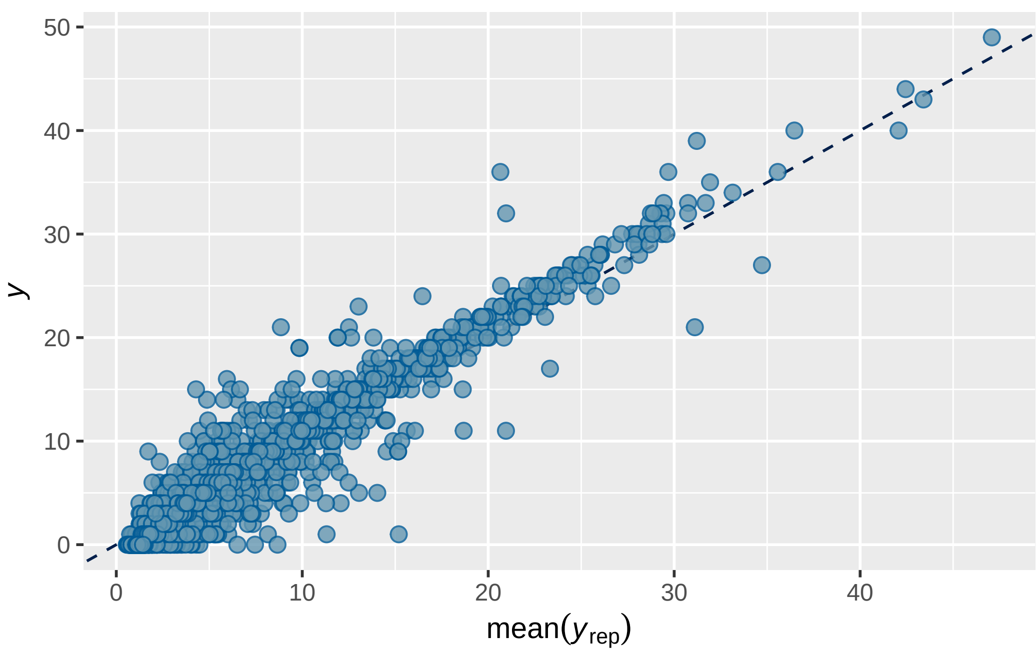

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 2B longitudinal ADHD present output

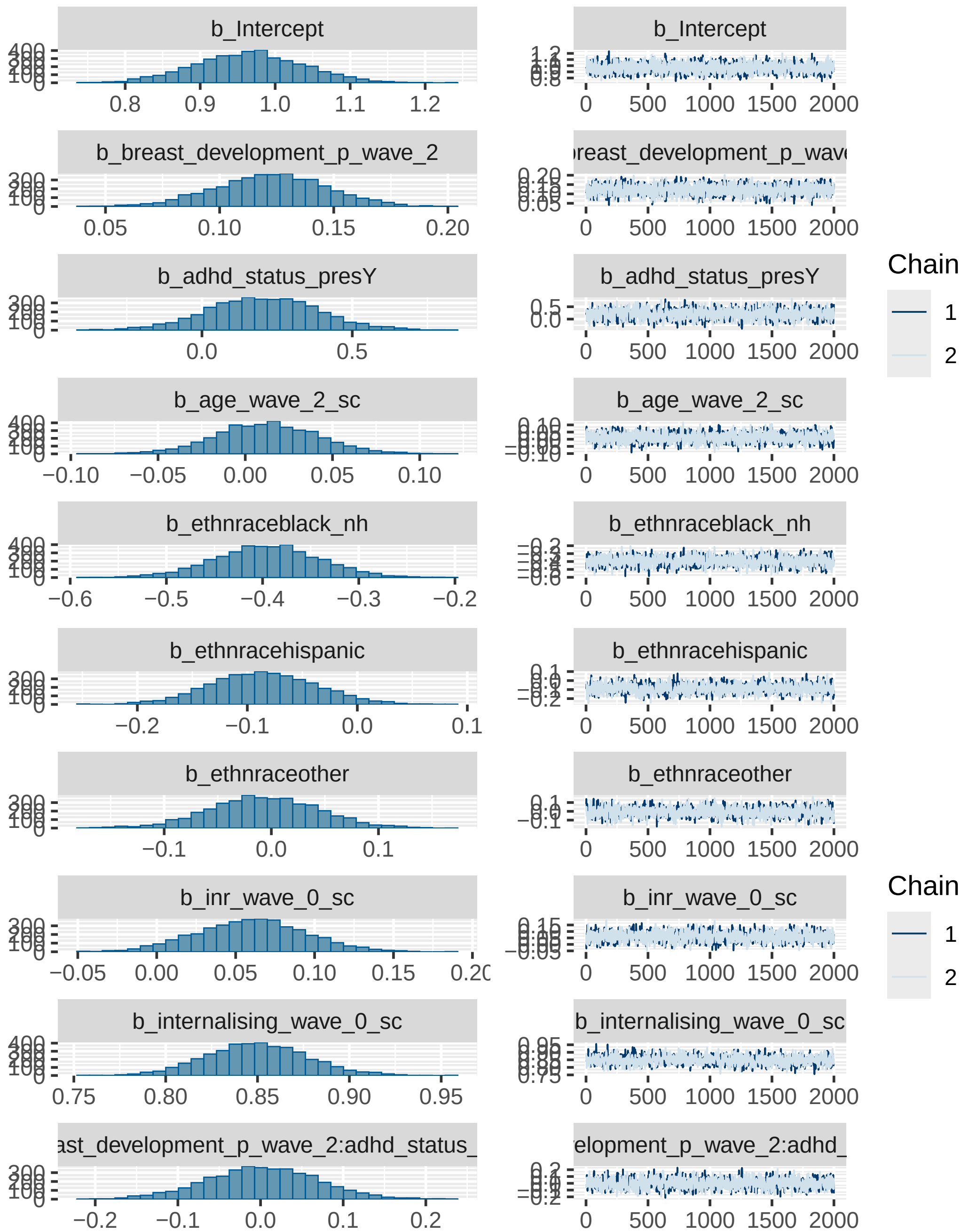
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8274823 0.004586237 0.8183964 0.8362702


 Family: poisson
  Links: mu = log
Formula: internalising_wave_3 ~ breast_development_p_wave_2 * adhd_status_pres + age_wave_2_s
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.849     0.015    0.821    0.878 1.001     1081     2057

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.114     0.033    0.057    0.186 1.003      431      364

Regression Coefficients:
                                            Estimate Est.Error l-95% CI
Intercept                                      0.965     0.072    0.825
breast_development_p_wave_2                    0.123     0.024    0.074
adhd_status_presY                              0.204     0.183   -0.156
age_wave_2_sc                                  0.013     0.029   -0.046
ethnraceblack_nh                              -0.395     0.055   -0.505
ethnracehispanic                              -0.085     0.048   -0.179
ethnraceother                                 -0.007     0.051   -0.106
inr_wave_0_sc                                  0.060     0.035   -0.007
internalising_wave_0_sc                        0.850     0.027    0.797
breast_development_p_wave_2:adhd_status_presY  0.002     0.067   -0.131
                                            u-95% CI  Rhat Bulk_ESS Tail_ESS
Intercept                                      1.106 1.001     1261     1930
breast_development_p_wave_2                    0.169 1.000     1223     2060
adhd_status_presY                              0.571 1.002     1089     1992
age_wave_2_sc                                  0.071 1.001     1292     2223
ethnraceblack_nh                              -0.288 1.001     1174     2007
ethnracehispanic                               0.010 1.002     1073     1838
ethnraceother                                  0.092 1.001     1146     1819
inr_wave_0_sc                                  0.129 1.001     1134     2084
```

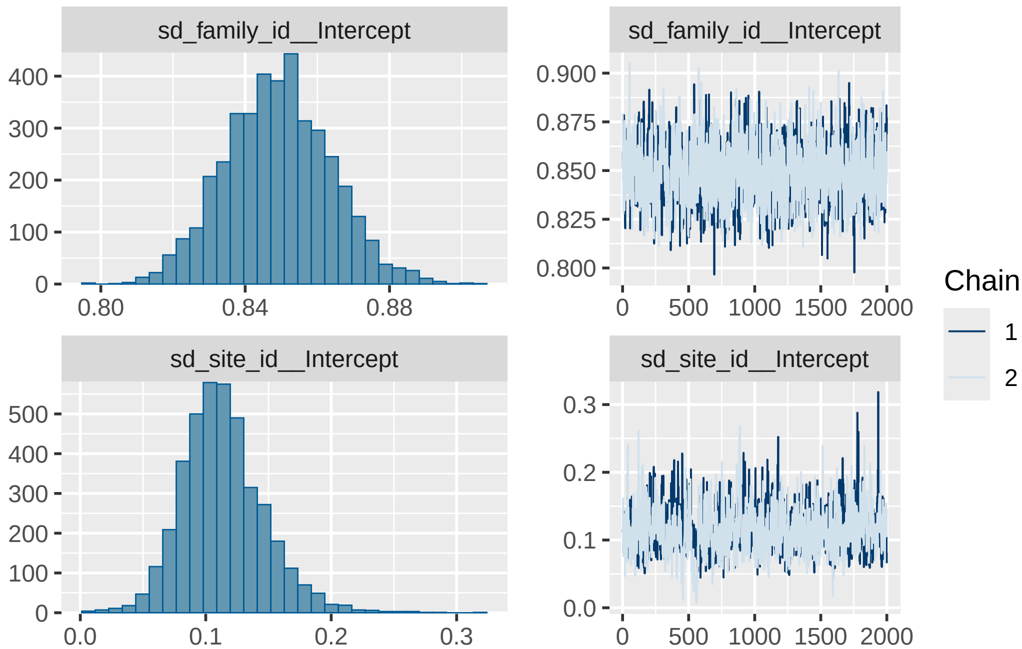
```
internalising_wave_0_sc                        0.905 1.015      880     1394
breast_development_p_wave_2:adhd_status_presY    0.133 1.001     1084     1958

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

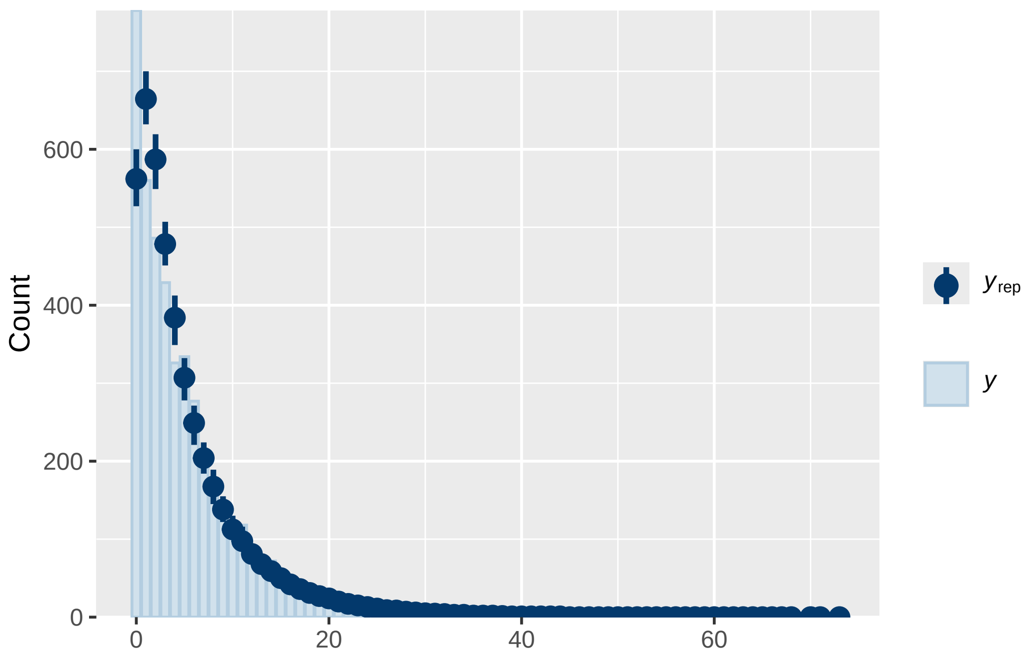
Model 2B longitudinal ADHD present diagnostics

```
                  prior     class                                       coef
           normal(0, 1)         b
           normal(0, 1)         b                          adhd_status_presY
           normal(0, 1)         b                              age_wave_2_sc
           normal(0, 1)         b                 breast_development_p_wave_2
           normal(0, 1)         b breast_development_p_wave_2:adhd_status_presY
           normal(0, 1)         b                           ethnraceblack_nh
           normal(0, 1)         b                           ethnracehispanic
           normal(0, 1)         b                              ethnraceother
           normal(0, 1)         b                              inr_wave_0_sc
           normal(0, 1)         b                    internalising_wave_0_sc
 student_t(3, 1.4, 2.5) Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                                  Intercept
   student_t(3, 0, 2.5)        sd
   student_t(3, 0, 2.5)        sd                                  Intercept
     group resp dpar nlpar lb ub tag       source
                                             user
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                          default
                              0           default
 family_id                    0      (vectorized)
 family_id                    0      (vectorized)
   site_id                    0      (vectorized)
```

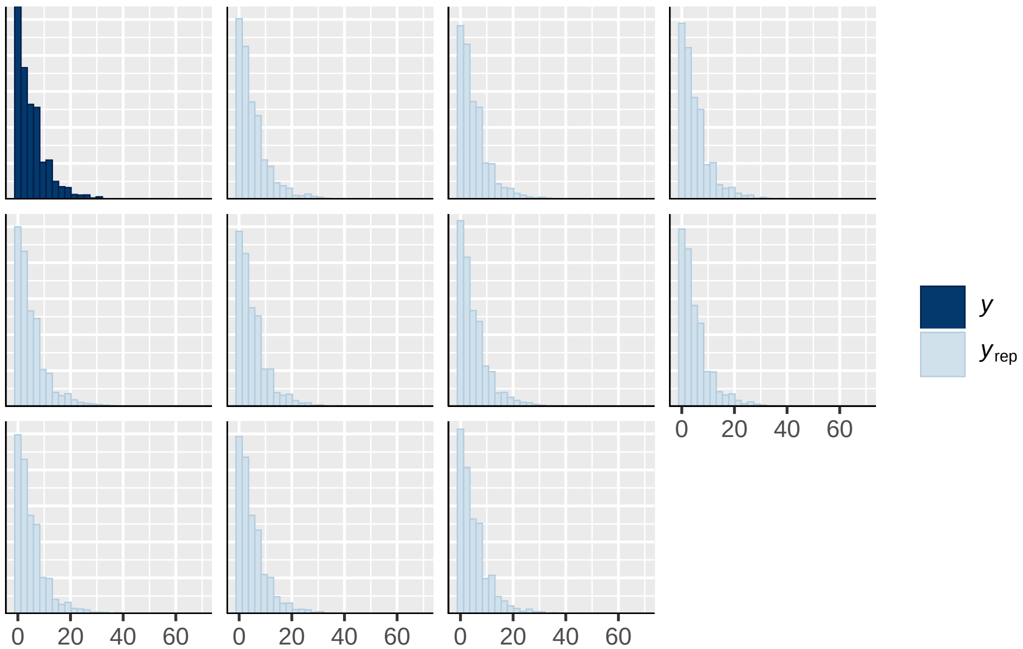
site_id 0 (vectorized)



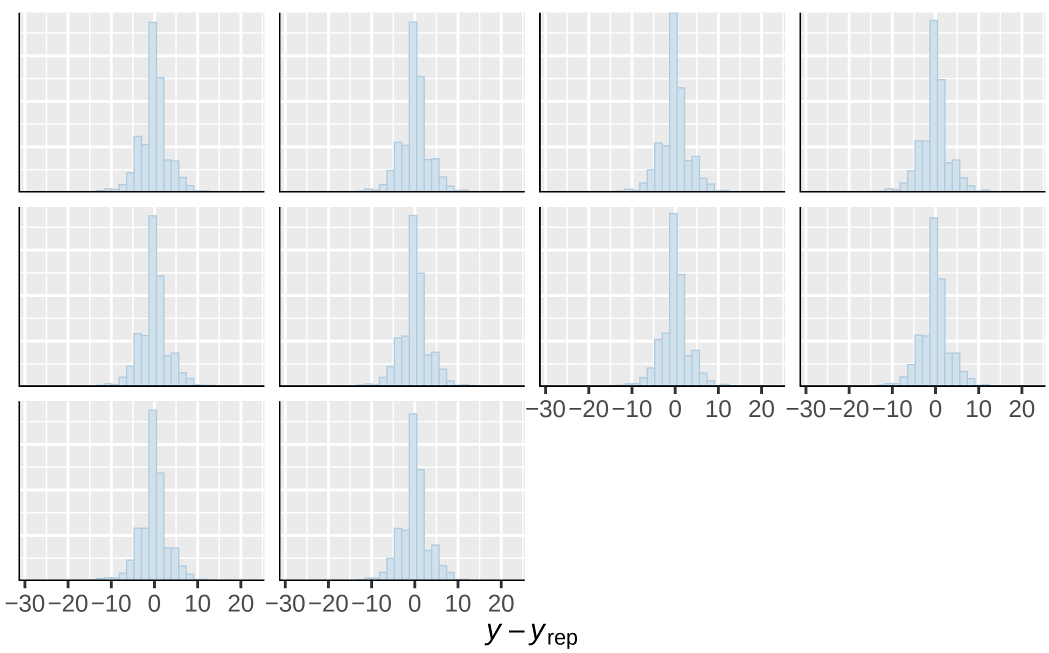




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

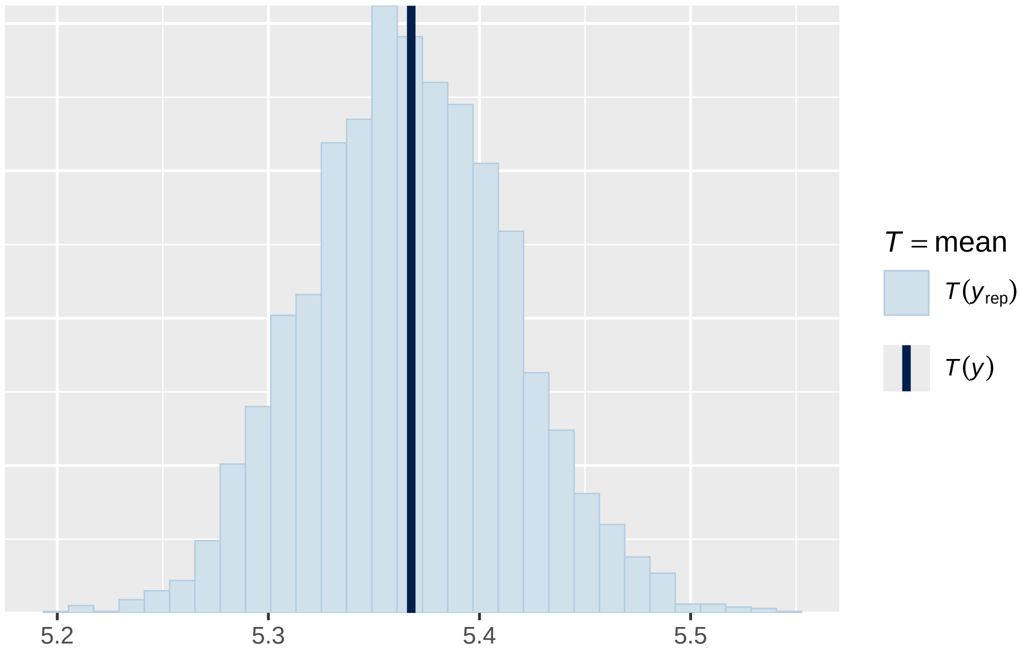



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

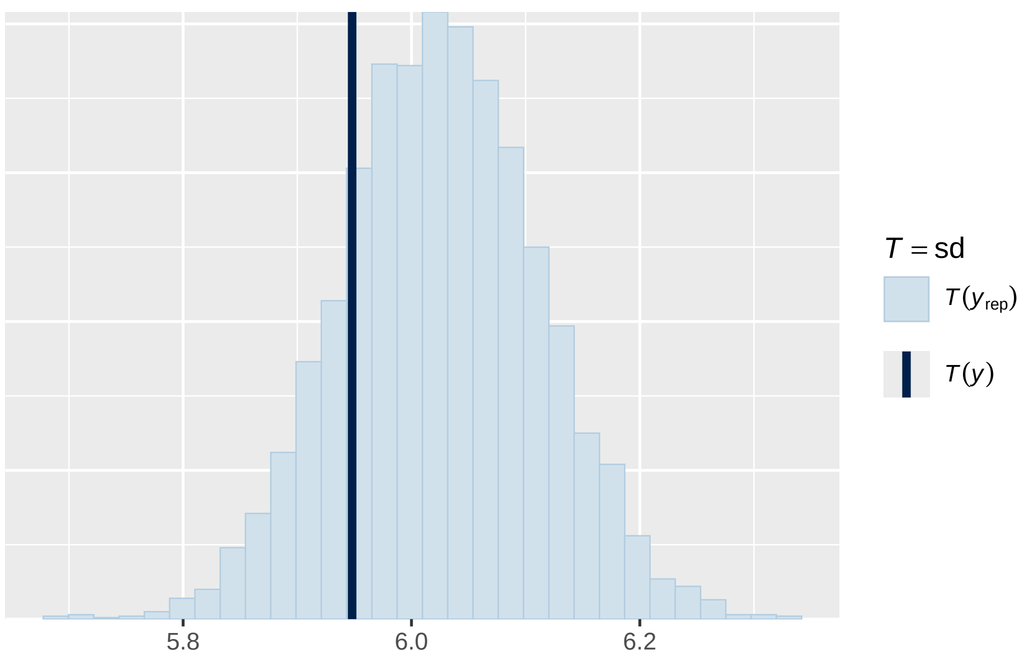



```
Using all posterior draws for ppc type 'stat' by default.
```

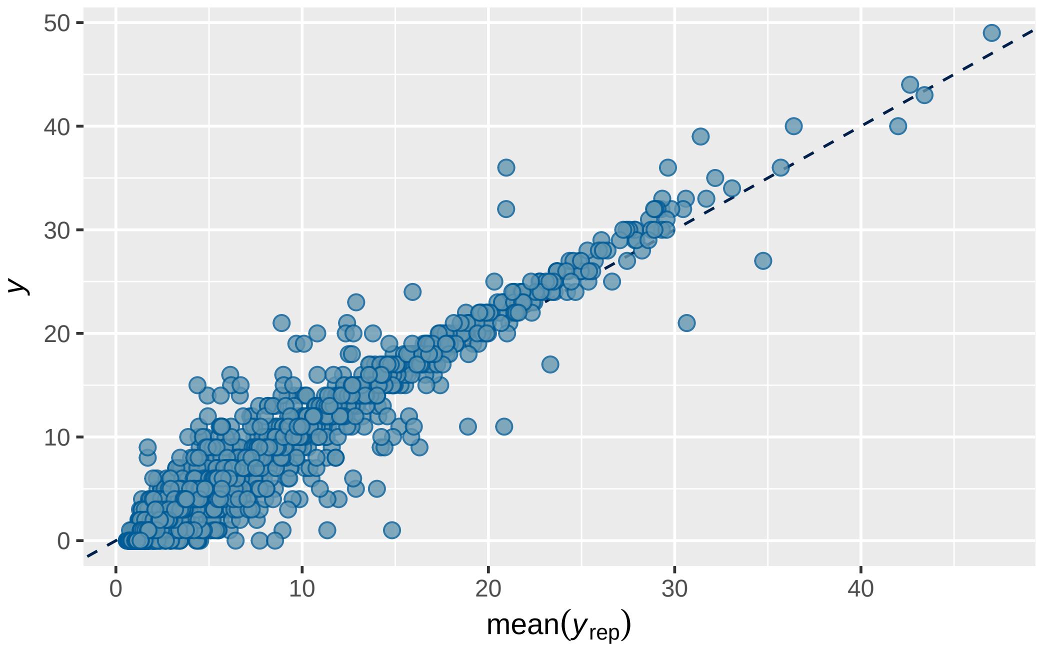
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 3 longitudinal ADHD present output

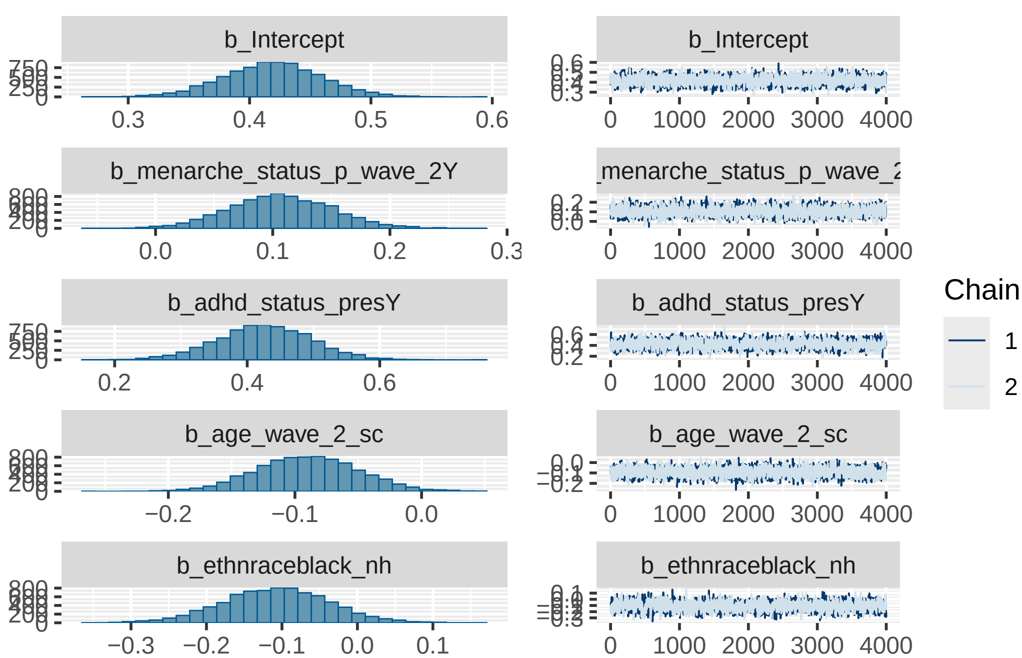
```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.8212615 0.00533859 0.8102805 0.8313783


 Family: poisson
  Links: mu = log
Formula: externalising_wave_3 ~ menarche_status_p_wave_2 * adhd_status_pres + +age_wave_2_sc
   Data: analytic_df_wide (Number of observations: 4496)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3915)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.026     0.021    0.986    1.067 1.002     1495     3351

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.070     0.037    0.006    0.151 1.001      602     1363
```

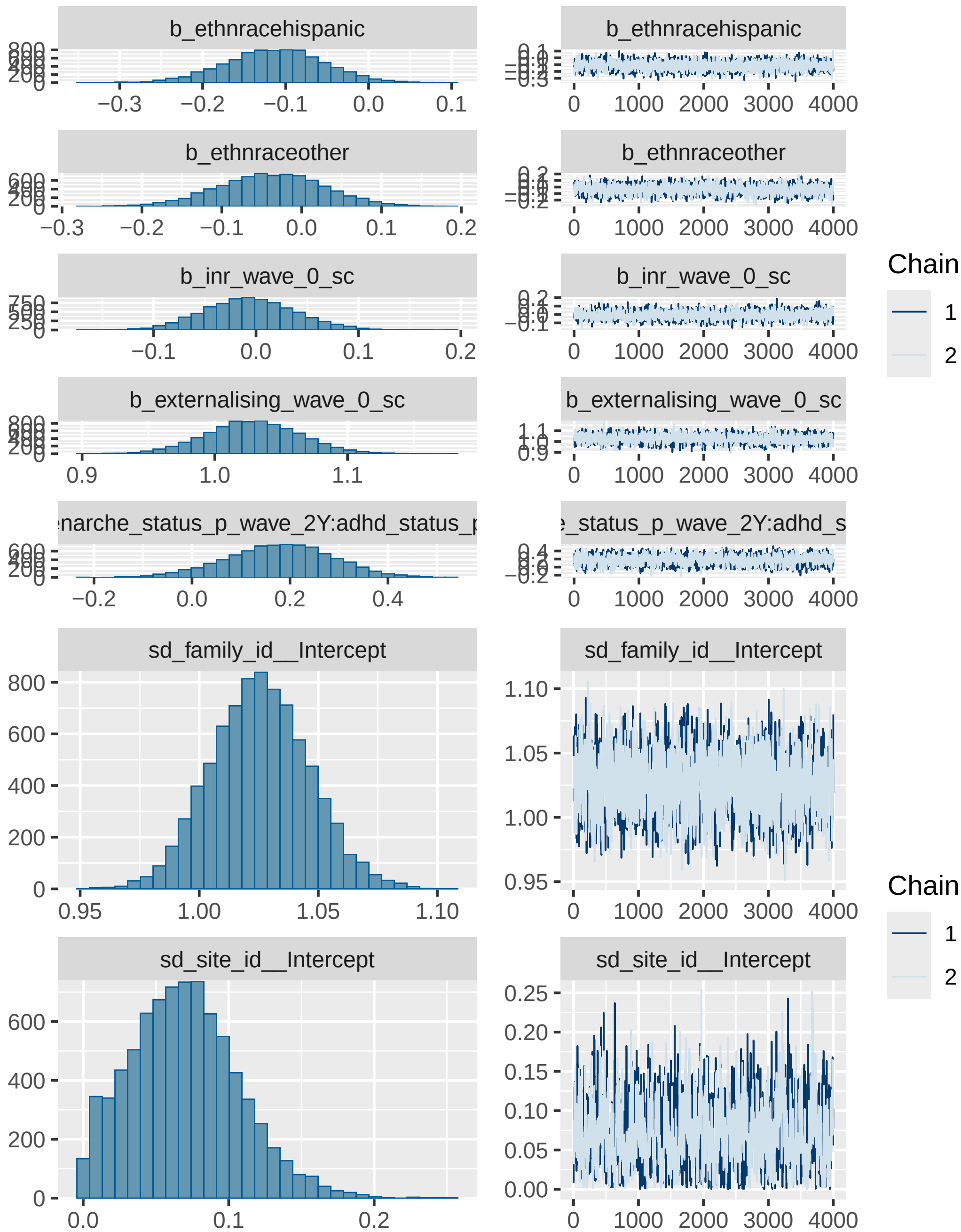
```
Regression Coefficients:
                                         Estimate Est.Error l-95% CI
Intercept                                   0.419     0.040    0.339
menarche_status_p_wave_2Y                   0.107     0.044    0.023
adhd_status_presY                           0.425     0.071    0.285
age_wave_2_sc                              -0.090     0.040   -0.169
ethnraceblack_nh                           -0.108     0.071   -0.248
ethnracehispanic                           -0.113     0.059   -0.231
ethnraceother                              -0.038     0.065   -0.167
inr_wave_0_sc                              -0.006     0.045   -0.091
externalising_wave_0_sc                     1.029     0.035    0.961
menarche_status_p_wave_2Y:adhd_status_presY    0.179     0.108   -0.037
                                         u-95% CI  Rhat Bulk_ESS Tail_ESS
Intercept                                   0.499 1.000     2211     3678
menarche_status_p_wave_2Y                   0.194 1.001     2205     3690
adhd_status_presY                           0.564 1.001     1595     2521
age_wave_2_sc                              -0.012 1.001     2276     3214
ethnraceblack_nh                            0.030 1.003     1493     2467
ethnracehispanic                           -0.001 1.000     2006     3547
ethnraceother                               0.089 1.001     1677     3299
inr_wave_0_sc                               0.085 1.000     1889     3855
externalising_wave_0_sc                     1.097 1.003     1583     2501
menarche_status_p_wave_2Y:adhd_status_presY    0.384 1.002     1491     2666

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

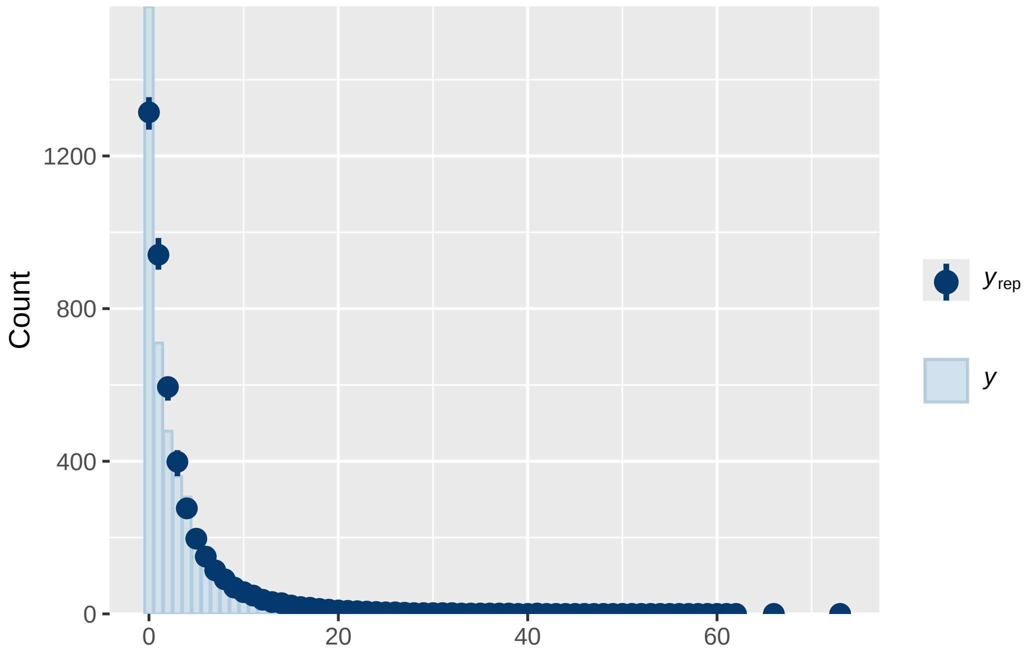
Model 3 longitudinal ADHD present diagnostics

```
       prior     class                                        coef
normal(0, 1)         b
normal(0, 1)         b                           adhd_status_presY
normal(0, 1)         b                               age_wave_2_sc
normal(0, 1)         b                            ethnraceblack_nh
normal(0, 1)         b                            ethnracehispanic
normal(0, 1)         b                               ethnraceother
normal(0, 1)         b                     externalising_wave_0_sc
normal(0, 1)         b                               inr_wave_0_sc
normal(0, 1)         b                   menarche_status_p_wave_2Y
normal(0, 1)         b menarche_status_p_wave_2Y:adhd_status_presY
```

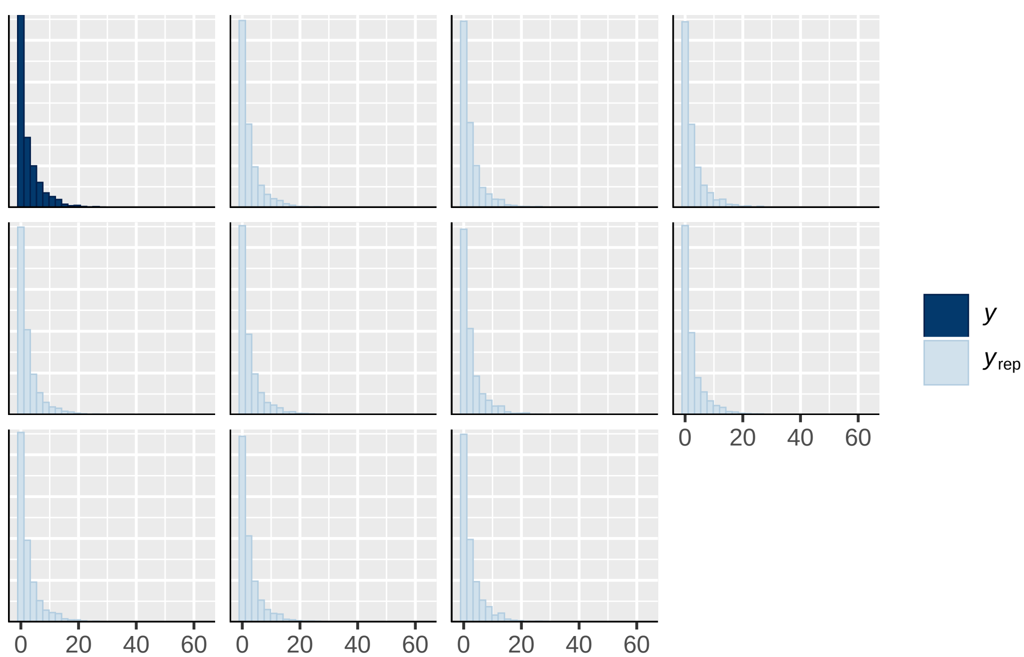
```
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                                            Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd                                            Intercept
    group resp dpar nlpar lb ub tag        source
                                             user
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                     (vectorized)
                                          default
                             0            default
family_id                    0       (vectorized)
family_id                    0       (vectorized)
  site_id                    0       (vectorized)
  site_id                    0       (vectorized)
```



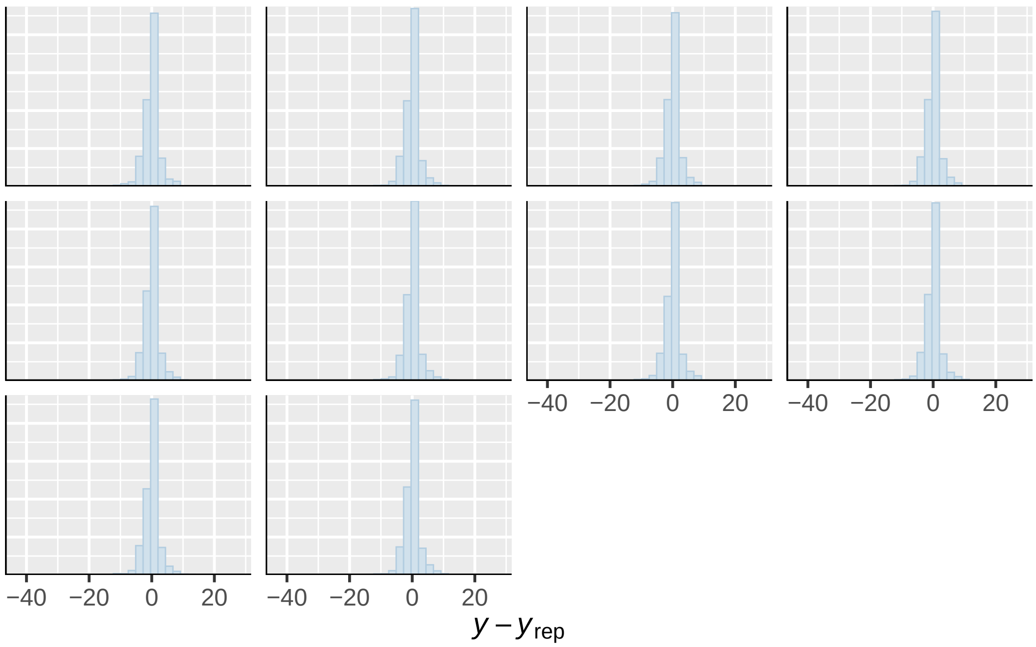

b_ethnracehispanic
b_ethnracehispanic
b_ethnraceother
b_ethnraceother
b_inr_wave_0_sc
b_inr_wave_0_sc
b_externalising_wave_0_sc
b_externalising_wave_0_sc
narche_status_p_wave_2Y:adhd_status_p
_status_p_wave_2Y:adhd_s
sd_family_id__Intercept
sd_family_id__Intercept
sd_site_id__Intercept
sd_site_id__Intercept
0 1000 2000 3000 4000
Chain
1
2
Chain
1
2



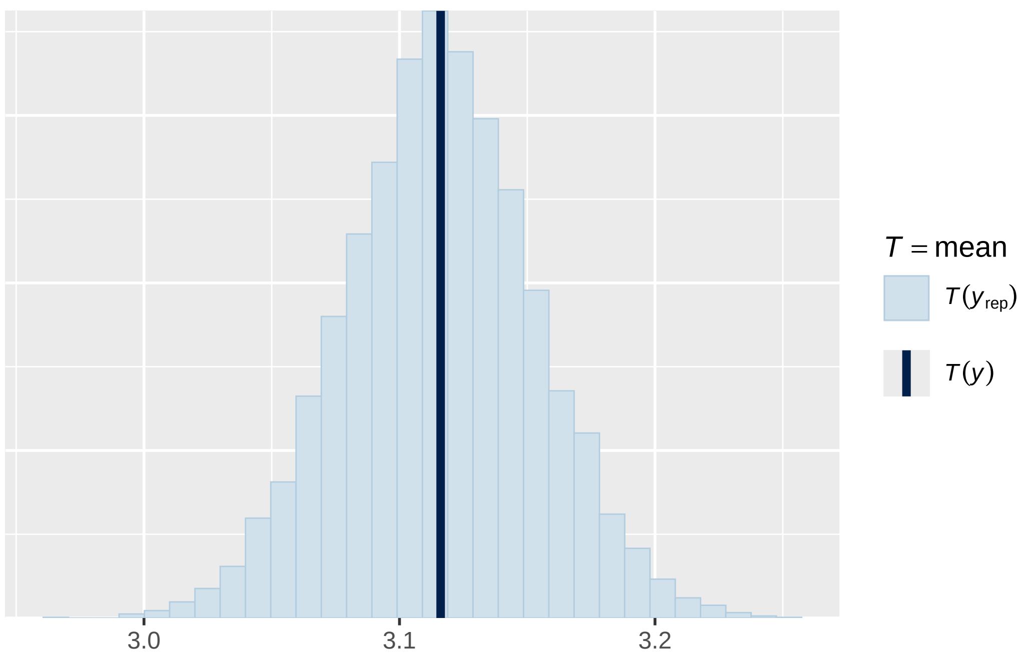

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



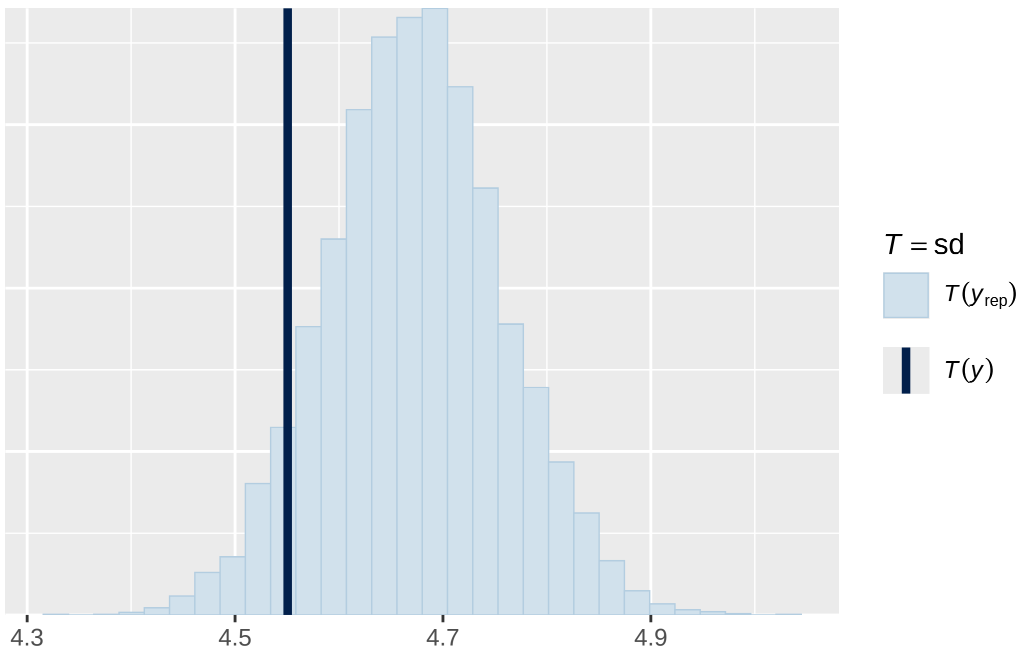

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



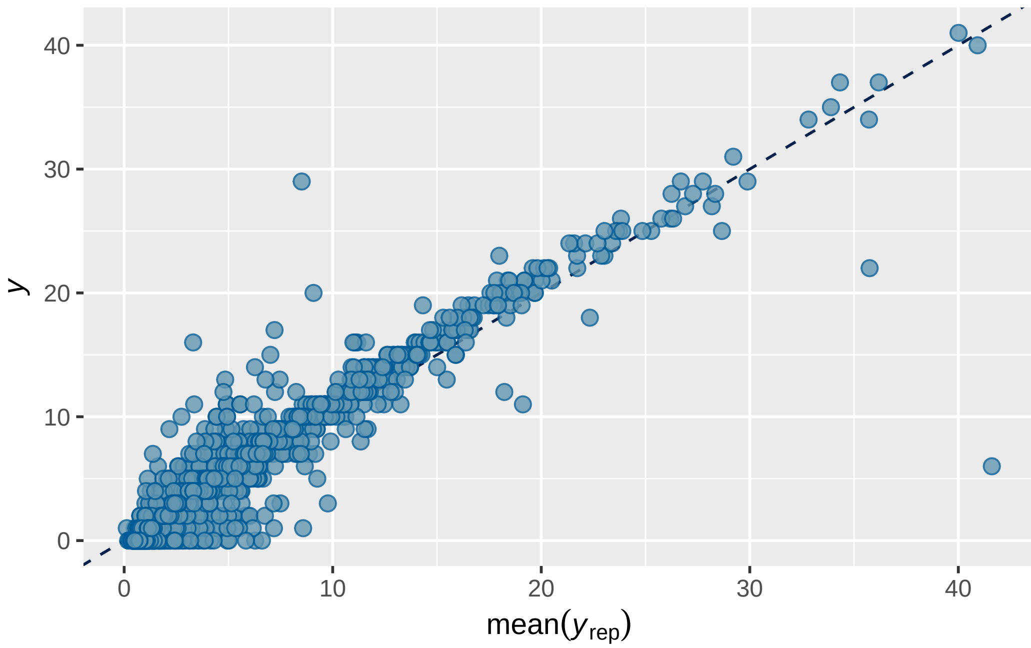

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 3B longitudinal ADHD present output

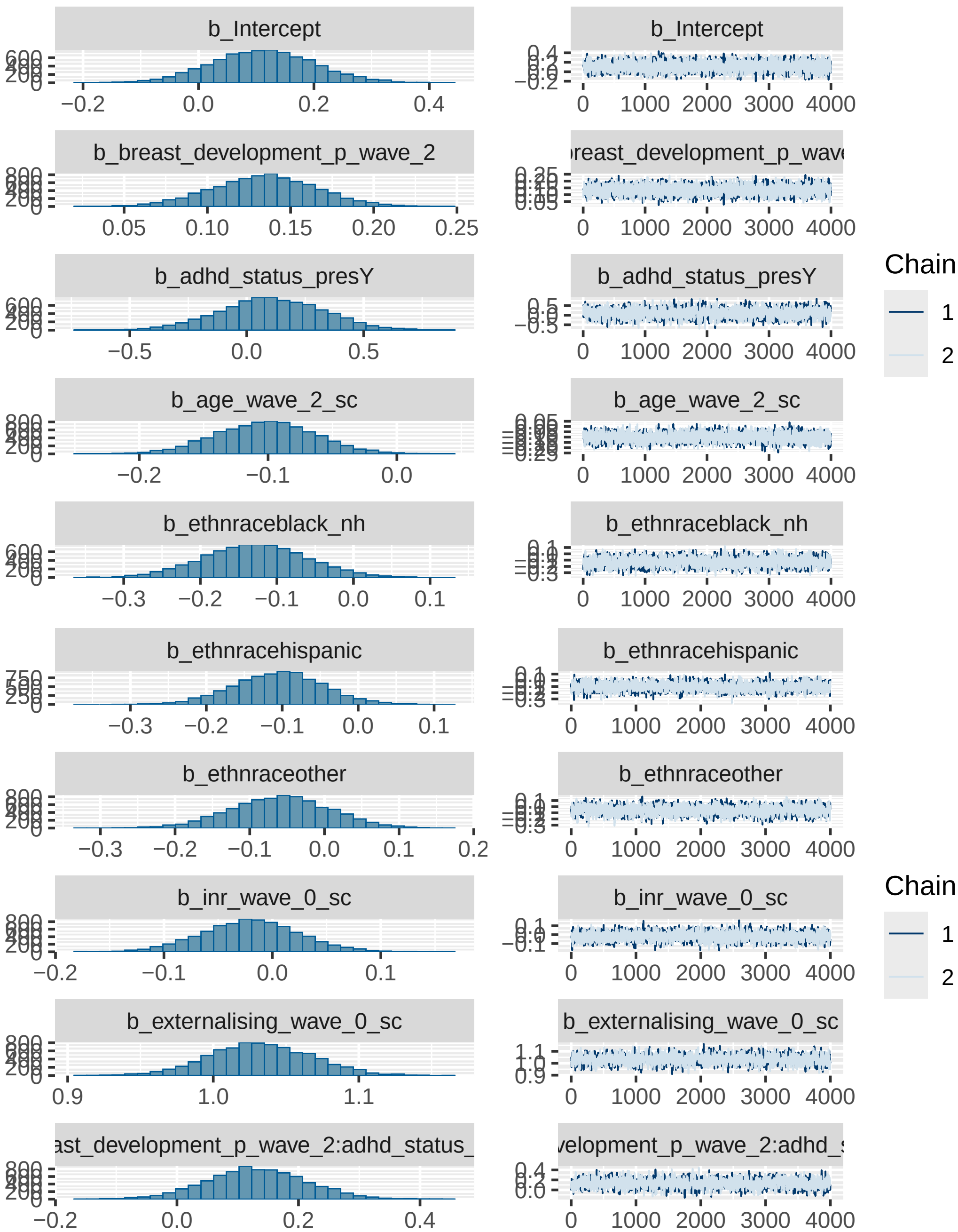
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8238293 0.005235983 0.8129815 0.8334161


 Family: poisson
  Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2 * adhd_status_pres + age_wave_2_s
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.034     0.021    0.994    1.074 1.003     1919     3560

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.070     0.038    0.006    0.149 1.003      603     1245

Regression Coefficients:
                                            Estimate Est.Error l-95% CI
Intercept                                      0.112     0.089   -0.060
breast_development_p_wave_2                    0.134     0.031    0.073
adhd_status_presY                              0.107     0.222   -0.331
age_wave_2_sc                                 -0.102     0.039   -0.178
ethnraceblack_nh                              -0.128     0.068   -0.260
ethnracehispanic                              -0.105     0.058   -0.219
ethnraceother                                 -0.059     0.067   -0.191
inr_wave_0_sc                                 -0.020     0.044   -0.107
externalising_wave_0_sc                        1.033     0.035    0.965
breast_development_p_wave_2:adhd_status_presY  0.131     0.081   -0.028
                                            u-95% CI  Rhat Bulk_ESS Tail_ESS
Intercept                                      0.289 1.000     2241     3971
breast_development_p_wave_2                    0.196 1.001     2257     4052
adhd_status_presY                              0.537 1.001     1455     2891
age_wave_2_sc                                 -0.025 1.001     1864     3311
ethnraceblack_nh                               0.005 1.001     1444     2701
ethnracehispanic                               0.010 1.002     1820     3539
ethnraceother                                  0.069 1.001     1609     3337
inr_wave_0_sc                                  0.068 1.001     1776     3324
```

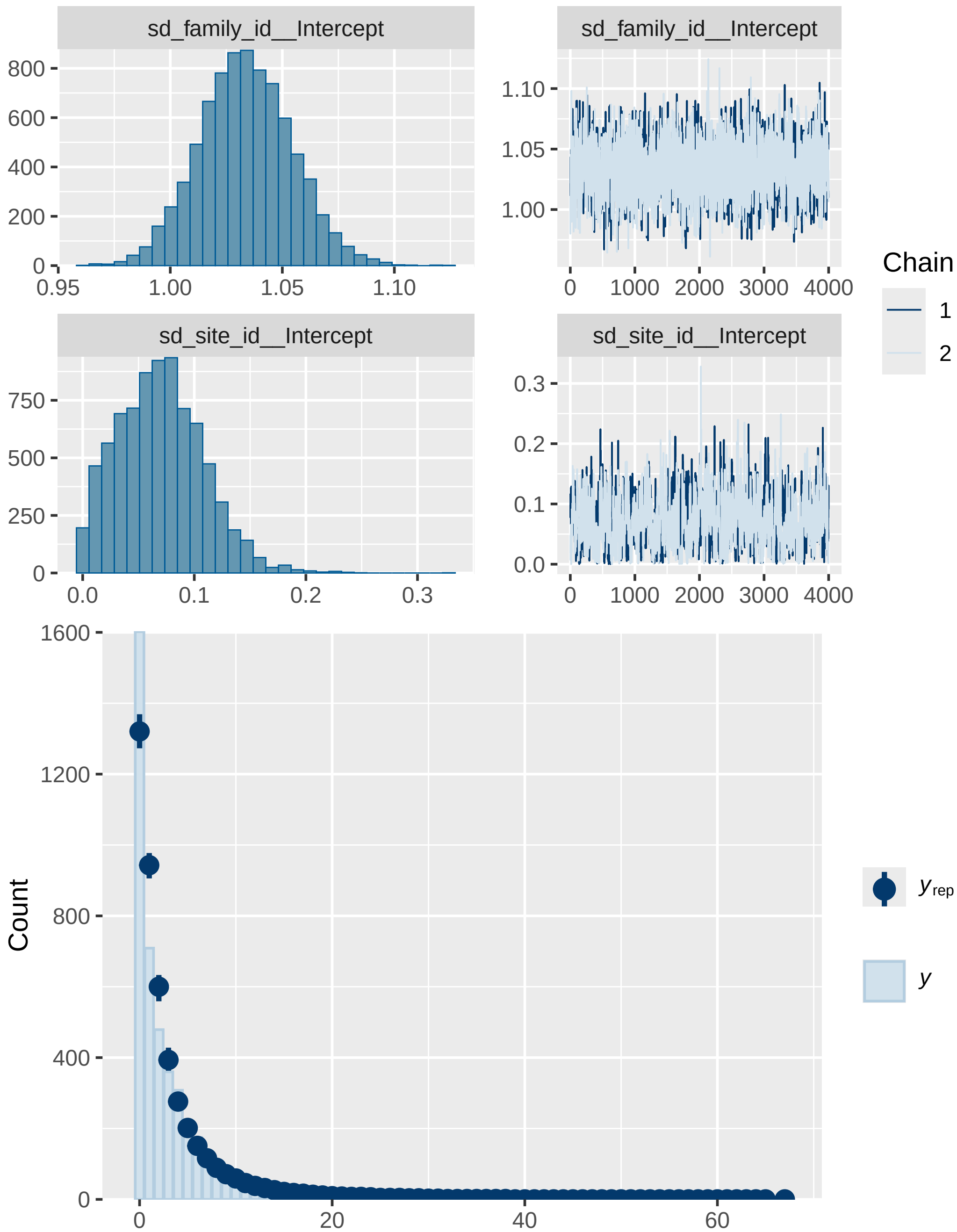
```
externalising_wave_0_sc                           1.100 1.003     1240     2642
breast_development_p_wave_2:adhd_status_presY    0.289 1.001     1403     2937

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 3B longitudinal ADHD present diagnostics

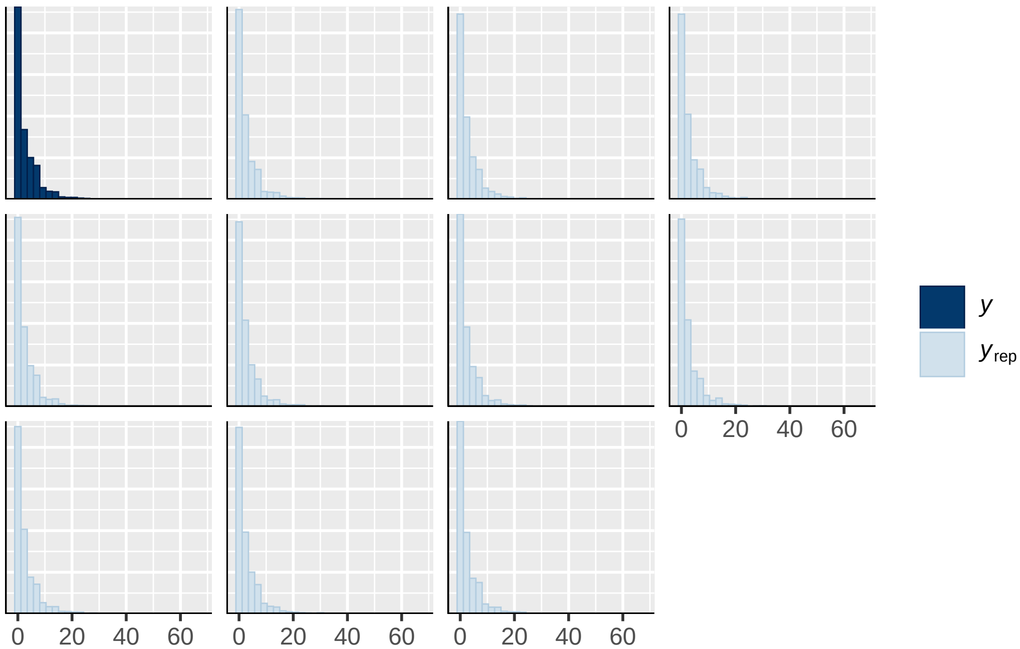
```
              prior     class                                          coef
       normal(0, 1)         b
       normal(0, 1)         b                              adhd_status_presY
       normal(0, 1)         b                                  age_wave_2_sc
       normal(0, 1)         b                    breast_development_p_wave_2
       normal(0, 1)         b breast_development_p_wave_2:adhd_status_presY
       normal(0, 1)         b                               ethnraceblack_nh
       normal(0, 1)         b                               ethnracehispanic
       normal(0, 1)         b                                  ethnraceother
       normal(0, 1)         b                        externalising_wave_0_sc
       normal(0, 1)         b                                  inr_wave_0_sc
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                                      Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                                      Intercept
     group resp dpar nlpar lb ub tag        source
                                              user
                                      (vectorized)
                                      (vectorized)
                                      (vectorized)
                                      (vectorized)
                                      (vectorized)
                                      (vectorized)
                                      (vectorized)
                                      (vectorized)
                                      (vectorized)
                                           default
                              0            default
 family_id                    0       (vectorized)
 family_id                    0       (vectorized)
   site_id                    0       (vectorized)
```

site_id 0 (vectorized)

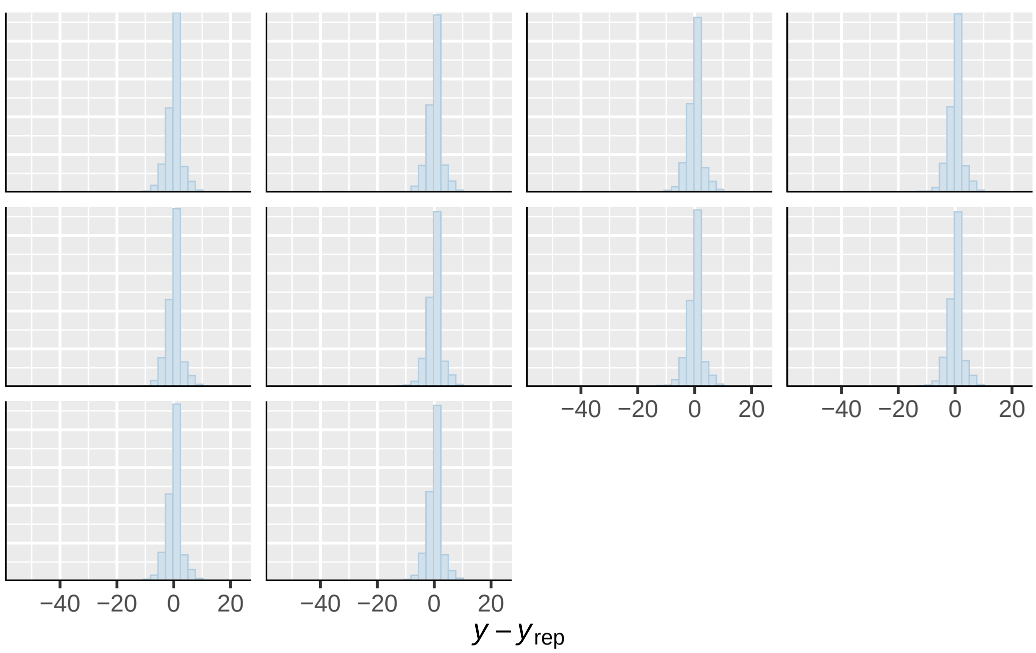




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

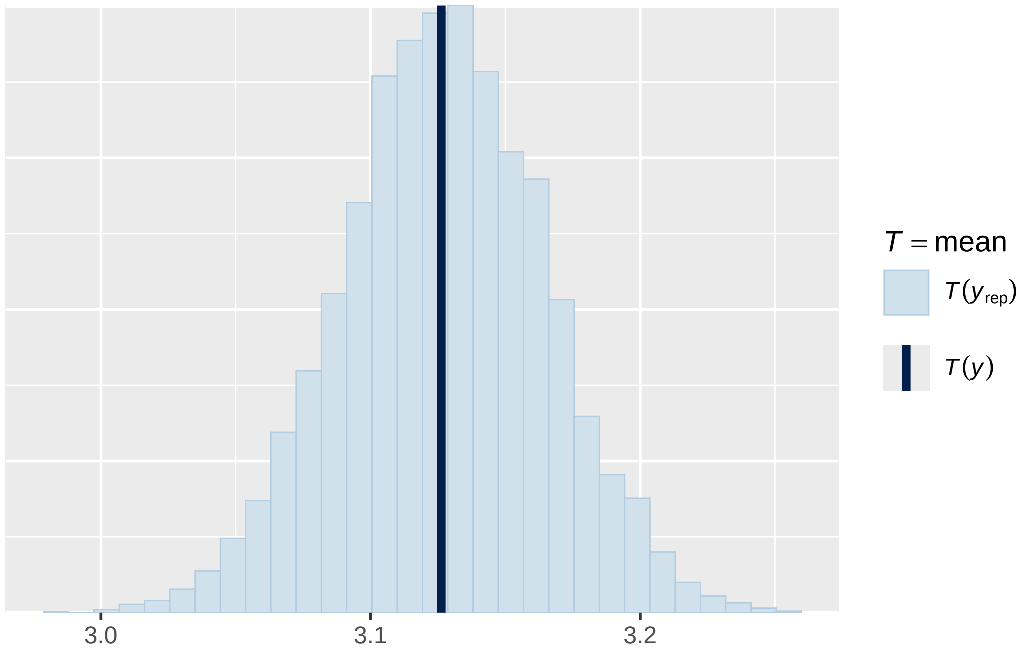


Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

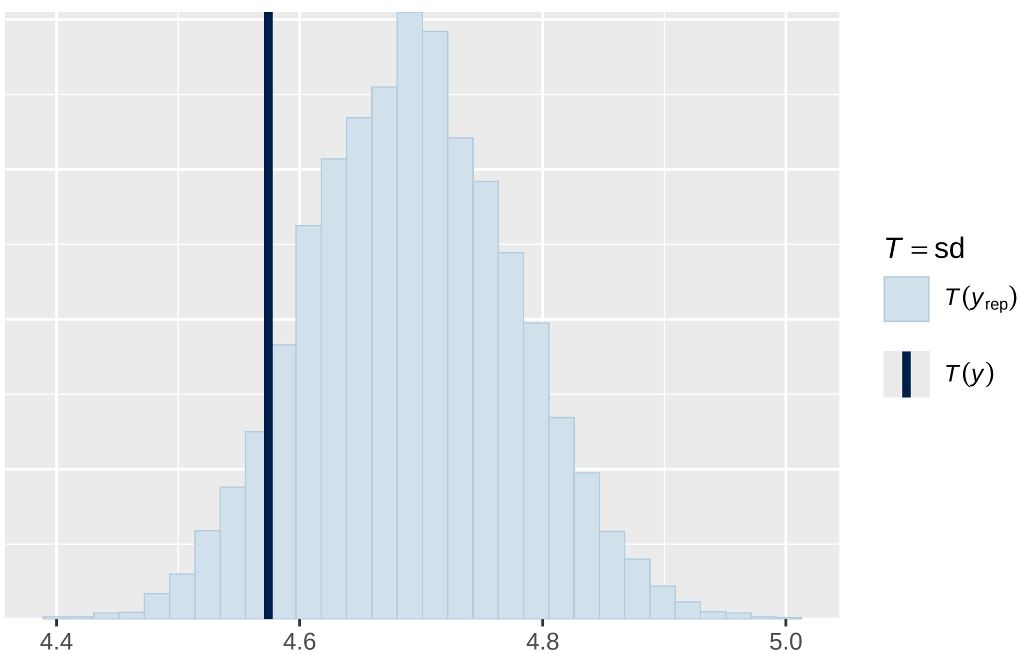


Using all posterior draws for ppc type 'stat' by default.

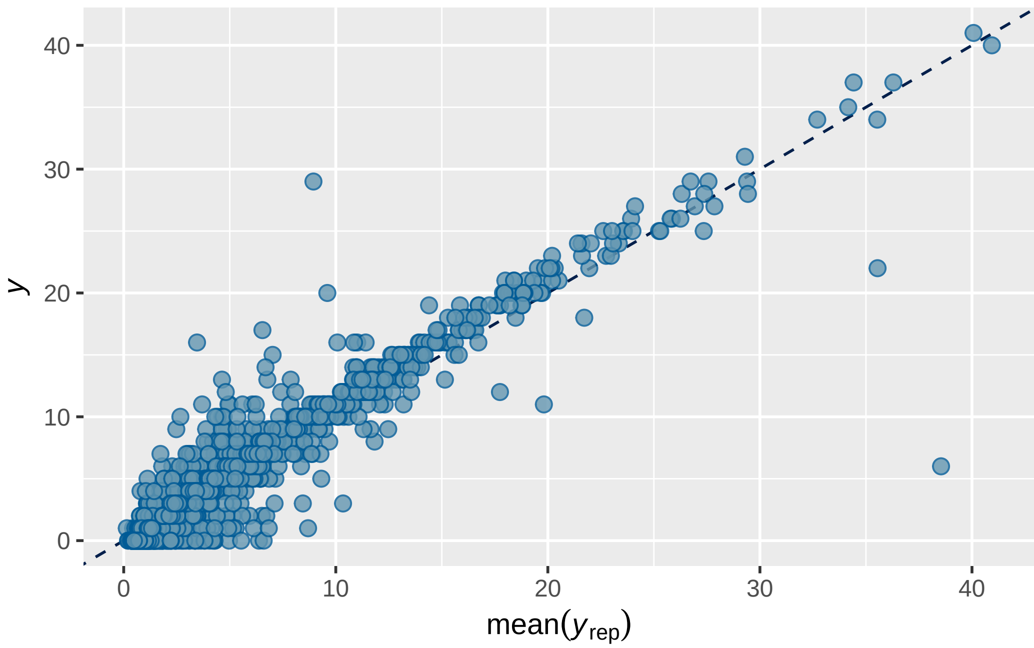
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 1 longitudinal observed output

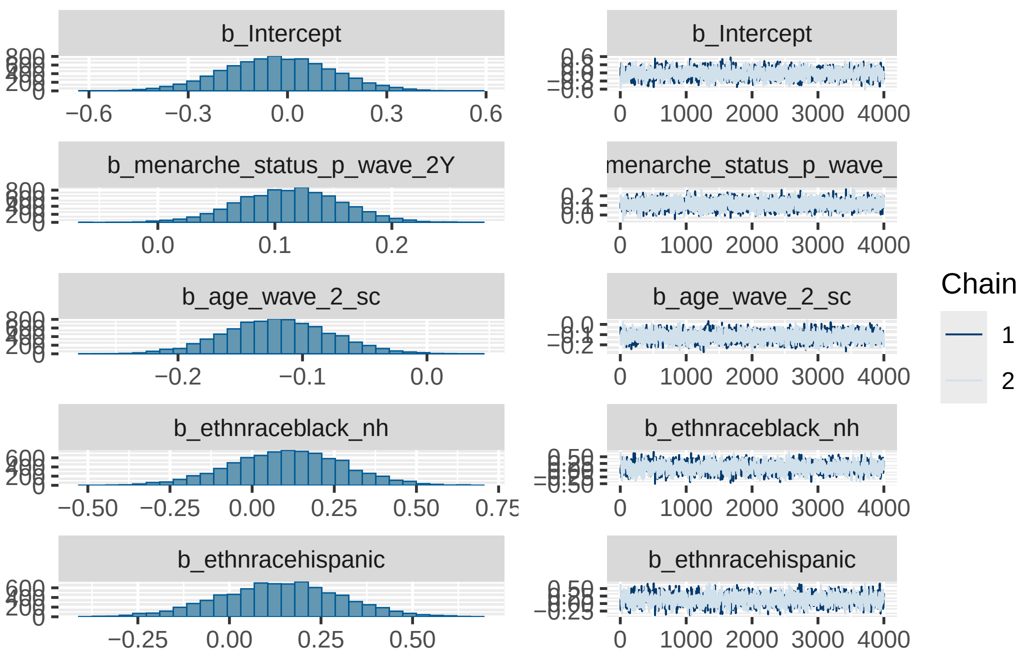
```
   Estimate   Est.Error      Q2.5     Q97.5
R2 0.739703 0.007972974 0.7235204 0.7545925


 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave_
   Data: obs_df_wide (Number of observations: 4043)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3513)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.878     0.022    0.836    0.920 1.001     2291     4325

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.076     0.039    0.007    0.157 1.004      577      872
```

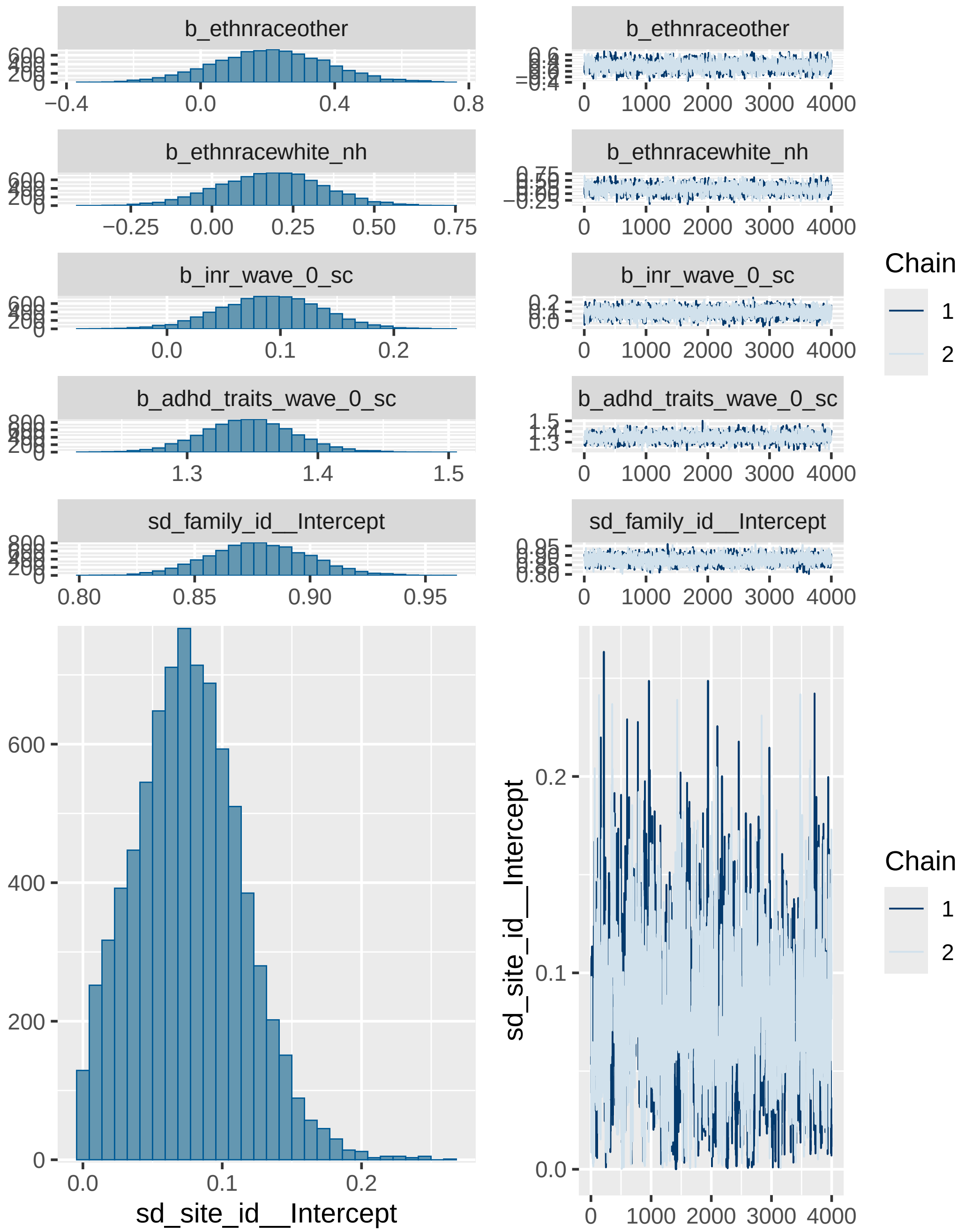
```
Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                   -0.029     0.162   -0.347    0.289 1.000     1338
menarche_status_p_wave_2Y    0.113     0.044    0.027    0.197 1.000     2368
age_wave_2_sc               -0.120     0.043   -0.204   -0.036 1.001     2169
ethnraceblack_nh             0.107     0.171   -0.236    0.442 1.000     1313
ethnracehispanic             0.139     0.167   -0.193    0.464 1.001     1320
ethnraceother                0.210     0.169   -0.124    0.544 1.000     1386
ethnracewhite_nh             0.181     0.161   -0.137    0.495 1.000     1305
inr_wave_0_sc                0.091     0.045    0.002    0.178 1.000     2203
adhd_traits_wave_0_sc        1.347     0.035    1.278    1.418 1.002     1498
                          Tail_ESS
Intercept                     2348
menarche_status_p_wave_2Y     3964
age_wave_2_sc                 3917
ethnraceblack_nh              2618
ethnracehispanic              2387
ethnraceother                 2341
ethnracewhite_nh              2164
inr_wave_0_sc                 3842
adhd_traits_wave_0_sc         2589

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 1 longitudinal observed diagnostics

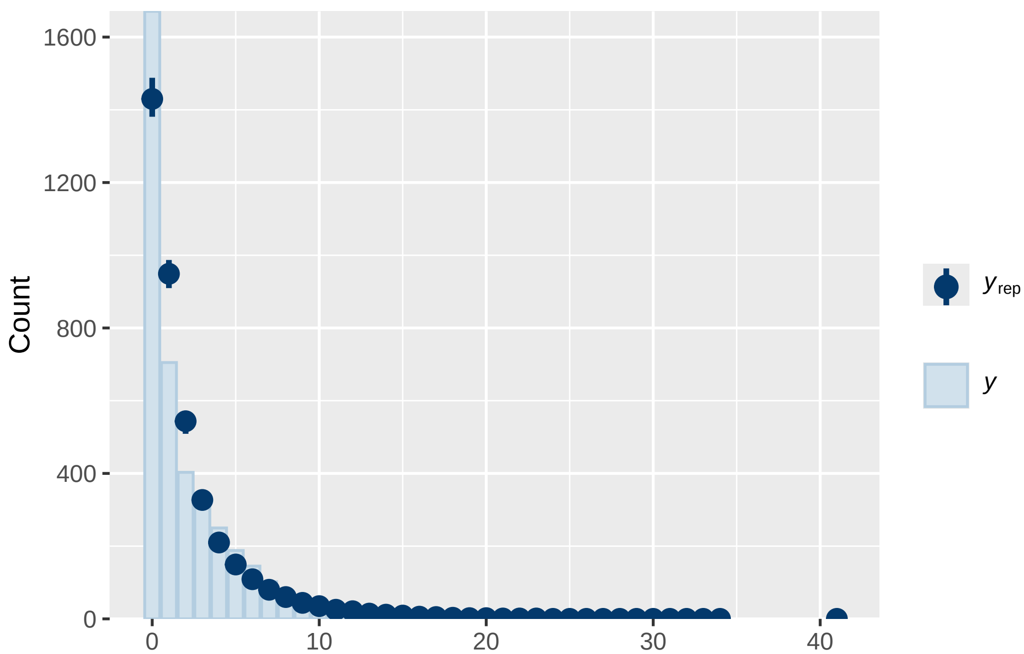
```
                prior     class                      coef     group resp dpar
         normal(0, 1)         b
         normal(0, 1)         b     adhd_traits_wave_0_sc
         normal(0, 1)         b             age_wave_2_sc
         normal(0, 1)         b          ethnraceblack_nh
         normal(0, 1)         b          ethnracehispanic
         normal(0, 1)         b             ethnraceother
         normal(0, 1)         b          ethnracewhite_nh
         normal(0, 1)         b             inr_wave_0_sc
         normal(0, 1)         b menarche_status_p_wave_2Y
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd                           family_id
```

```
student_t(3, 0, 2.9)        sd                   Intercept family_id
student_t(3, 0, 2.9)        sd                             site_id
student_t(3, 0, 2.9)        sd                   Intercept   site_id
nlpar lb ub tag       source
                        user
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                     default
       0             default
       0        (vectorized)
       0        (vectorized)
       0        (vectorized)
       0        (vectorized)
```

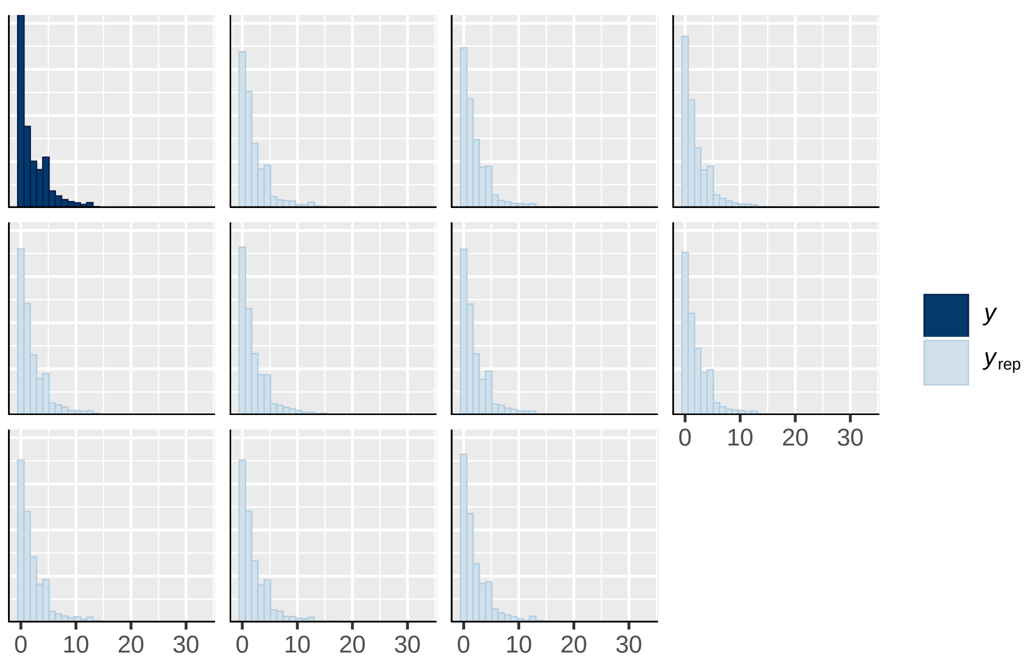




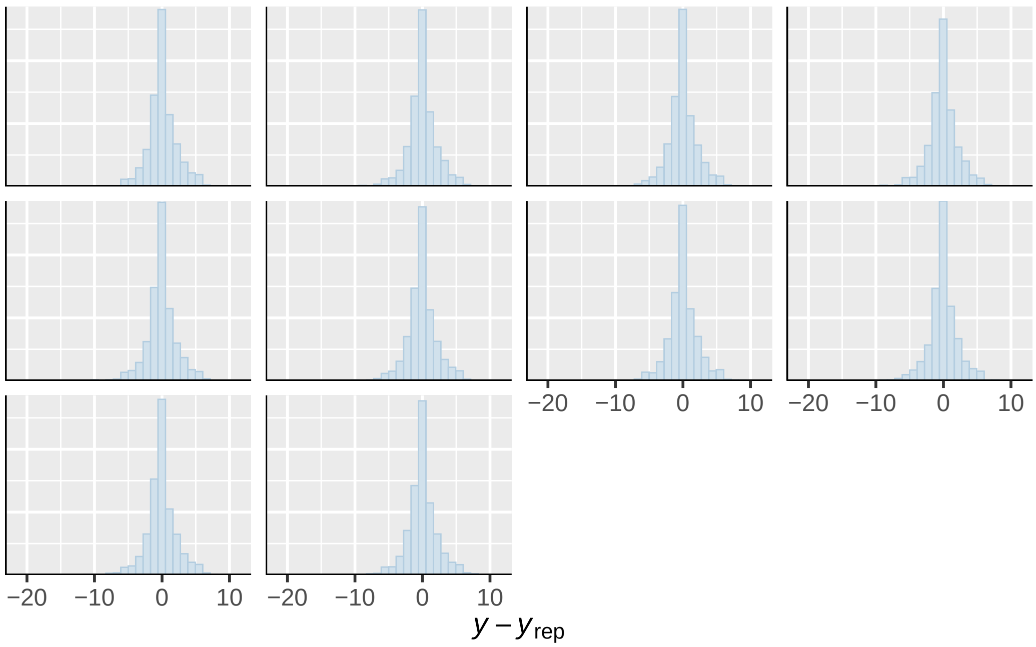
b_ethnraceother
b_ethnracewhite_nh
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
sd_family_id__Intercept
sd_site_id__Intercept
Chain
1
2



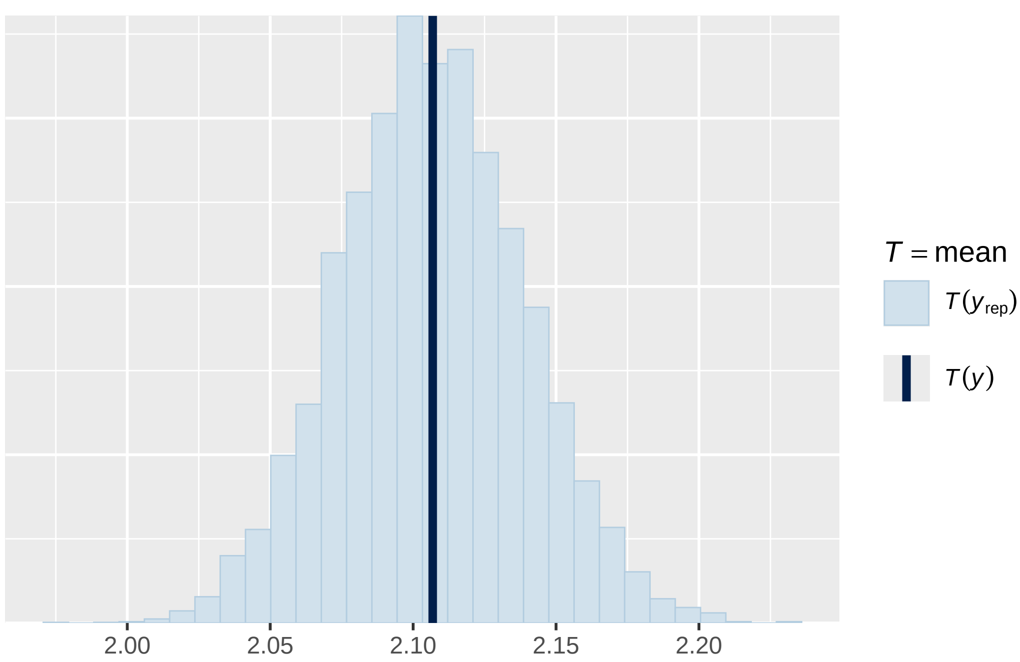

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



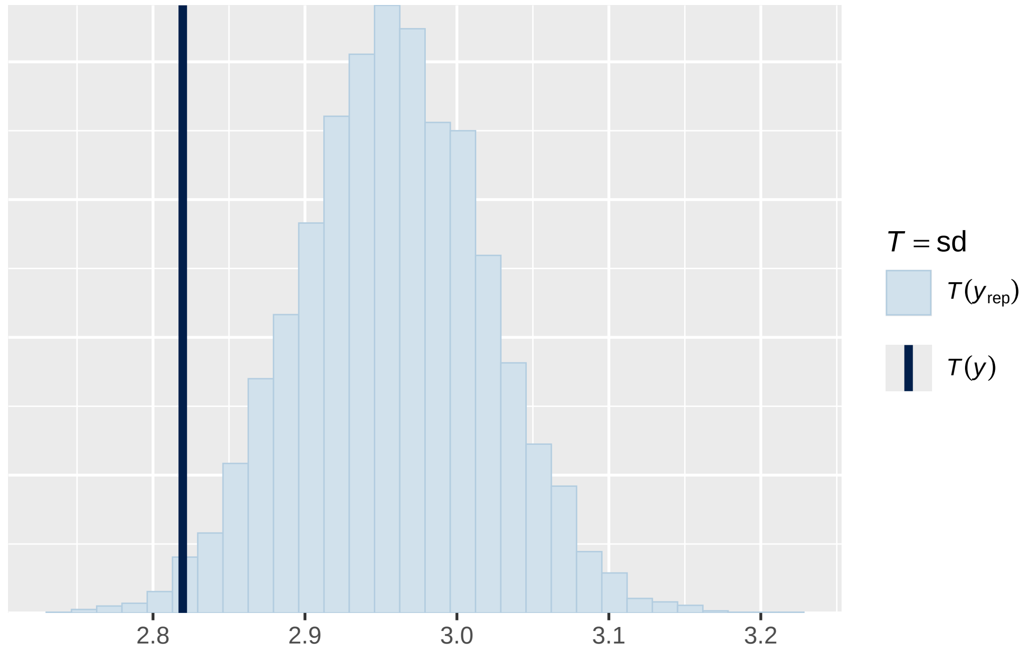

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



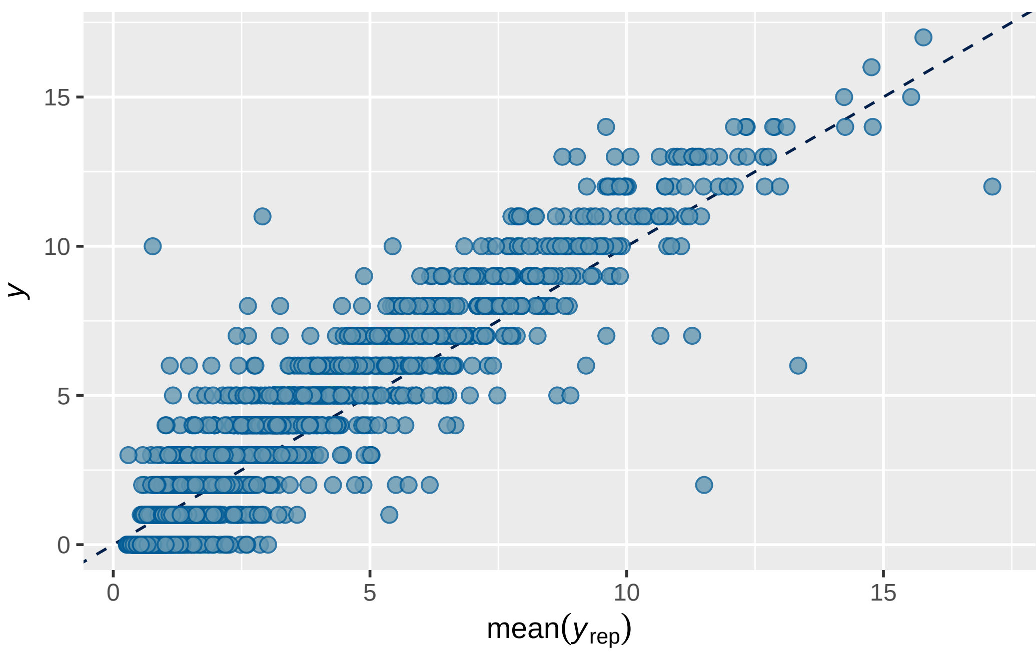

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



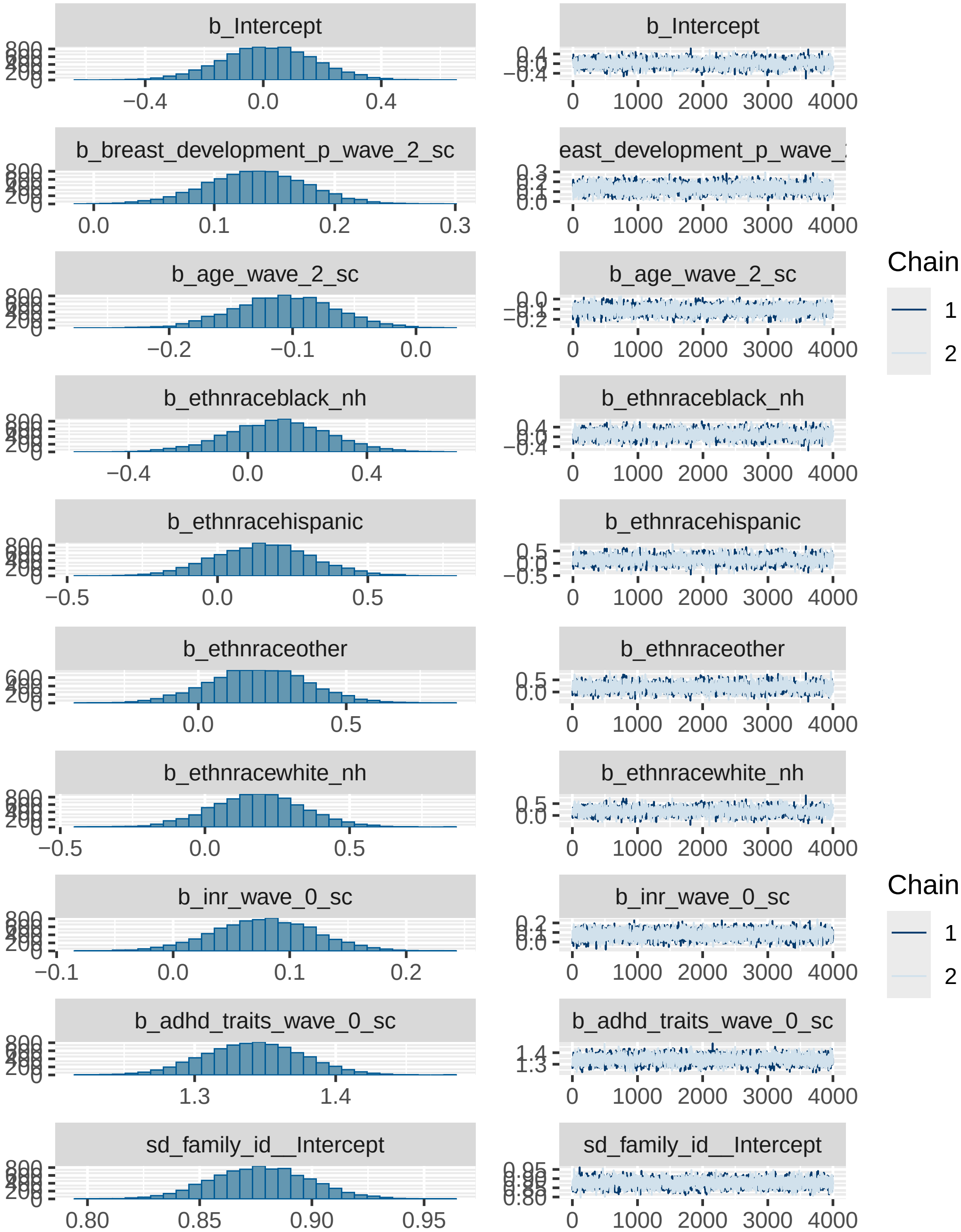
Model 1B longitudinal observed output

```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7393405 0.007815485 0.7235745 0.7541467


 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + in
   Data: obs_df_wide (Number of observations: 4065)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3531)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.877     0.022    0.835    0.921 1.000     2244     3432

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.083     0.037    0.011    0.158 1.001      839     1246

Regression Coefficients:
                              Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                        0.017     0.158   -0.295    0.325 1.000
breast_development_p_wave_2_sc    0.135     0.042    0.051    0.217 1.000
age_wave_2_sc                   -0.106     0.041   -0.186   -0.027 1.000
ethnraceblack_nh                 0.102     0.168   -0.234    0.434 1.000
ethnracehispanic                 0.154     0.164   -0.164    0.479 1.000
ethnraceother                    0.204     0.168   -0.122    0.534 1.000
ethnracewhite_nh                 0.184     0.158   -0.128    0.496 1.000
inr_wave_0_sc                    0.079     0.044   -0.008    0.163 1.000
adhd_traits_wave_0_sc            1.342     0.035    1.273    1.411 1.001
                              Bulk_ESS Tail_ESS
Intercept                         1379     3095
breast_development_p_wave_2_sc     2735     4430
age_wave_2_sc                     2358     4013
ethnraceblack_nh                  1320     2801
ethnracehispanic                  1386     3053
ethnraceother                     1404     2800
ethnracewhite_nh                  1360     3056
inr_wave_0_sc                     2669     4402
adhd_traits_wave_0_sc             1976     3678
```

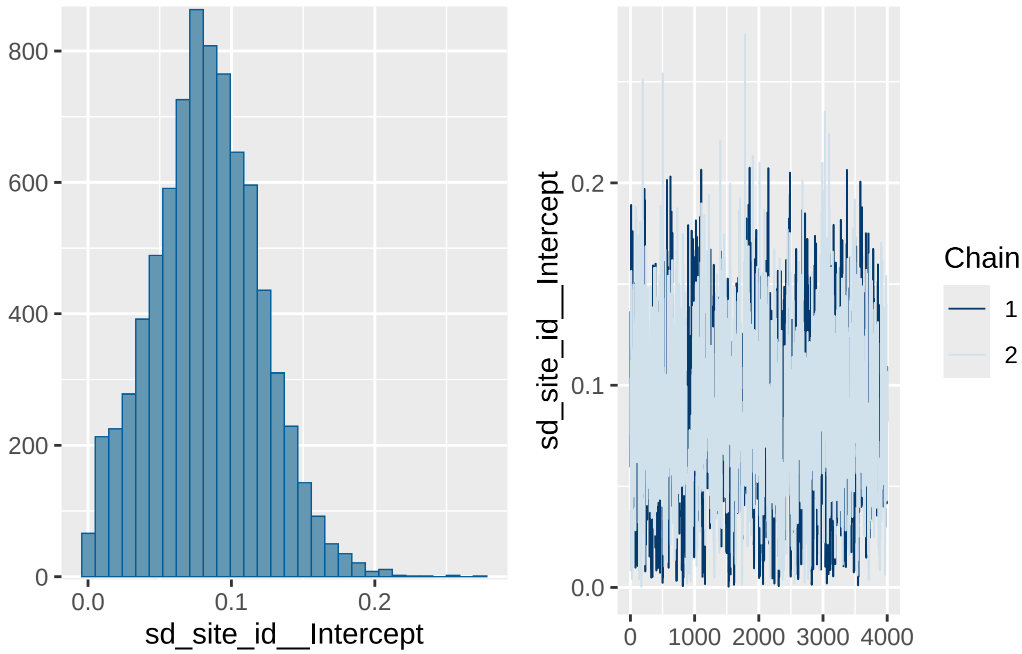
```
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

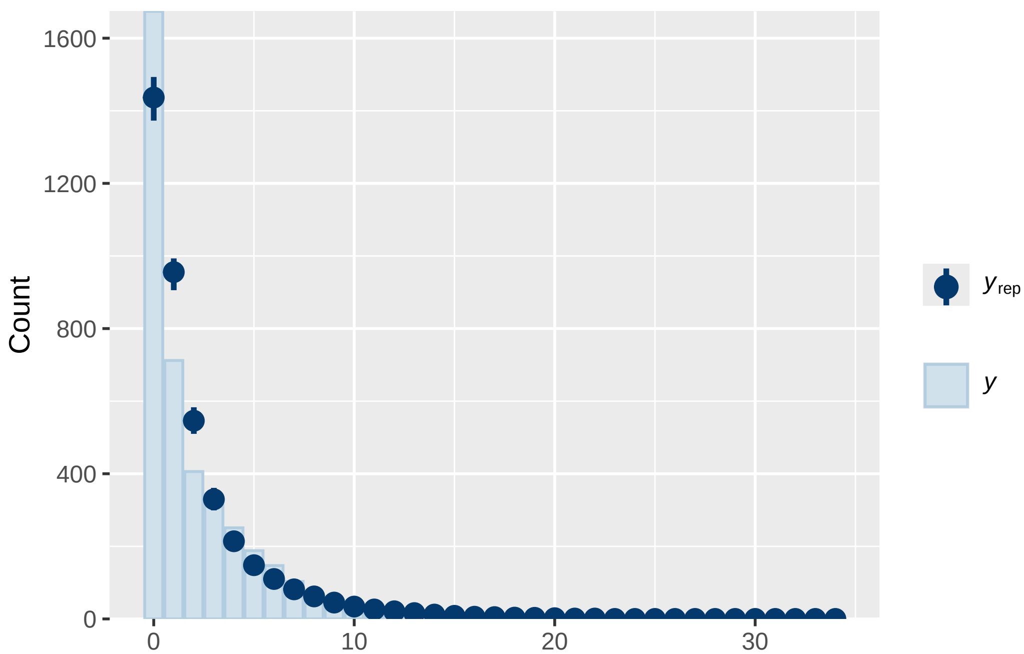
Model 1B longitudinal observed diagnostics

```
                prior     class                            coef     group resp
         normal(0, 1)         b
         normal(0, 1)         b           adhd_traits_wave_0_sc
         normal(0, 1)         b                   age_wave_2_sc
         normal(0, 1)         b breast_development_p_wave_2_sc
         normal(0, 1)         b                ethnraceblack_nh
         normal(0, 1)         b                ethnracehispanic
         normal(0, 1)         b                   ethnraceother
         normal(0, 1)         b                ethnracewhite_nh
         normal(0, 1)         b                   inr_wave_0_sc
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                                 family_id
 student_t(3, 0, 2.7)        sd                       Intercept family_id
 student_t(3, 0, 2.7)        sd                                   site_id
 student_t(3, 0, 2.7)        sd                       Intercept   site_id
 dpar nlpar lb ub tag       source
                              user
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                           default
             0             default
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
```

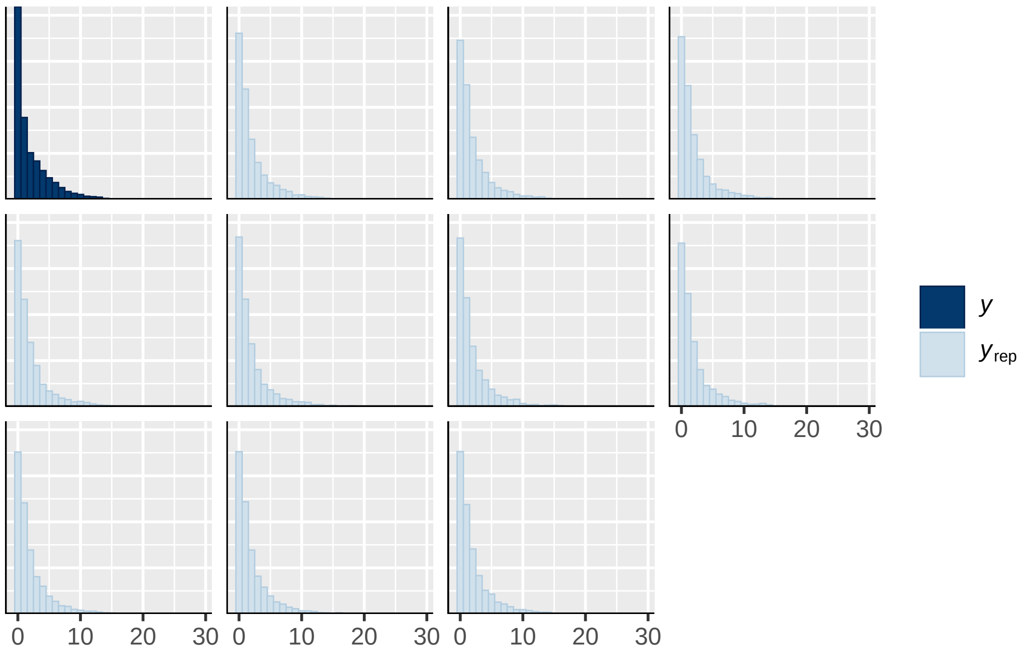


b_Intercept
b_breast_development_p_wave_2_sc
b_age_wave_2_sc
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnraceother
b_ethnracewhite_nh
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
sd_family_id__Intercept
Chain
1
2

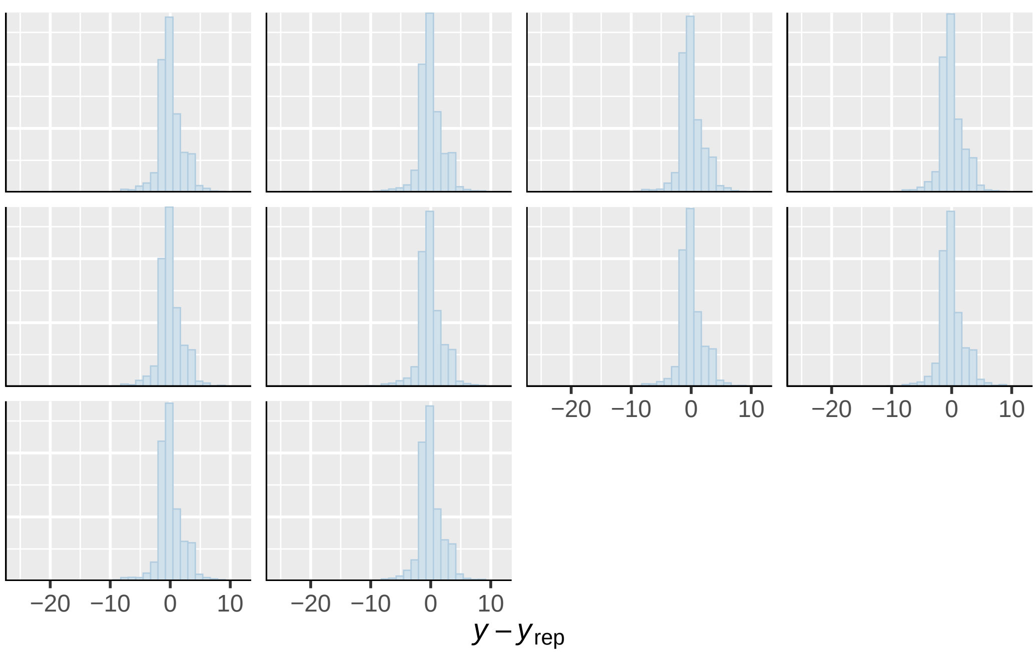




`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

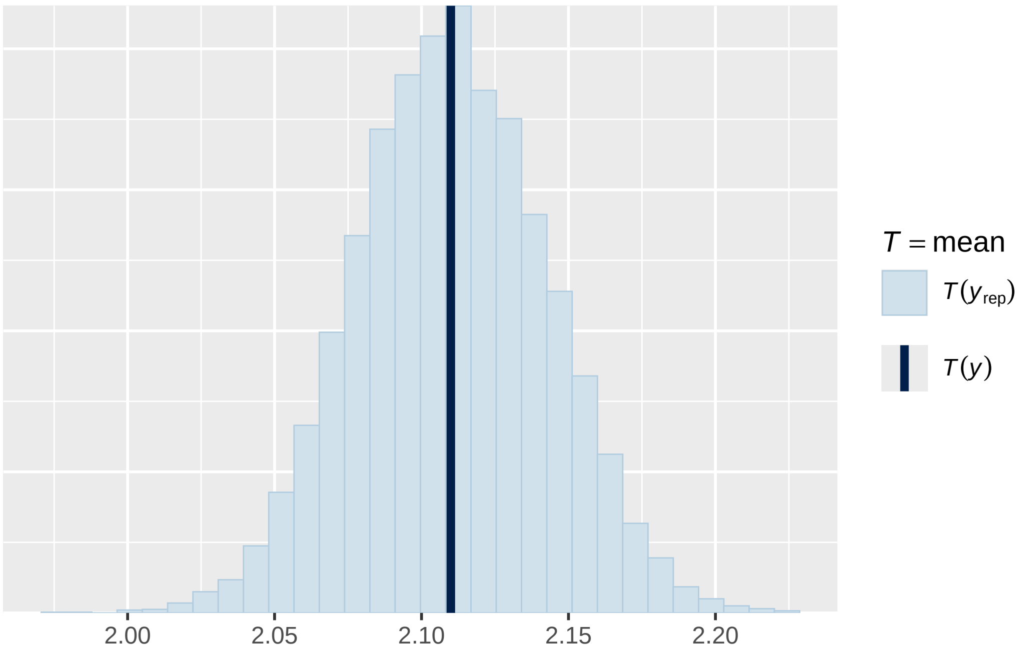



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

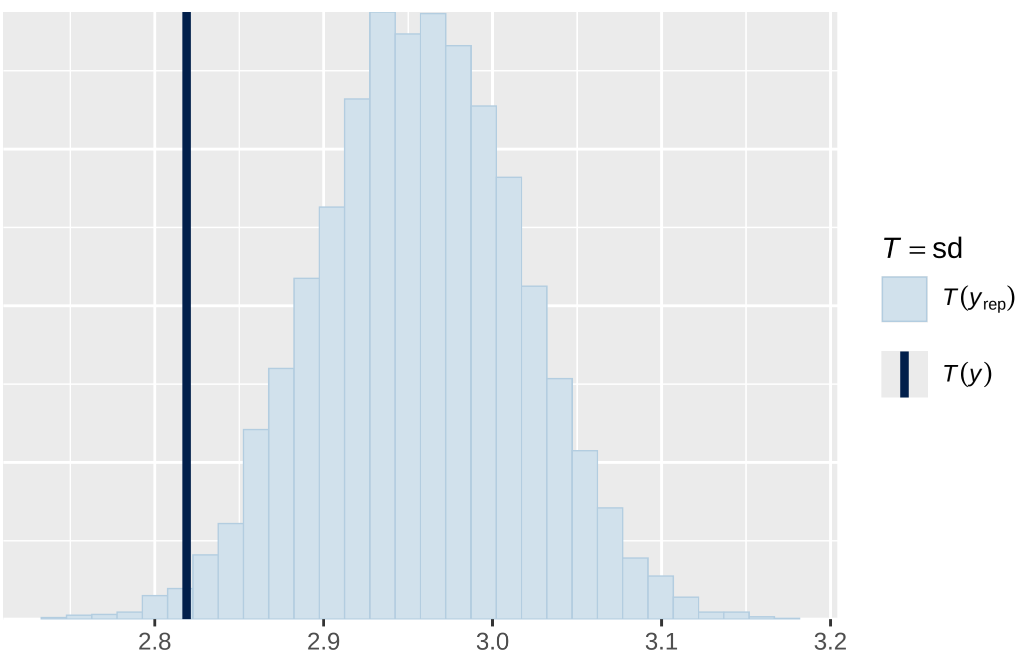



```
Using all posterior draws for ppc type 'stat' by default.
```

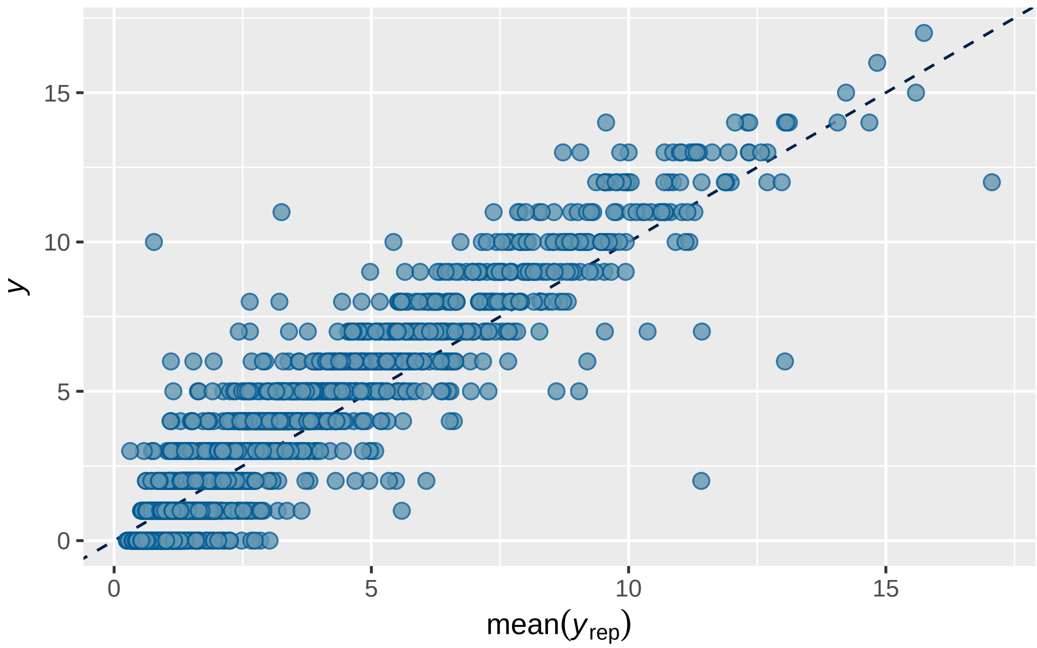
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5B longitudinal observed output

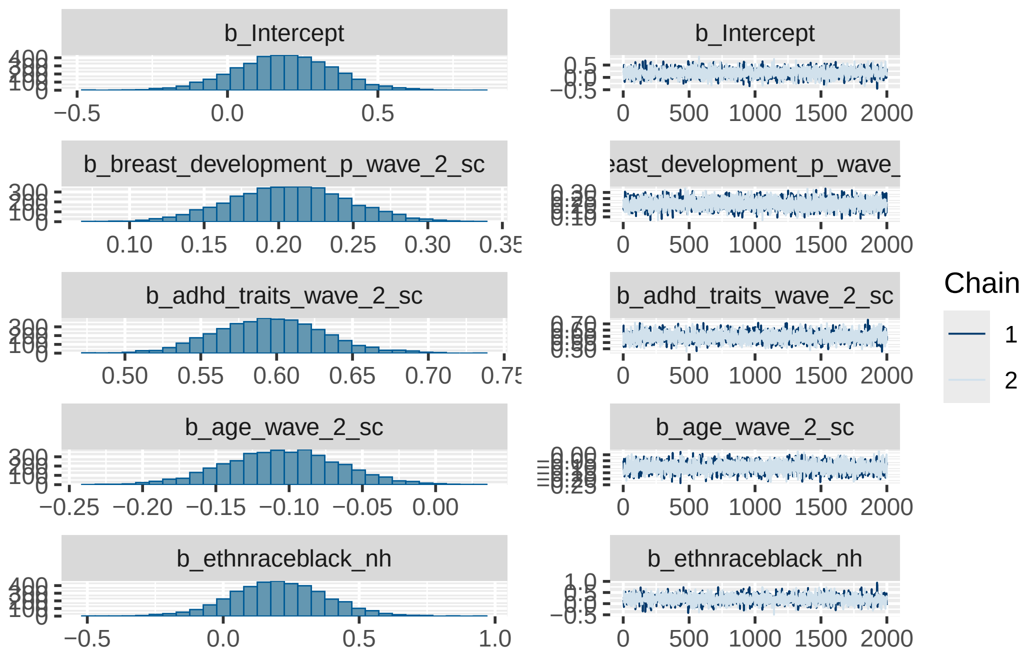
```
    Estimate   Est.Error      Q2.5    Q97.5
R2 0.8261396 0.005376451 0.8150095 0.836055


 Family: poisson
  Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age_
   Data: obs_df_wide (Number of observations: 4064)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3530)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.956     0.020    0.916    0.995 1.000     1600     2516

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.086     0.039    0.012    0.169 1.002      439      865
```

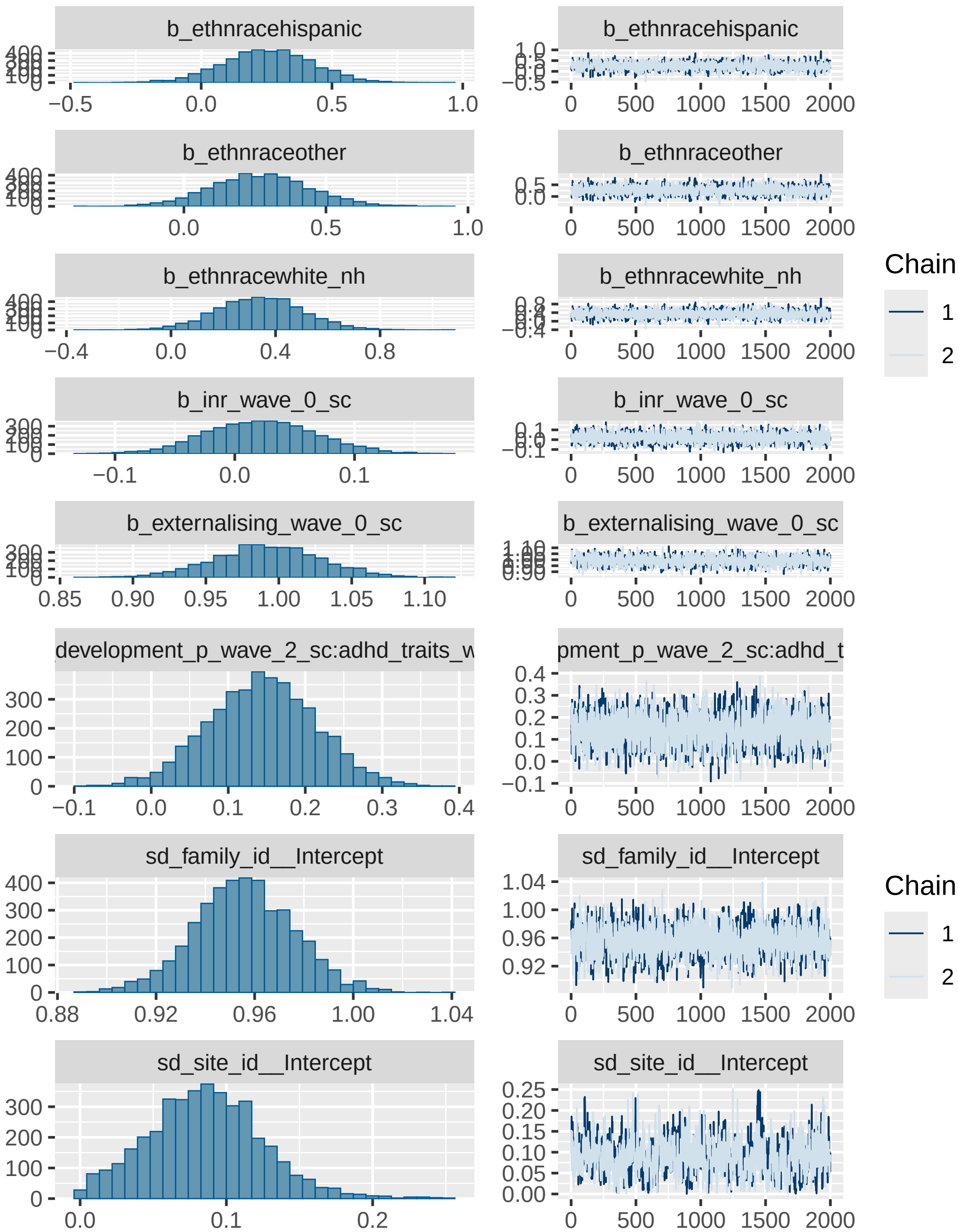
```
Regression Coefficients:
                                                  Estimate Est.Error
Intercept                                            0.188     0.165
breast_development_p_wave_2_sc                       0.207     0.040
adhd_traits_wave_2_sc                                0.596     0.036
age_wave_2_sc                                       -0.104     0.039
ethnraceblack_nh                                     0.206     0.175
ethnracehispanic                                     0.256     0.171
ethnraceother                                        0.271     0.173
ethnracewhite_nh                                     0.341     0.165
inr_wave_0_sc                                        0.023     0.046
externalising_wave_0_sc                              0.992     0.036
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc 0.144     0.069
                                                  l-95% CI u-95% CI  Rhat
Intercept                                           -0.133    0.512 1.001
breast_development_p_wave_2_sc                       0.131    0.284 1.000
adhd_traits_wave_2_sc                                0.528    0.671 1.002
age_wave_2_sc                                       -0.182   -0.028 1.001
ethnraceblack_nh                                    -0.143    0.546 1.001
ethnracehispanic                                    -0.075    0.582 1.001
ethnraceother                                       -0.059    0.614 1.000
ethnracewhite_nh                                     0.018    0.663 1.000
inr_wave_0_sc                                       -0.064    0.114 1.000
externalising_wave_0_sc                              0.920    1.062 1.000
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc 0.009    0.280 1.000
                                                  Bulk_ESS Tail_ESS
Intercept                                             1994     2260
breast_development_p_wave_2_sc                        2797     3168
adhd_traits_wave_2_sc                                 2173     2524
age_wave_2_sc                                         2560     2998
ethnraceblack_nh                                      1813     2078
ethnracehispanic                                      1981     2417
ethnraceother                                         1907     1991
ethnracewhite_nh                                      1985     2262
inr_wave_0_sc                                         2291     2922
externalising_wave_0_sc                               1832     2599
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc  2069     2820

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

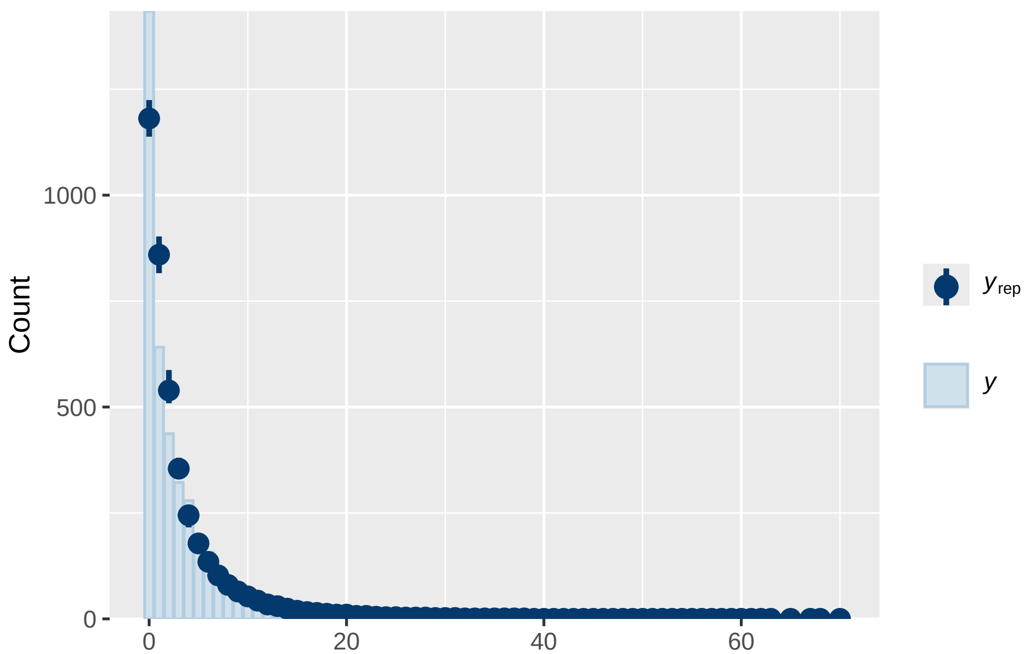
Model 5B longitudinal observed diagnostics

```
               prior     class
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
                                                  coef     group resp dpar nlpar

                                 adhd_traits_wave_2_sc
                                         age_wave_2_sc
                      breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
                                    ethnraceblack_nh
                                    ethnracehispanic
                                       ethnraceother
                                    ethnracewhite_nh
                               externalising_wave_0_sc
                                         inr_wave_0_sc


                                                       family_id
                                             Intercept family_id
                                                         site_id
                                             Intercept   site_id
lb ub tag       source
                  user
          (vectorized)
          (vectorized)
```

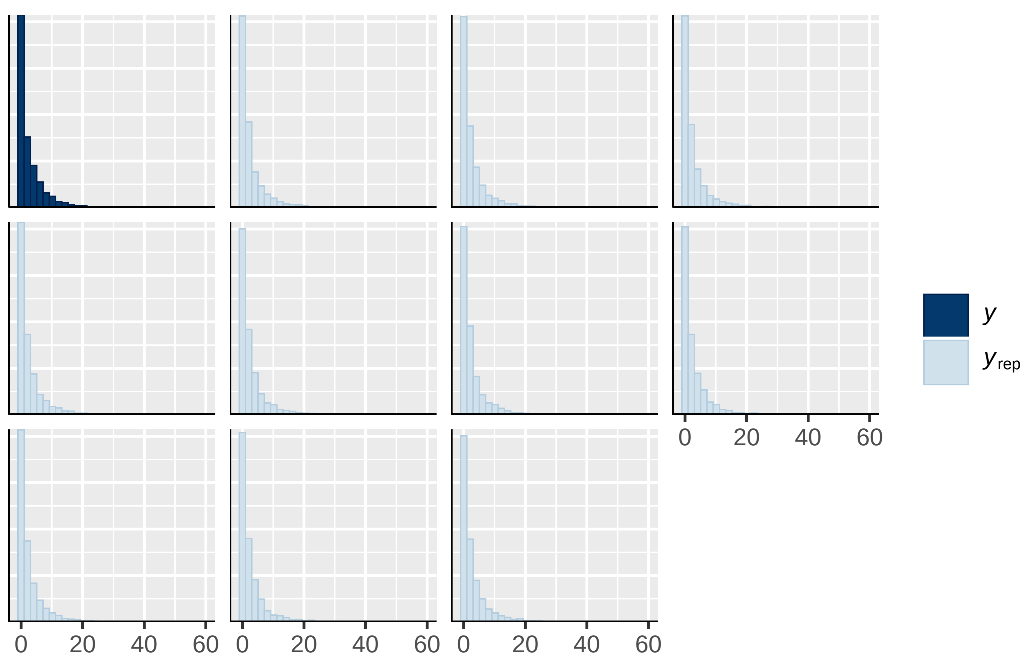
```
             (vectorized)
             (vectorized)
             (vectorized)
             (vectorized)
             (vectorized)
             (vectorized)
             (vectorized)
             (vectorized)
                  default
0                 default
0            (vectorized)
0            (vectorized)
0            (vectorized)
0            (vectorized)
```



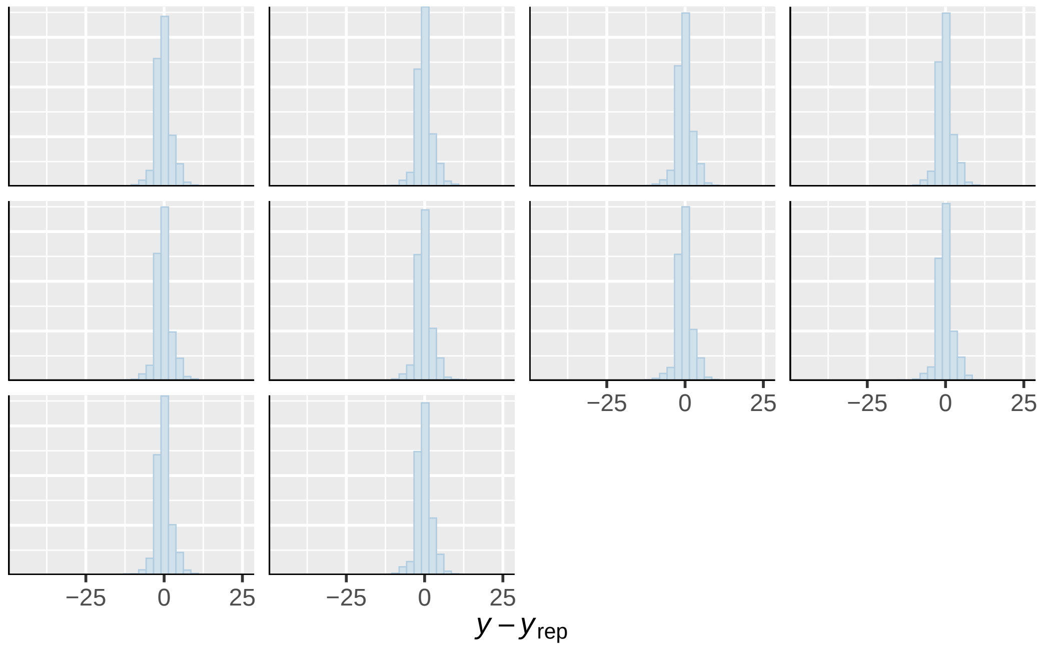

b_ethnracehispanic
b_ethnracehispanic
b_ethnraceother
b_ethnraceother
b_ethnracewhite_nh
b_ethnracewhite_nh
b_inr_wave_0_sc
b_inr_wave_0_sc
b_externalising_wave_0_sc
b_externalising_wave_0_sc
development_p_wave_2_sc:adhd_traits_w
pment_p_wave_2_sc:adhd_t
sd_family_id__Intercept
sd_family_id__Intercept
sd_site_id__Intercept
sd_site_id__Intercept
Chain
1
2
Chain
1
2



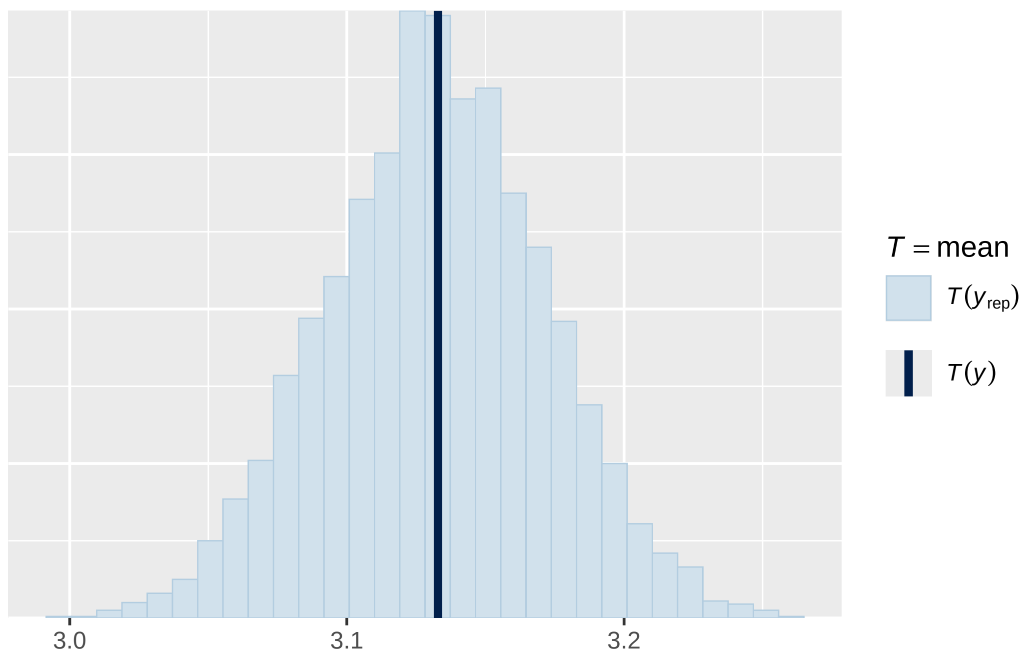

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



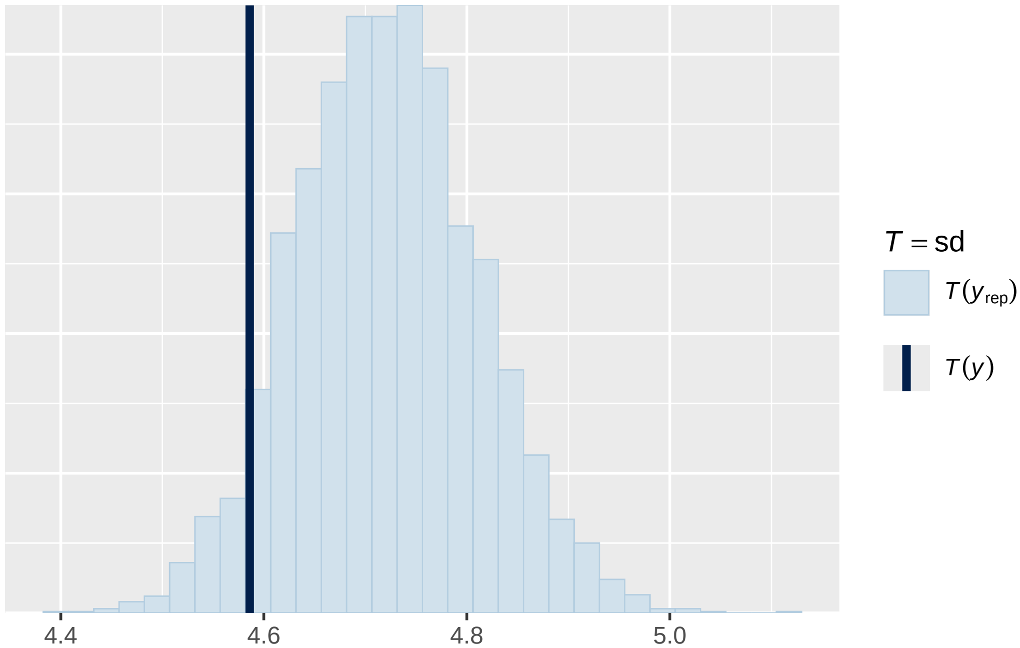

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



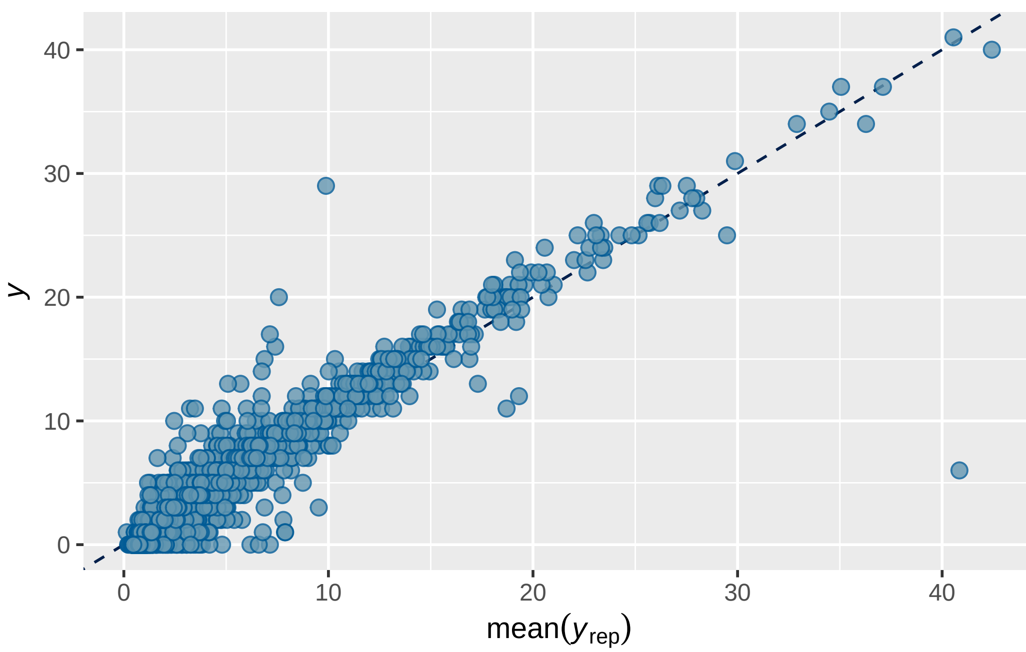

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



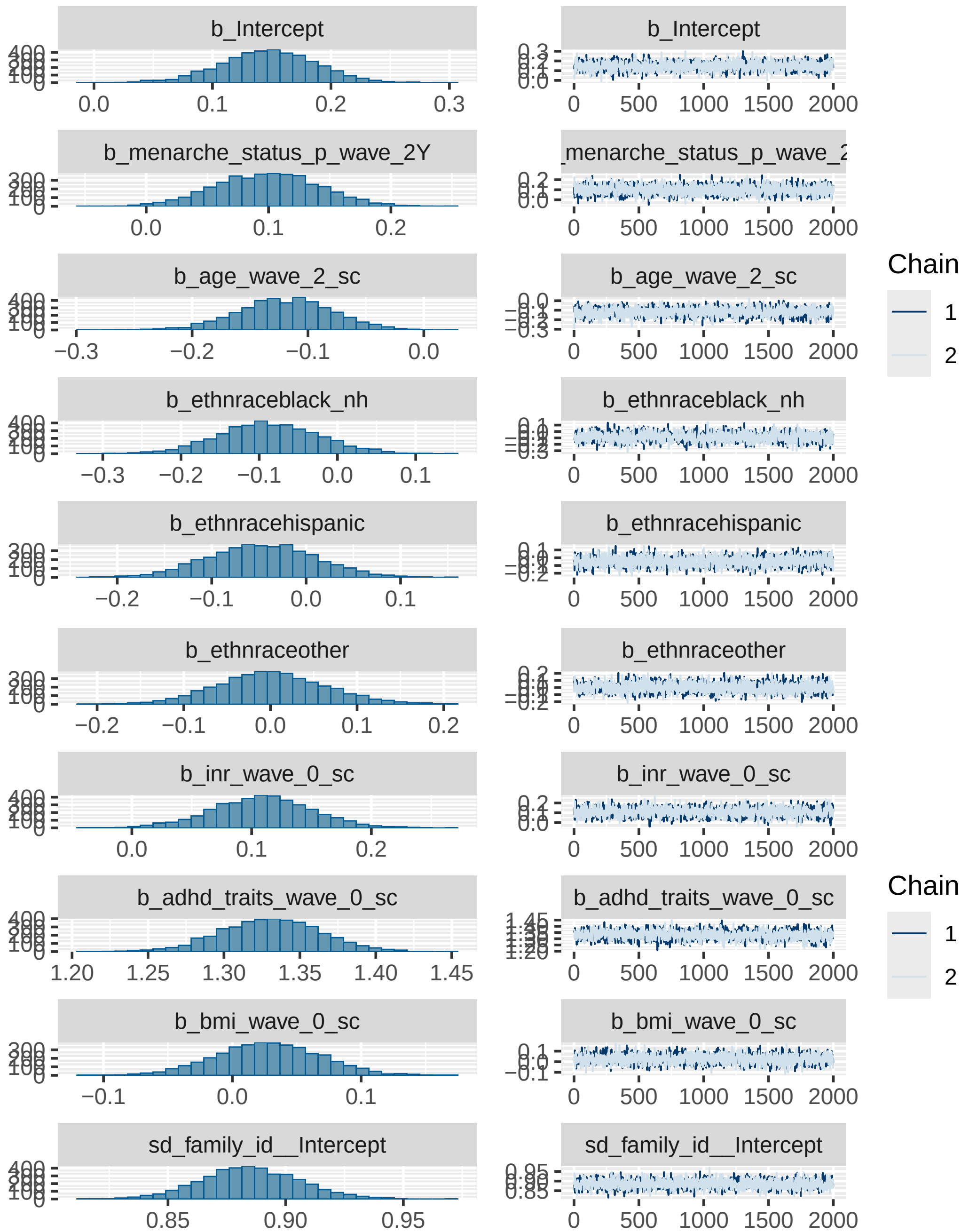
Model 1 longitudinal BMI output

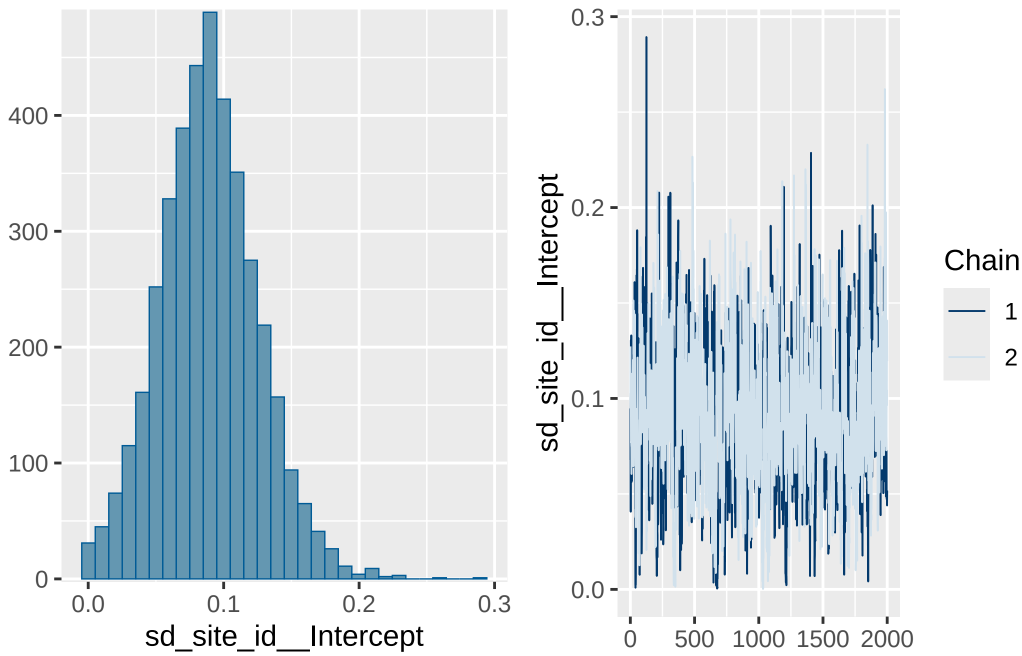
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7360488 0.007489116 0.7209785 0.7505381


 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave_
   Data: analytic_df_wide (Number of observations: 4496)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3915)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.885     0.021    0.844    0.927 1.002     1412     2182

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.089     0.036    0.019    0.164 1.001      555      595

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                    0.147     0.040    0.067    0.224 1.001     1879
menarche_status_p_wave_2Y    0.099     0.042    0.017    0.181 1.001     1900
age_wave_2_sc               -0.120     0.041   -0.200   -0.040 1.001     1712
ethnraceblack_nh            -0.088     0.064   -0.213    0.041 1.000     1642
ethnracehispanic            -0.047     0.057   -0.157    0.064 1.000     1851
ethnraceother                0.002     0.063   -0.119    0.133 1.001     1562
inr_wave_0_sc                0.109     0.042    0.025    0.191 1.001     1573
adhd_traits_wave_0_sc        1.331     0.033    1.266    1.395 1.006     1239
bmi_wave_0_sc                0.028     0.041   -0.050    0.107 1.002     1879
                          Tail_ESS
Intercept                     2160
menarche_status_p_wave_2Y     2718
age_wave_2_sc                 2418
ethnraceblack_nh              2631
ethnracehispanic              2863
ethnraceother                 2195
inr_wave_0_sc                 2221
adhd_traits_wave_0_sc         2174
bmi_wave_0_sc                 2422
```

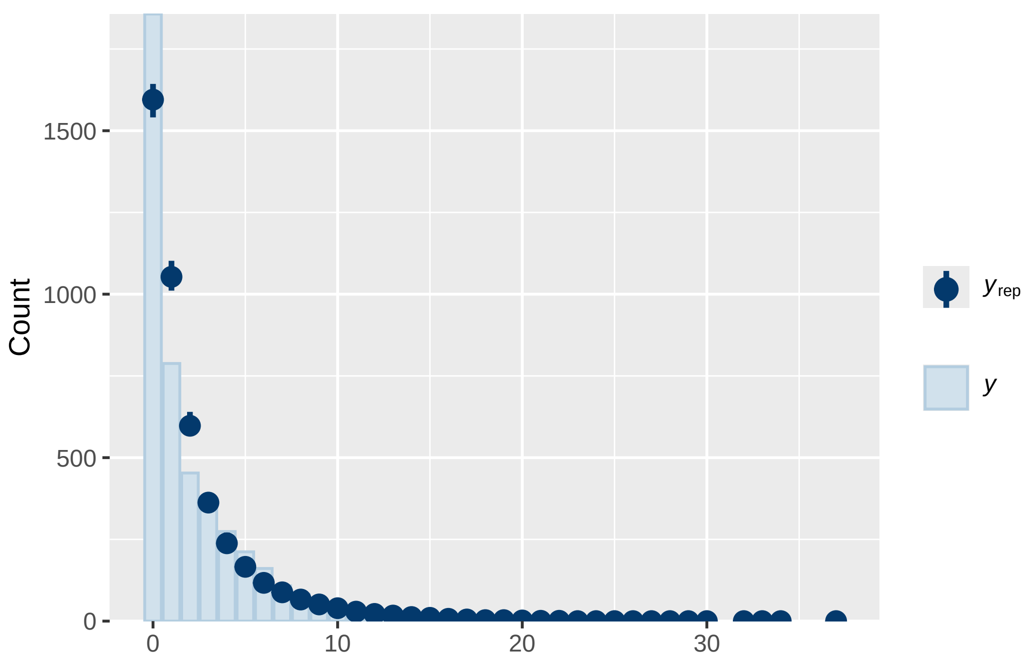
```
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 1 longitudinal BMI diagnostics

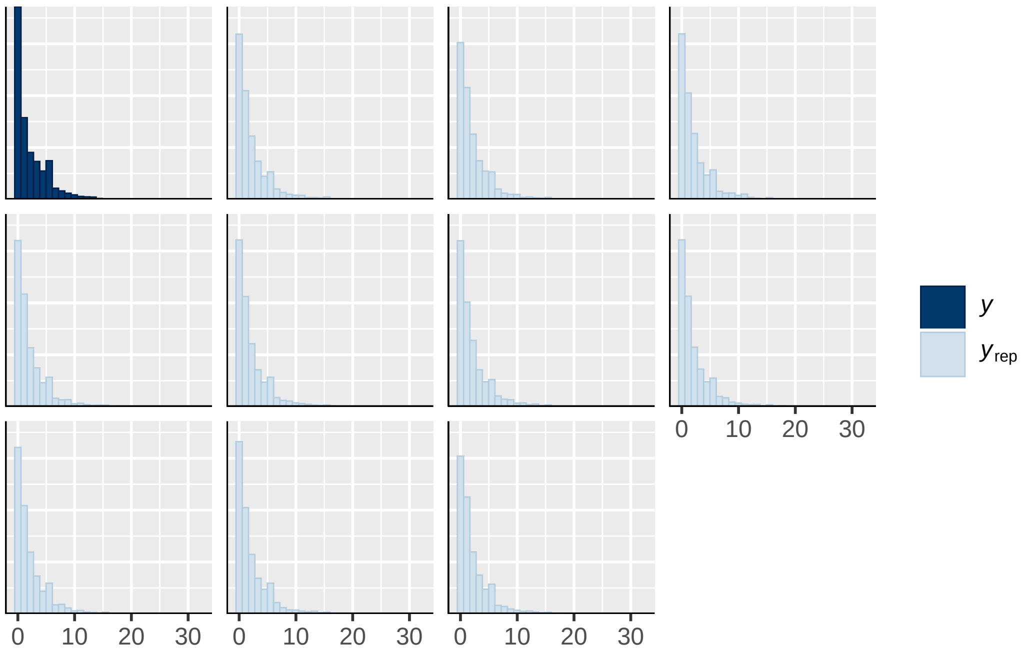
```
                prior     class                      coef     group resp dpar
         normal(0, 1)         b                                              
         normal(0, 1)         b     adhd_traits_wave_0_sc                    
         normal(0, 1)         b             age_wave_2_sc                    
         normal(0, 1)         b             bmi_wave_0_sc                    
         normal(0, 1)         b          ethnraceblack_nh                    
         normal(0, 1)         b          ethnracehispanic                    
         normal(0, 1)         b             ethnraceother                    
         normal(0, 1)         b             inr_wave_0_sc                    
         normal(0, 1)         b menarche_status_p_wave_2Y                    
 student_t(3, 0, 2.7) Intercept                                              
 student_t(3, 0, 2.7)        sd                                              
 student_t(3, 0, 2.7)        sd                           family_id          
 student_t(3, 0, 2.7)        sd                 Intercept family_id          
 student_t(3, 0, 2.7)        sd                             site_id          
 student_t(3, 0, 2.7)        sd                 Intercept   site_id          
 nlpar lb ub tag       source
                         user
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                      default
        0             default
        0        (vectorized)
        0        (vectorized)
        0        (vectorized)
        0        (vectorized)
```


b_Intercept
b_menarche_status_p_wave_2Y
b_age_wave_2_sc
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnraceother
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
b_bmi_wave_0_sc
sd_family_id__Intercept
Chain
1
2

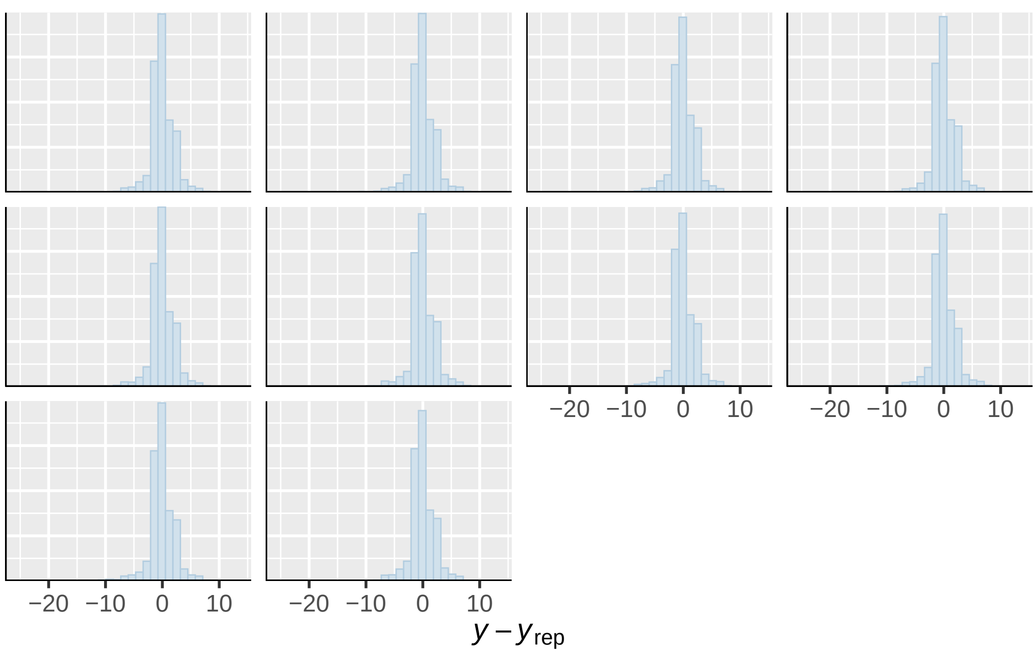





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

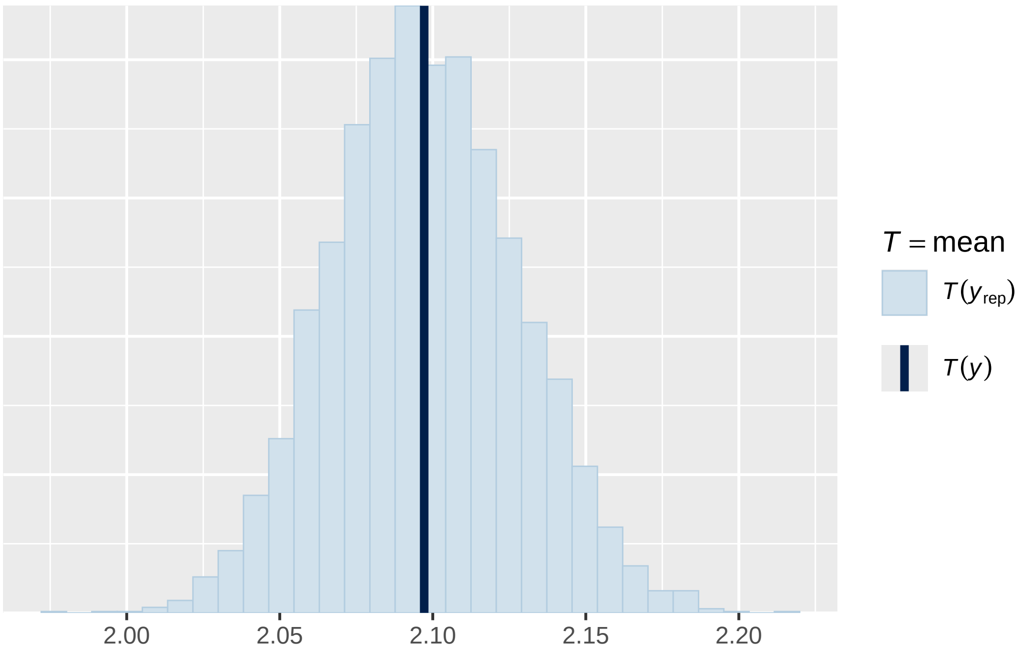



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

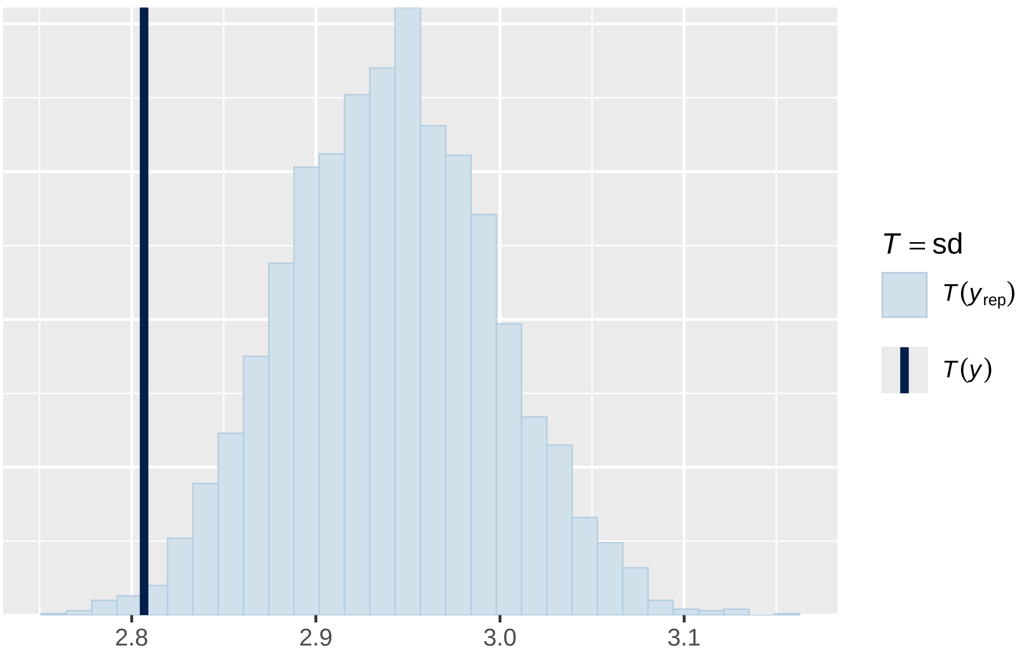



```
Using all posterior draws for ppc type 'stat' by default.
```

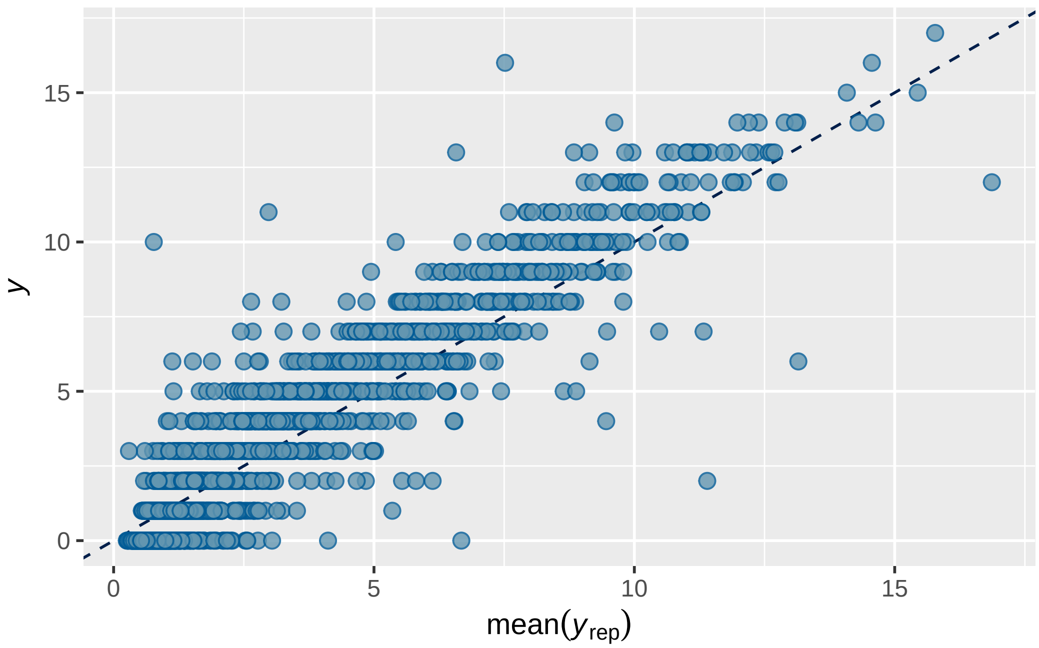
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 1B longitudinal BMI output

```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7358724 0.007396846 0.721296 0.7497154
```

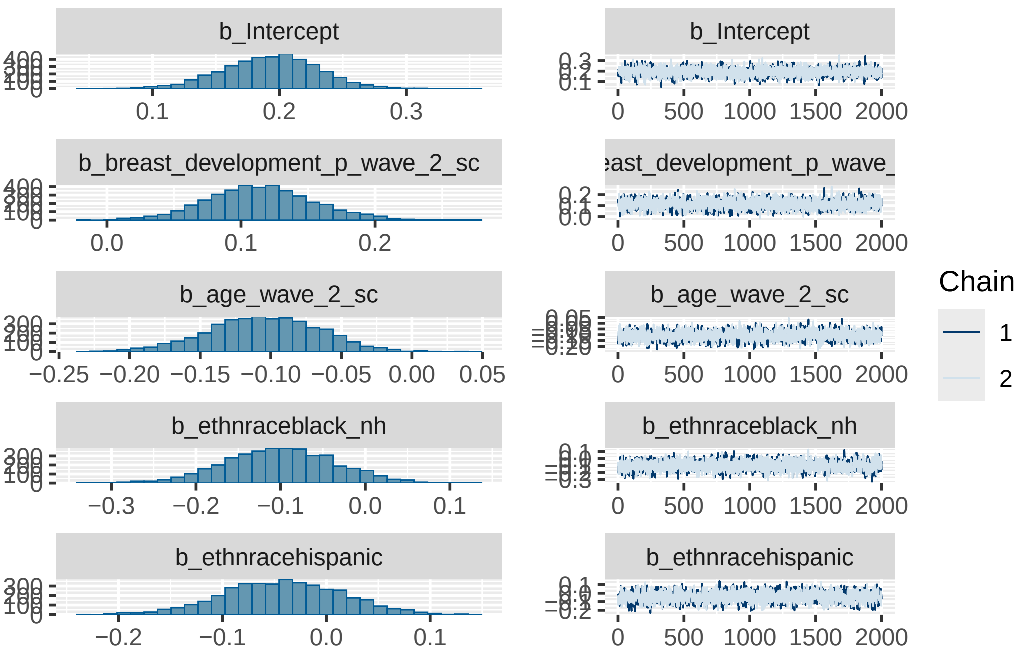
```
 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + in
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.885     0.021    0.846    0.929 1.000     1313     1736

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.092     0.037    0.021    0.169 1.009      609      679
```

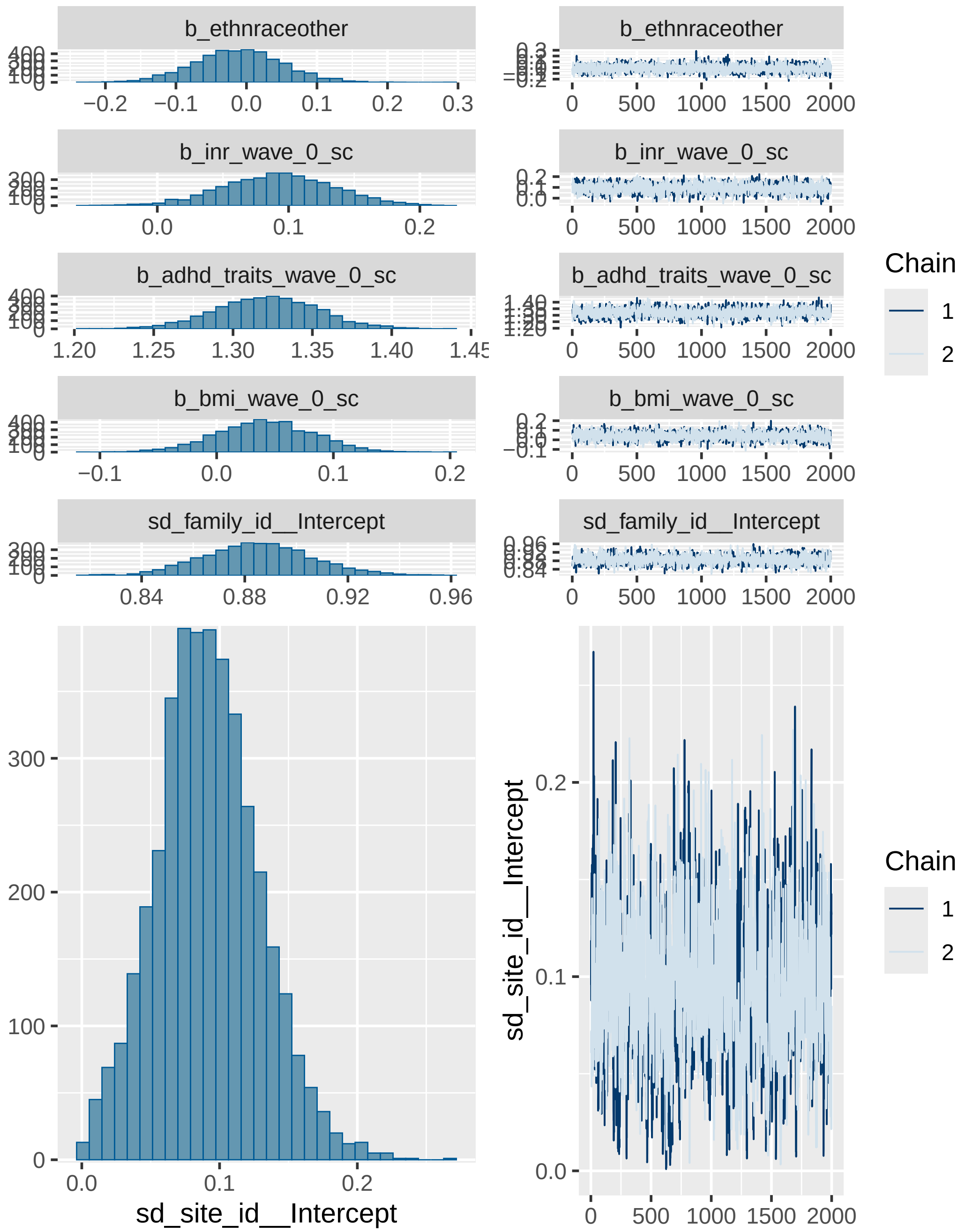
```
Regression Coefficients:
                            Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                      0.193     0.038    0.116    0.265 1.000
breast_development_p_wave_2_sc    0.115     0.040    0.038    0.196 1.001
age_wave_2_sc                 -0.105     0.039   -0.183   -0.031 1.000
ethnraceblack_nh              -0.100     0.065   -0.227    0.025 1.001
ethnracehispanic              -0.042     0.057   -0.156    0.071 1.001
ethnraceother                 -0.011     0.062   -0.131    0.112 1.001
inr_wave_0_sc                  0.092     0.042    0.011    0.174 1.000
adhd_traits_wave_0_sc          1.322     0.032    1.258    1.386 1.001
bmi_wave_0_sc                  0.042     0.041   -0.041    0.121 1.000
                            Bulk_ESS Tail_ESS
Intercept                       2353     2745
breast_development_p_wave_2_sc     2280     2800
age_wave_2_sc                   1932     2678
ethnraceblack_nh                1930     2513
ethnracehispanic                1778     2742
ethnraceother                   1938     2569
inr_wave_0_sc                   1850     2596
adhd_traits_wave_0_sc           1449     2200
bmi_wave_0_sc                   1985     2574

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

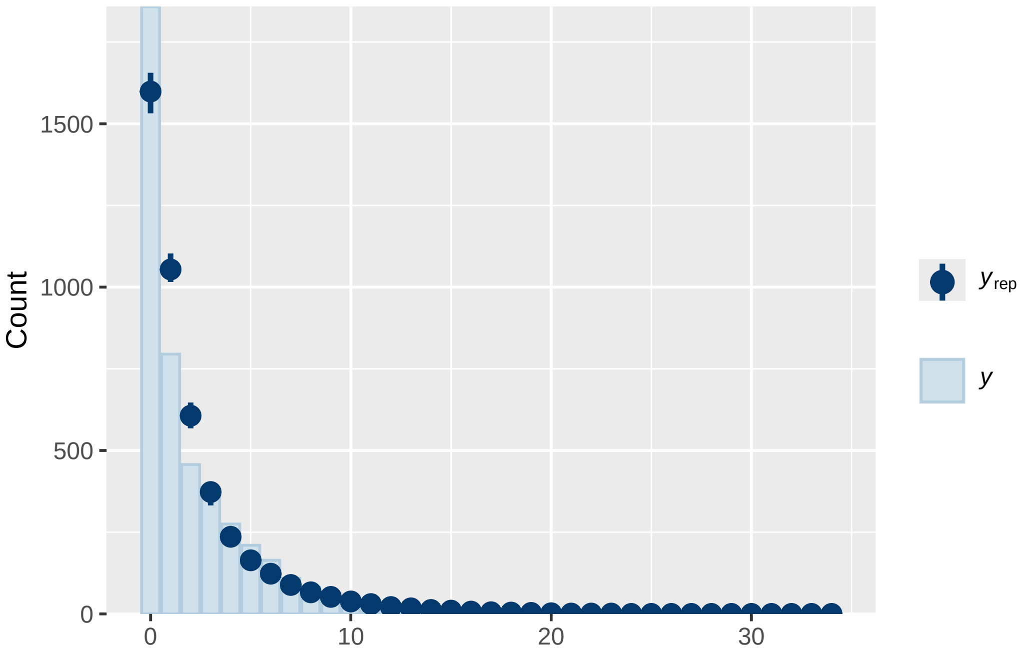
Model 1B longitudinal BMI diagnostics

```
               prior     class                           coef     group resp
        normal(0, 1)         b
        normal(0, 1)         b          adhd_traits_wave_0_sc
        normal(0, 1)         b                  age_wave_2_sc
        normal(0, 1)         b                  bmi_wave_0_sc
        normal(0, 1)         b breast_development_p_wave_2_sc
        normal(0, 1)         b               ethnraceblack_nh
        normal(0, 1)         b               ethnracehispanic
        normal(0, 1)         b                  ethnraceother
        normal(0, 1)         b                  inr_wave_0_sc
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                                family_id
```

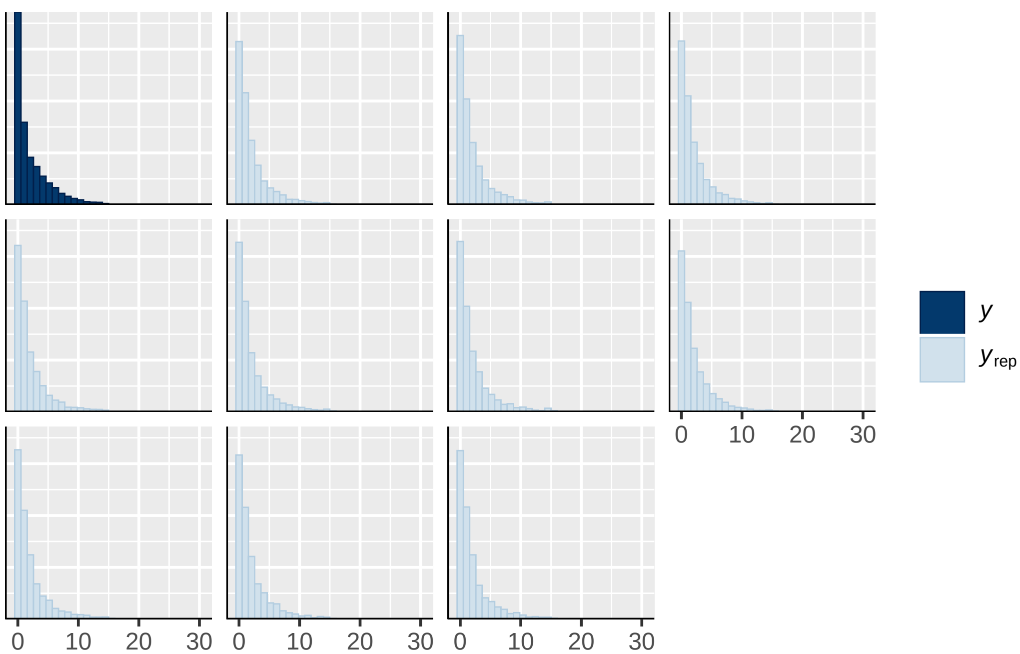
```
student_t(3, 0, 2.7)        sd                           Intercept family_id
student_t(3, 0, 2.7)        sd                                       site_id
student_t(3, 0, 2.7)        sd                           Intercept   site_id
dpar nlpar lb ub tag       source
                             user
                     (vectorized)
                     (vectorized)
                     (vectorized)
                     (vectorized)
                     (vectorized)
                     (vectorized)
                     (vectorized)
                     (vectorized)
                          default
            0             default
            0        (vectorized)
            0        (vectorized)
            0        (vectorized)
            0        (vectorized)
```



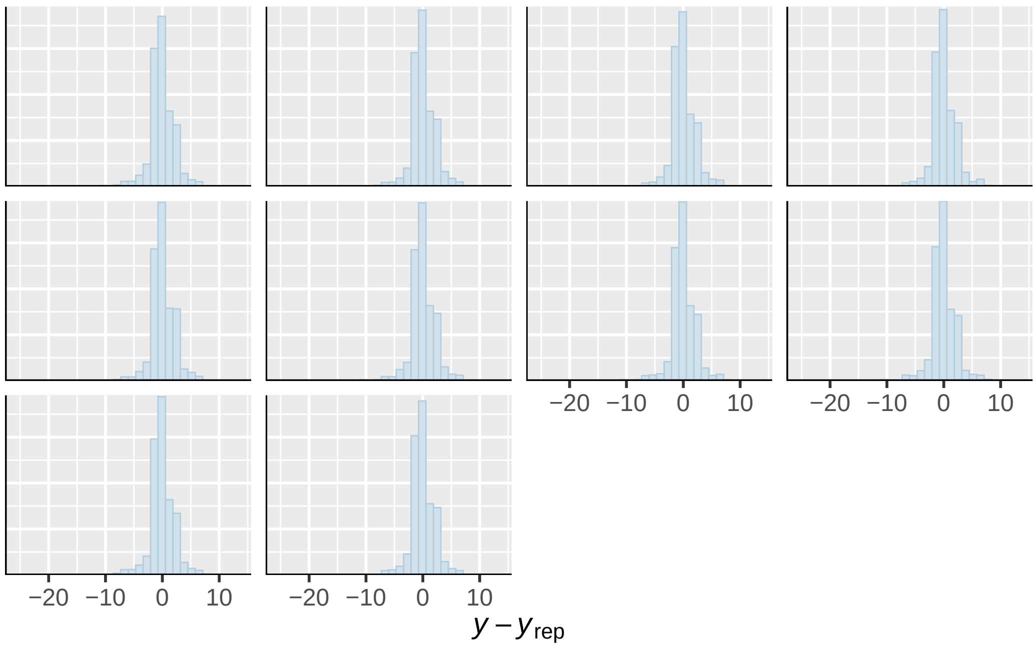

b_ethnraceother
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
b_bmi_wave_0_sc
sd_family_id__Intercept
sd_site_id__Intercept
Chain
1
2



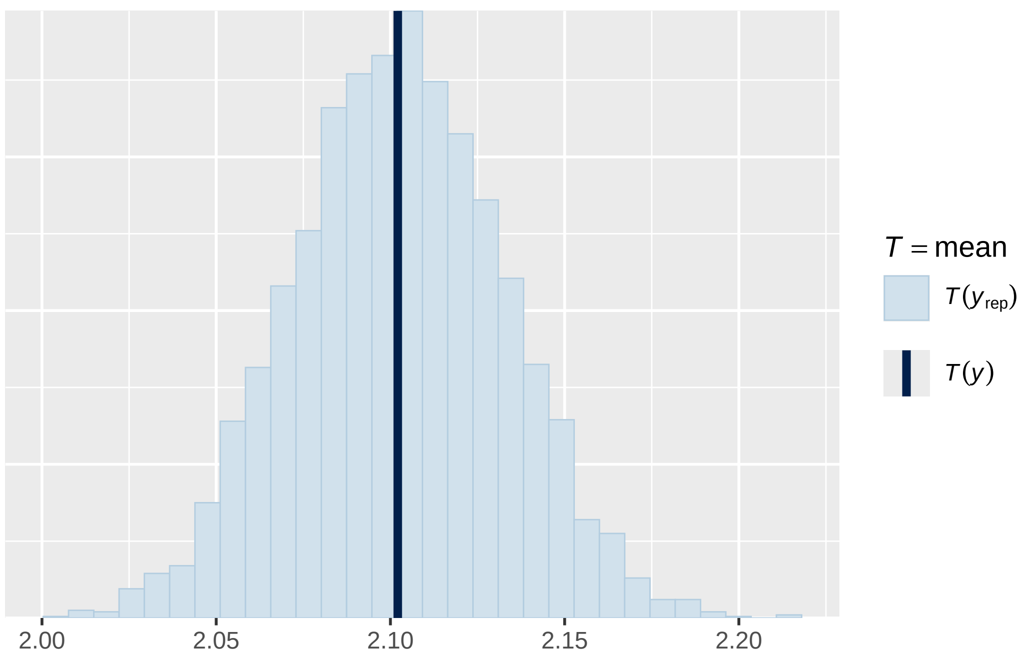

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



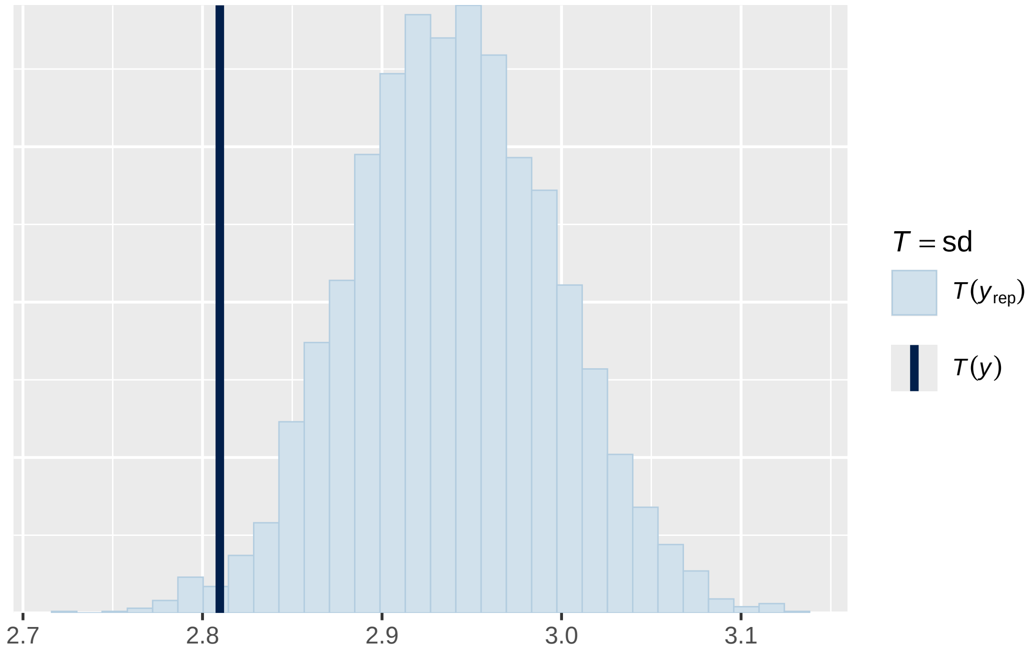

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



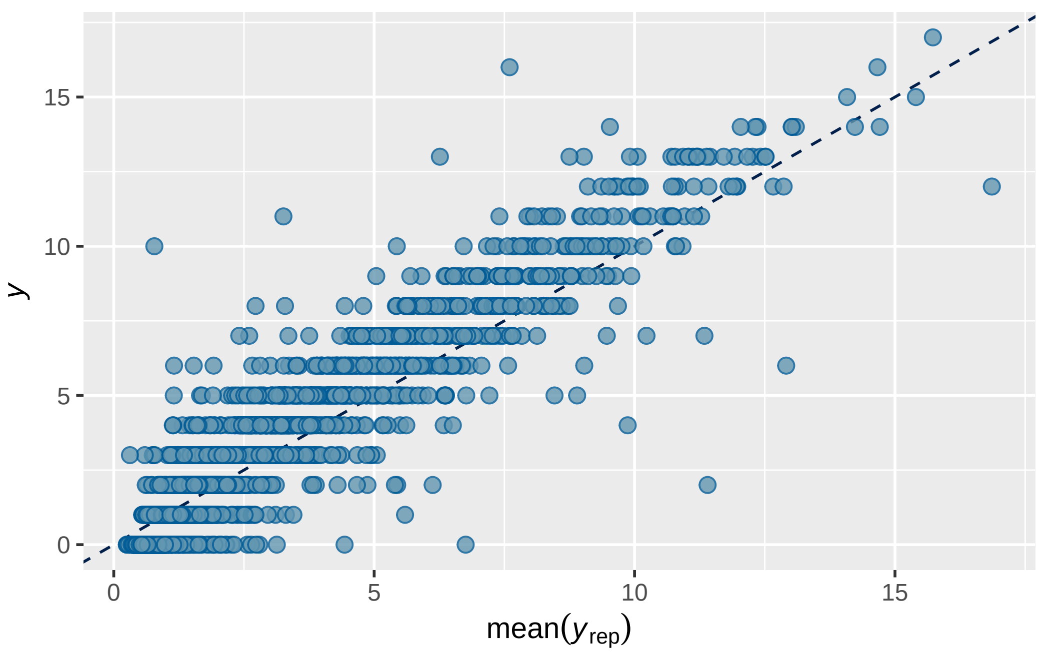

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5B longitudinal BMI output

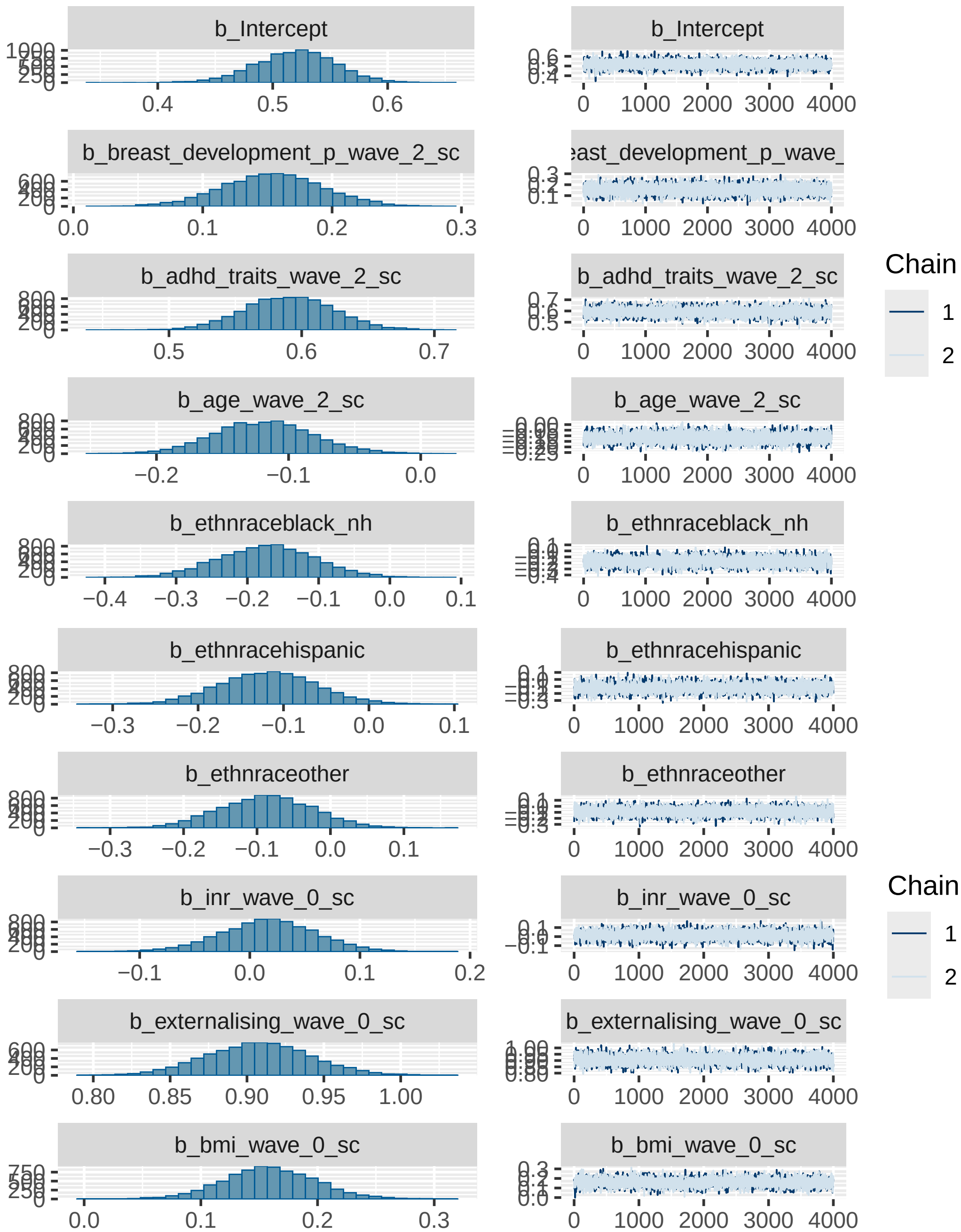
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8235553 0.005162047 0.8131239 0.8331237


 Family: poisson
  Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age_
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.986     0.020    0.948    1.025 1.000     2542     3803

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.067     0.038    0.004    0.146 1.000      717     1701

Regression Coefficients:
                                                     Estimate Est.Error
Intercept                                               0.520     0.035
breast_development_p_wave_2_sc                          0.154     0.039
adhd_traits_wave_2_sc                                   0.591     0.034
age_wave_2_sc                                          -0.118     0.038
ethnraceblack_nh                                       -0.173     0.067
ethnracehispanic                                       -0.118     0.058
ethnraceother                                          -0.087     0.063
inr_wave_0_sc                                           0.017     0.043
externalising_wave_0_sc                                 0.908     0.034
bmi_wave_0_sc                                           0.159     0.040
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc    0.170     0.066
                                                     l-95% CI u-95% CI  Rhat
Intercept                                               0.450    0.587 1.000
breast_development_p_wave_2_sc                          0.078    0.231 1.000
adhd_traits_wave_2_sc                                   0.525    0.657 1.000
age_wave_2_sc                                          -0.190   -0.042 1.000
ethnraceblack_nh                                       -0.305   -0.042 1.000
ethnracehispanic                                       -0.229   -0.003 1.000
ethnraceother                                          -0.209    0.035 1.001
```

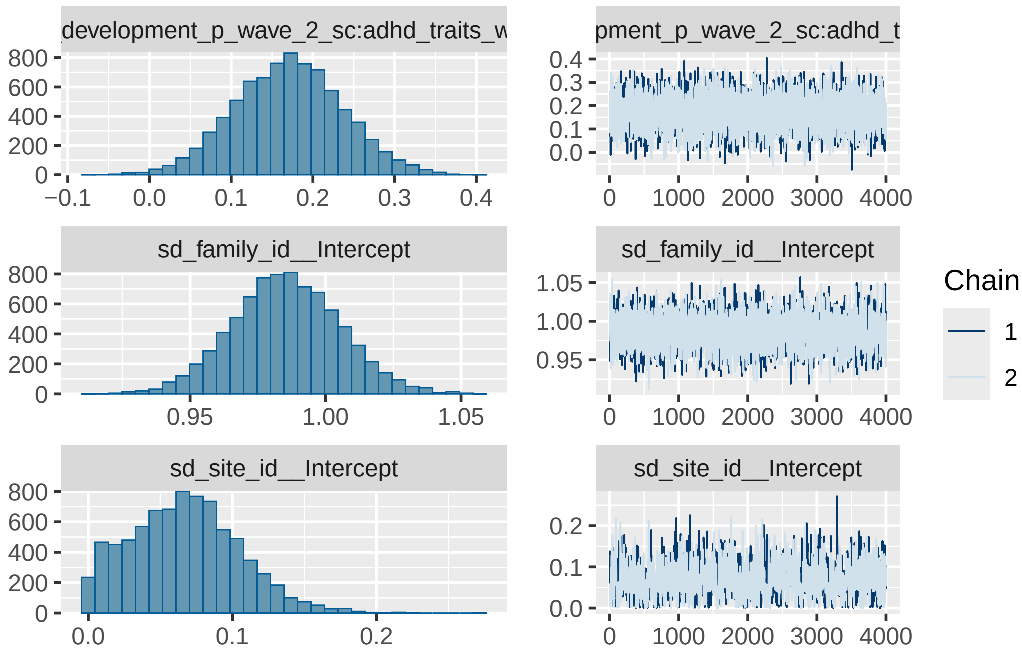
```
inr_wave_0_sc                                                -0.071    0.102 1.000
externalising_wave_0_sc                                       0.841    0.975 1.000
bmi_wave_0_sc                                                 0.082    0.239 1.000
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc          0.043    0.302 1.000
                                                           Bulk_ESS Tail_ESS
Intercept                                                      4214     5044
breast_development_p_wave_2_sc                                 3976     5238
adhd_traits_wave_2_sc                                          3207     4994
age_wave_2_sc                                                  4181     5455
ethnraceblack_nh                                               3215     4931
ethnracehispanic                                               3137     4979
ethnraceother                                                  3458     4916
inr_wave_0_sc                                                  3600     4754
externalising_wave_0_sc                                        2728     4140
bmi_wave_0_sc                                                  3302     4481
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc           2872     4850

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

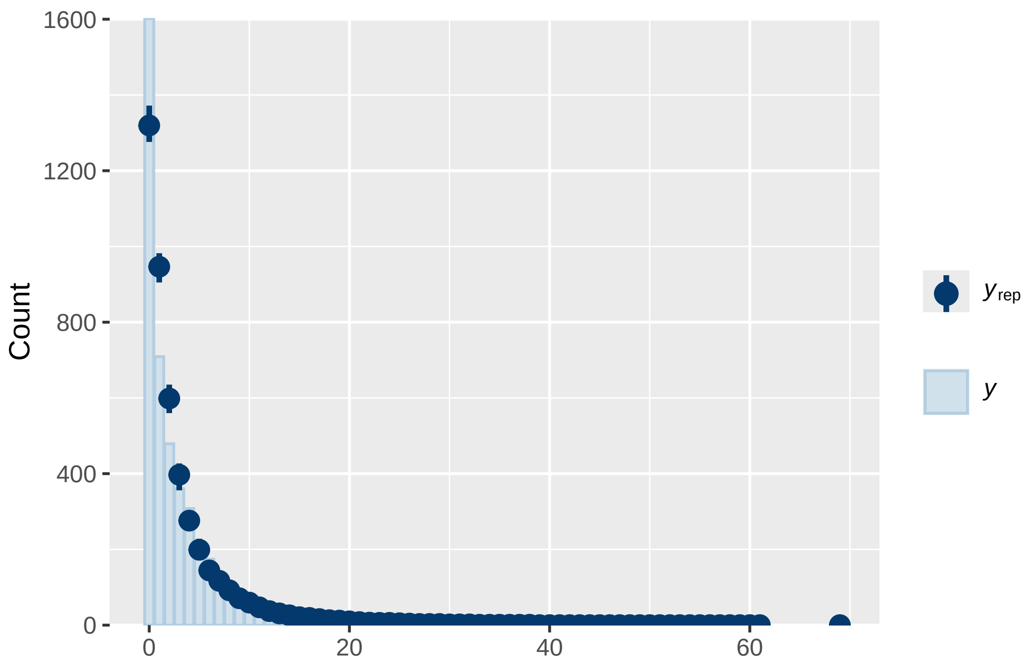
Model 5B longitudinal BMI diagnostics

```
                prior     class
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
                                                     coef     group resp dpar nlpar
```

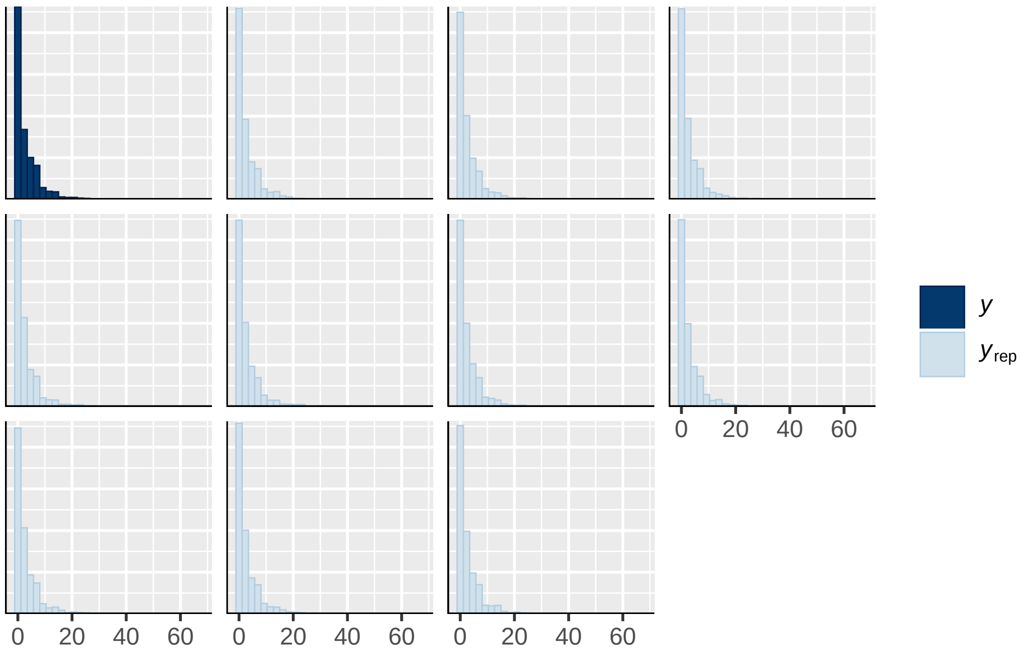
```
                                   adhd_traits_wave_2_sc
                                            age_wave_2_sc
                                            bmi_wave_0_sc
                         breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
                                         ethnraceblack_nh
                                         ethnracehispanic
                                            ethnraceother
                                 externalising_wave_0_sc
                                            inr_wave_0_sc


                                                                   family_id
                                                         Intercept family_id
                                                                     site_id
                                                         Intercept   site_id
lb ub tag       source
                  user
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
               default
 0             default
 0        (vectorized)
 0        (vectorized)
 0        (vectorized)
 0        (vectorized)
```


b_Intercept
b_breast_development_p_wave_2_sc
b_adhd_traits_wave_2_sc
b_age_wave_2_sc
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnraceother
b_inr_wave_0_sc
b_externalising_wave_0_sc
b_bmi_wave_0_sc
Chain
1
2
0 1000 2000 3000 4000

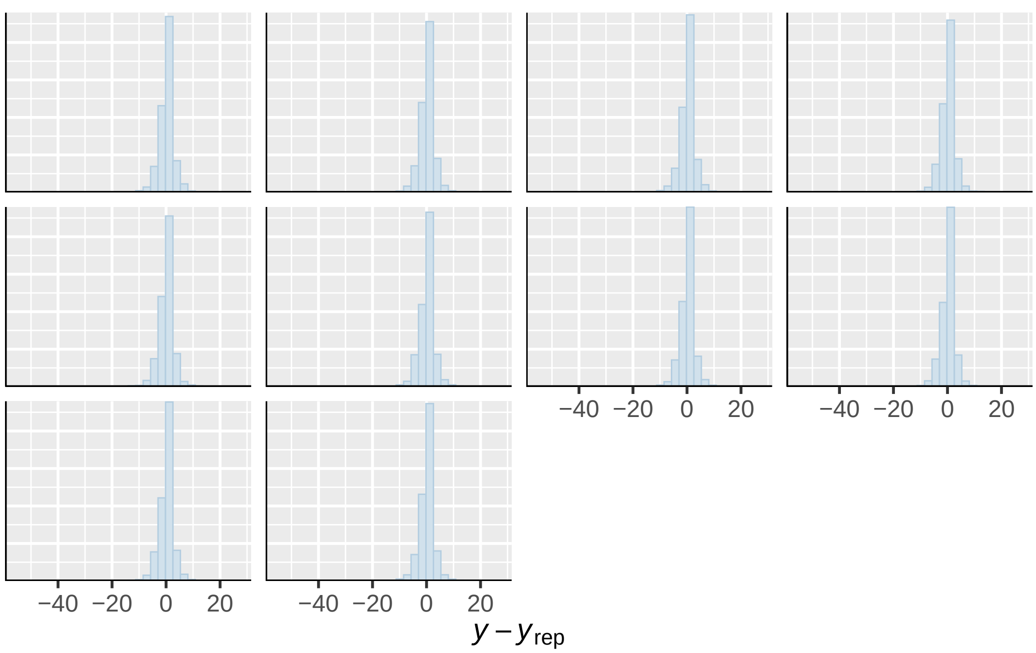





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

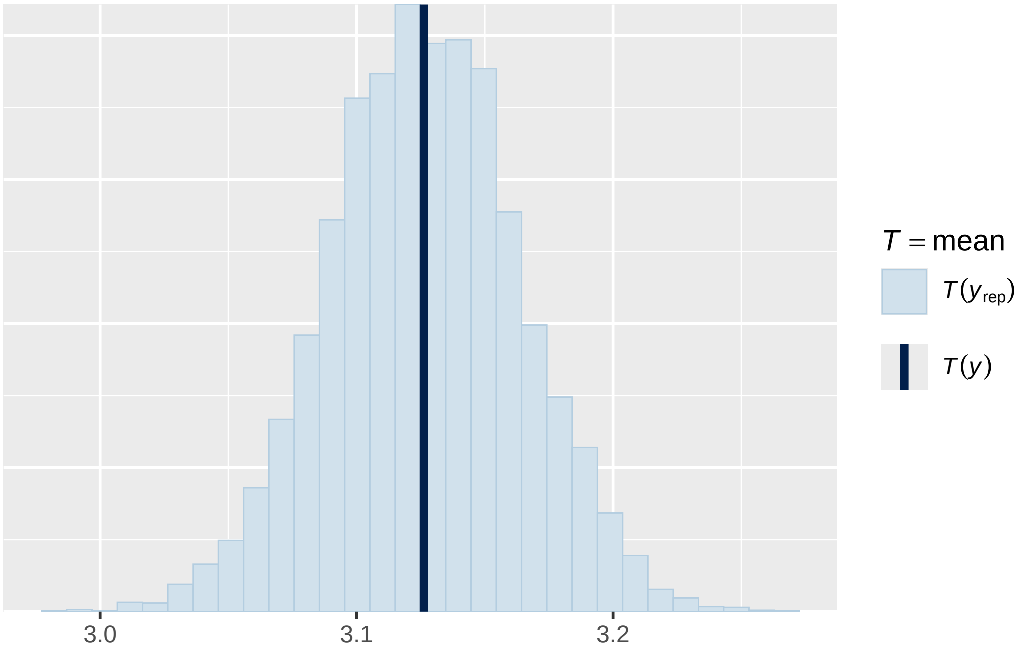



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

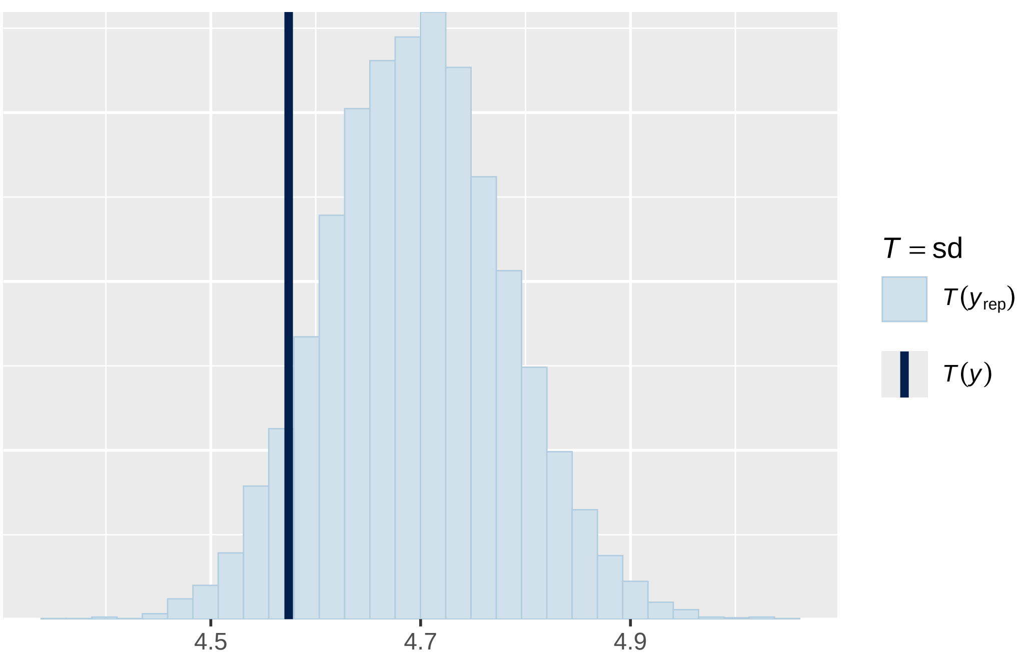



```
Using all posterior draws for ppc type 'stat' by default.
```

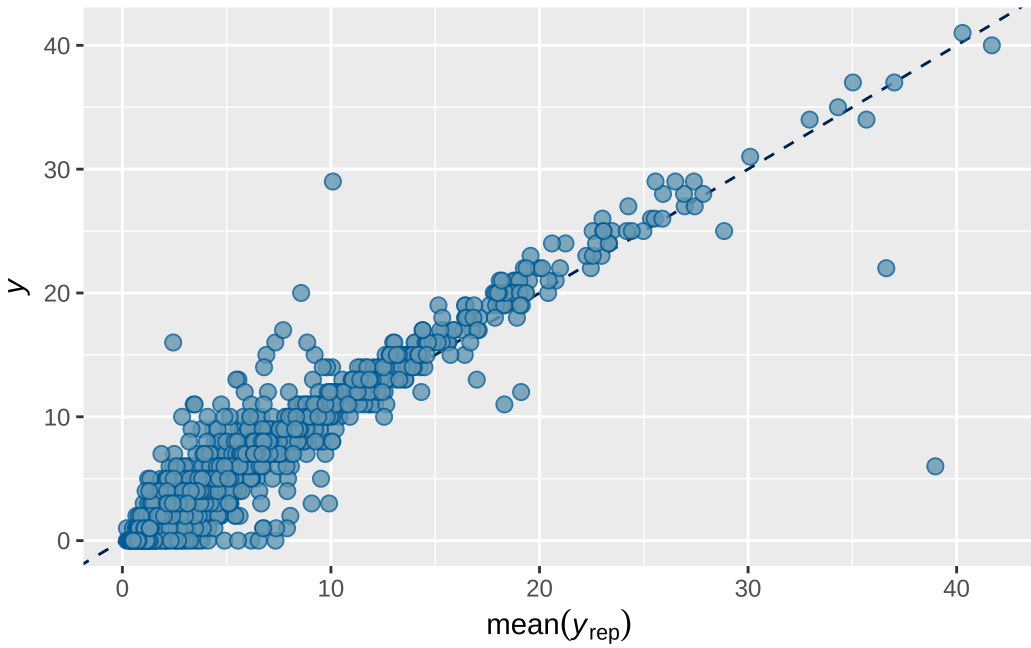
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 1 longitudinal stimulant output

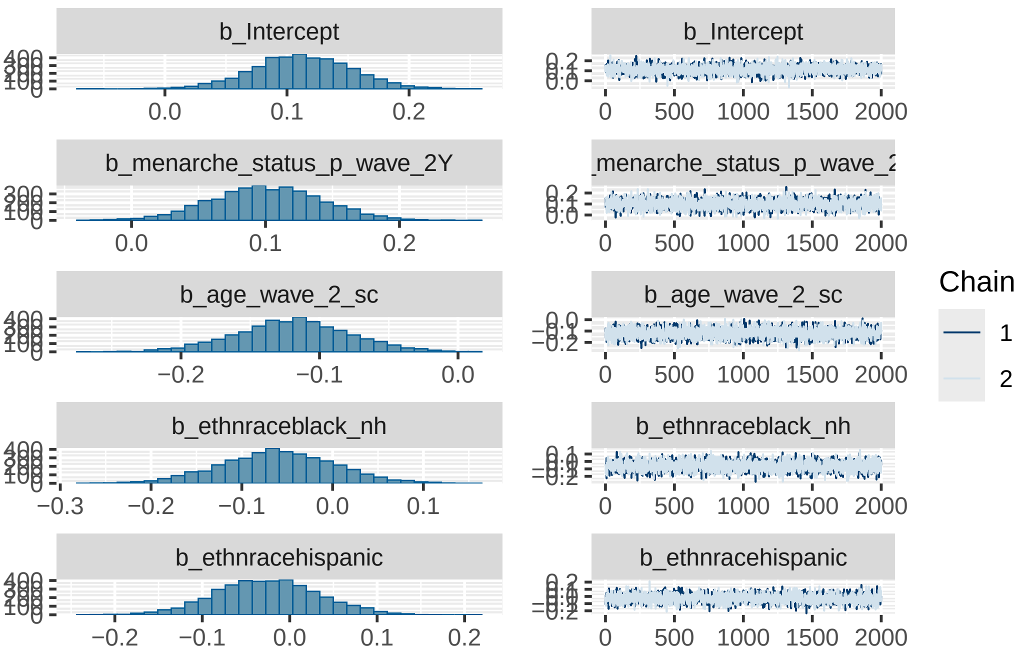
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7394205 0.007405415 0.7247865 0.7533296


 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave_
   Data: analytic_df_wide (Number of observations: 4496)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3915)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.871     0.021    0.832    0.913 1.001     1580     2613

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.082     0.036    0.012    0.158 1.001      663      964
```

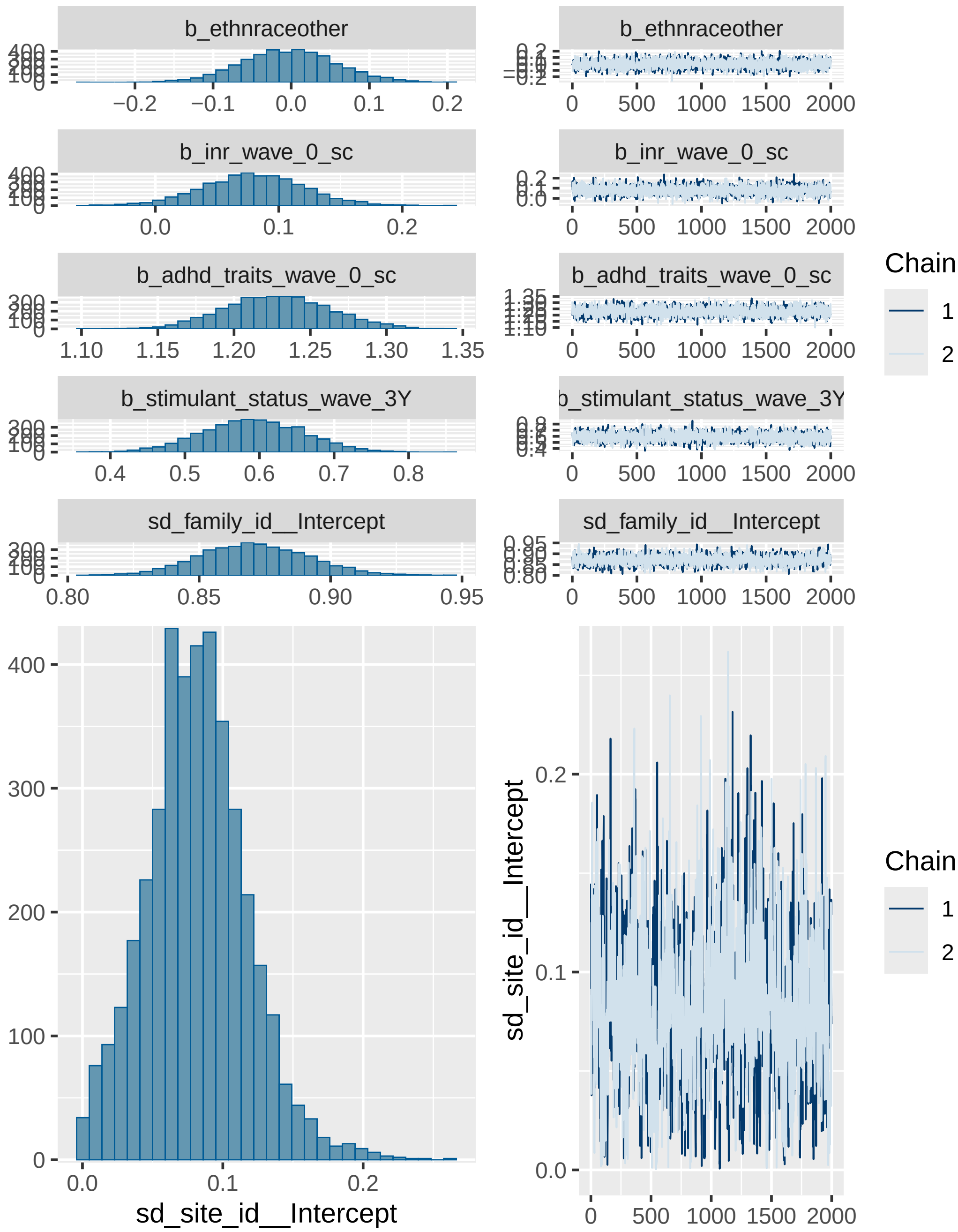
```
Regression Coefficients:
                         Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                   0.112     0.040    0.033    0.191 1.001     3038
menarche_status_p_wave_2Y   0.101     0.041    0.021    0.182 1.000     3011
age_wave_2_sc              -0.119     0.041   -0.201   -0.038 1.000     3452
ethnraceblack_nh           -0.060     0.063   -0.183    0.062 1.001     2584
ethnracehispanic           -0.026     0.059   -0.140    0.091 1.000     2499
ethnraceother              -0.003     0.061   -0.121    0.121 1.000     3267
inr_wave_0_sc               0.080     0.041   -0.001    0.161 1.001     2776
adhd_traits_wave_0_sc       1.229     0.034    1.164    1.298 1.001     2386
stimulant_status_wave_3Y    0.592     0.068    0.458    0.722 1.001     2203
                         Tail_ESS
Intercept                    3069
menarche_status_p_wave_2Y    3263
age_wave_2_sc                3191
ethnraceblack_nh             3439
ethnracehispanic             2786
ethnraceother                3443
inr_wave_0_sc                3077
adhd_traits_wave_0_sc        2675
stimulant_status_wave_3Y     2754

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

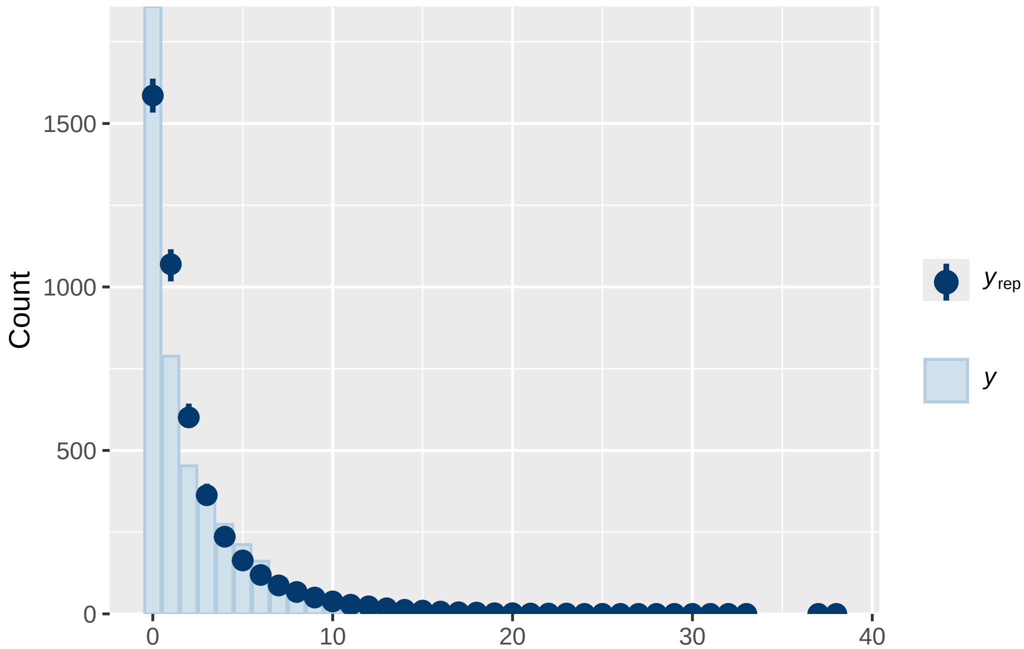
Model 1 stimulant diagnostics

```
              prior     class                      coef     group resp dpar
       normal(0, 1)         b                                              
       normal(0, 1)         b     adhd_traits_wave_0_sc                    
       normal(0, 1)         b             age_wave_2_sc                    
       normal(0, 1)         b          ethnraceblack_nh                    
       normal(0, 1)         b          ethnracehispanic                    
       normal(0, 1)         b             ethnraceother                    
       normal(0, 1)         b             inr_wave_0_sc                    
       normal(0, 1)         b menarche_status_p_wave_2Y                    
       normal(0, 1)         b  stimulant_status_wave_3Y                    
 student_t(3, 0, 2.7) Intercept                                            
 student_t(3, 0, 2.7)        sd                                            
 student_t(3, 0, 2.7)        sd                           family_id        
```

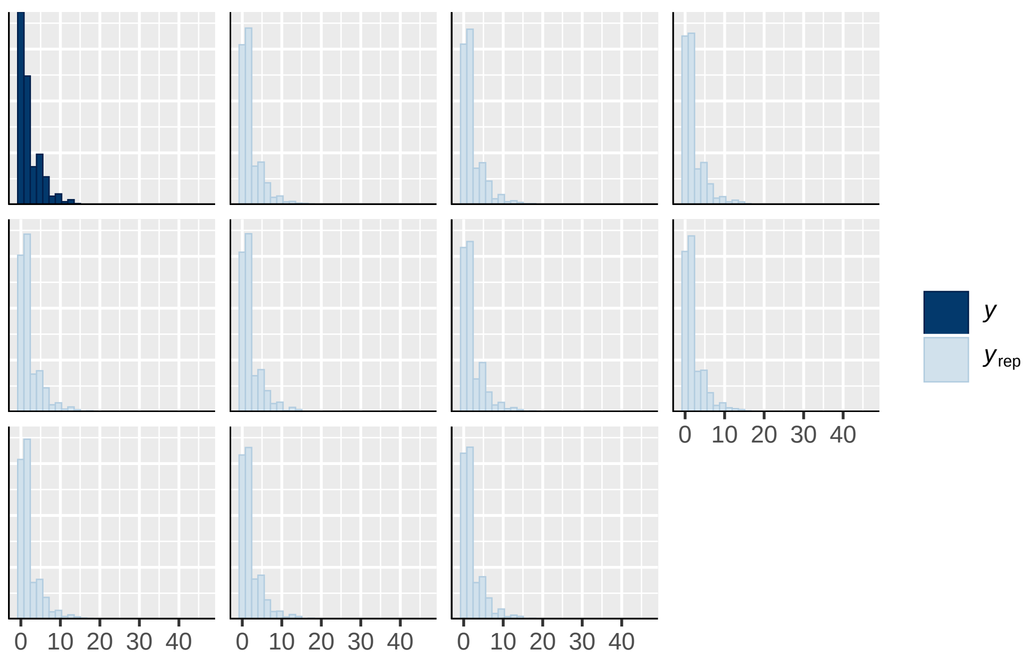
```
student_t(3, 0, 2.7)        sd                  Intercept family_id
student_t(3, 0, 2.7)        sd                              site_id
student_t(3, 0, 2.7)        sd                  Intercept   site_id
nlpar lb ub tag        source
                         user
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                      default
        0             default
        0        (vectorized)
        0        (vectorized)
        0        (vectorized)
        0        (vectorized)
```



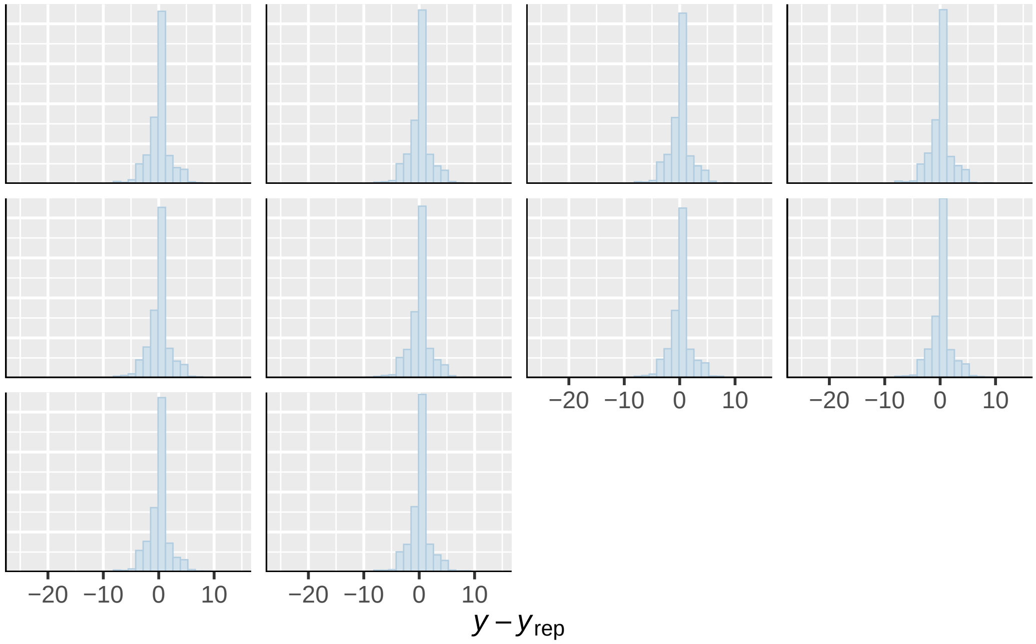

b_ethnraceother
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
b_stimulant_status_wave_3Y
sd_family_id__Intercept
sd_site_id__Intercept
Chain
1
2



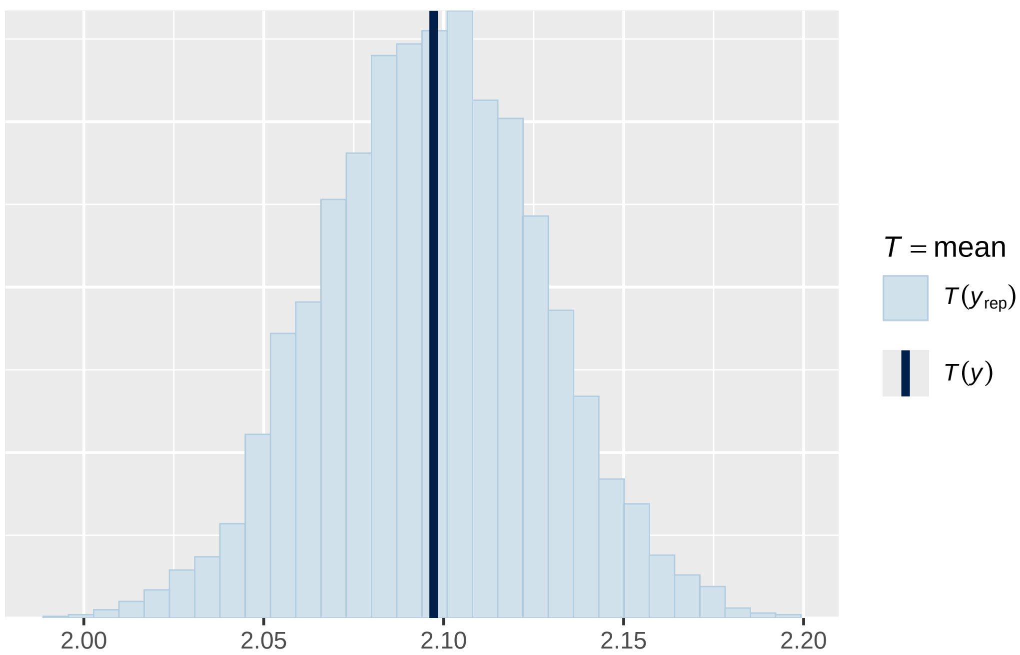
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



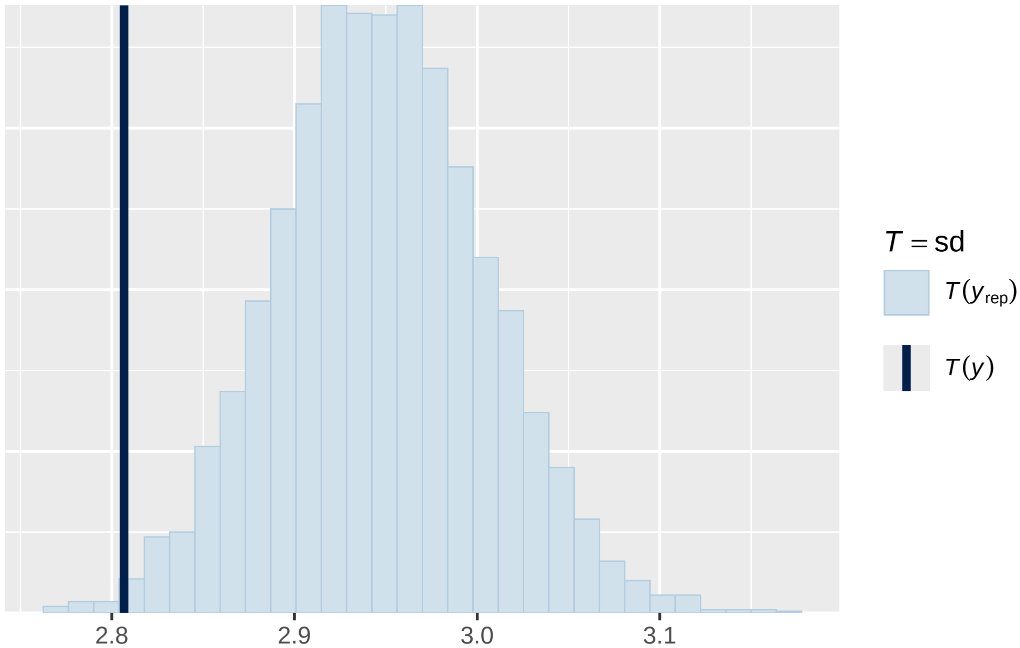
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.



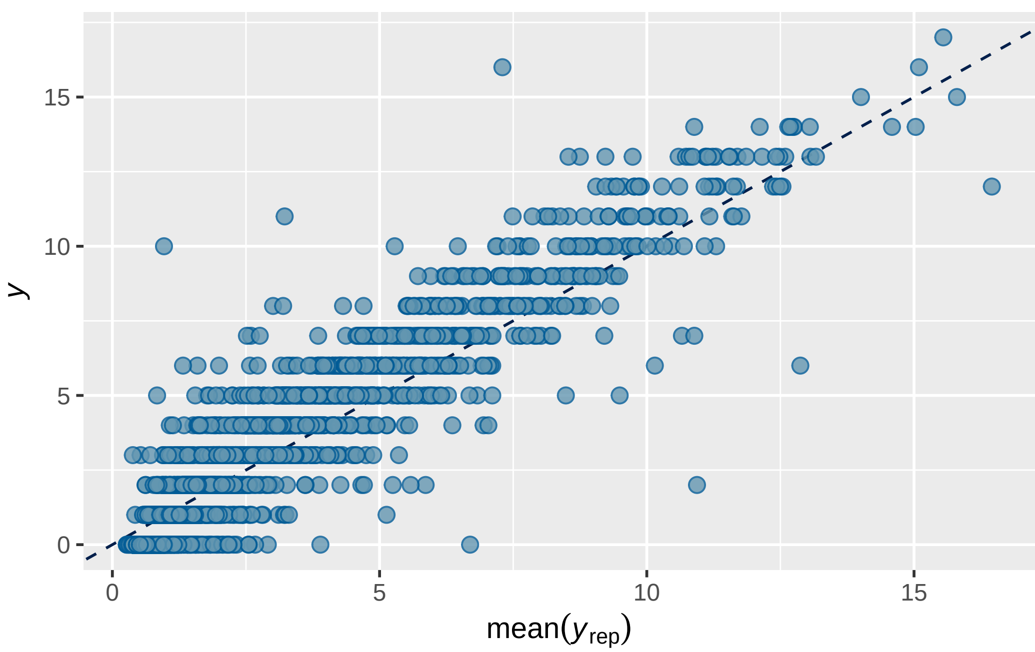

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



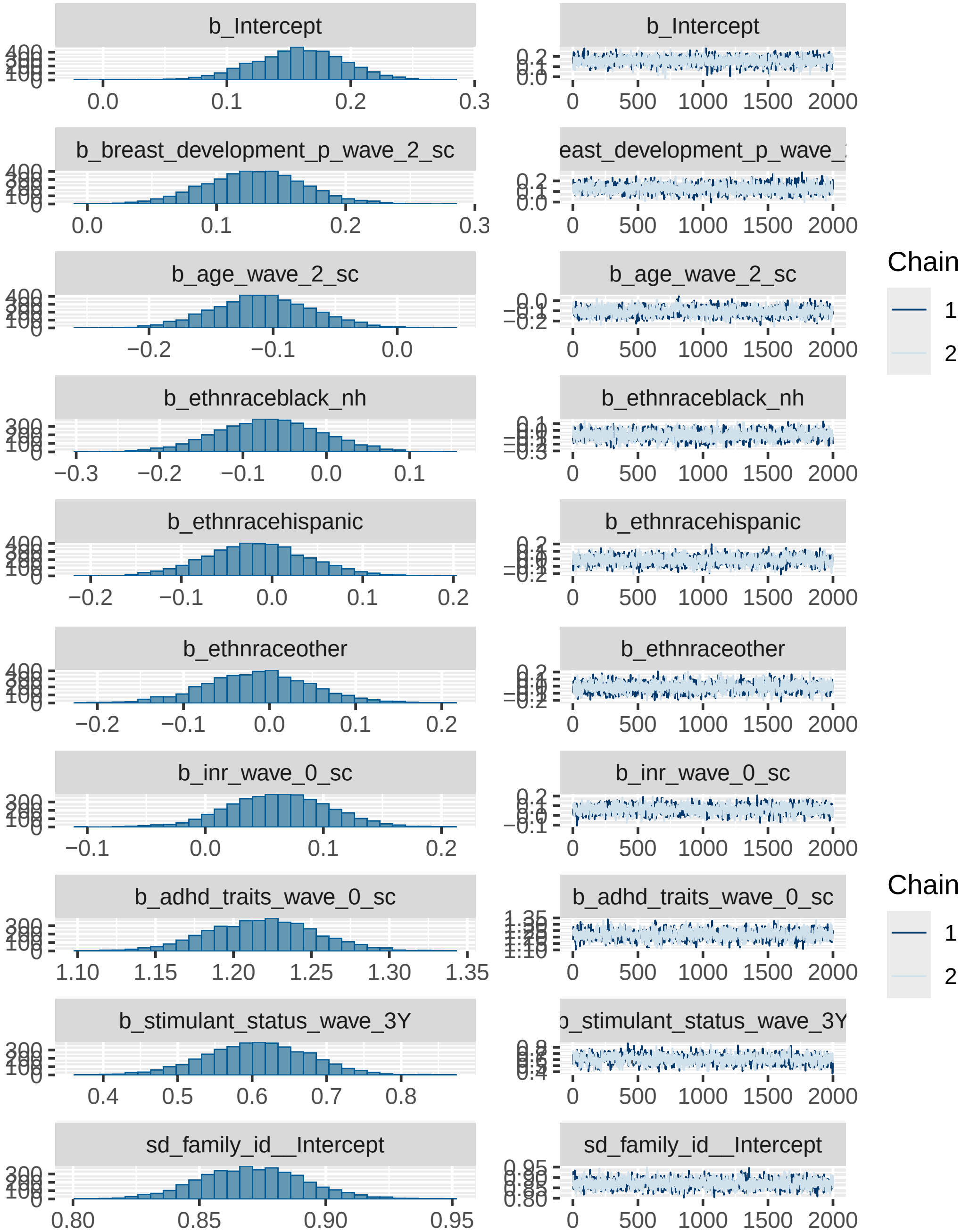
Model 1B longitudinal stimulant output

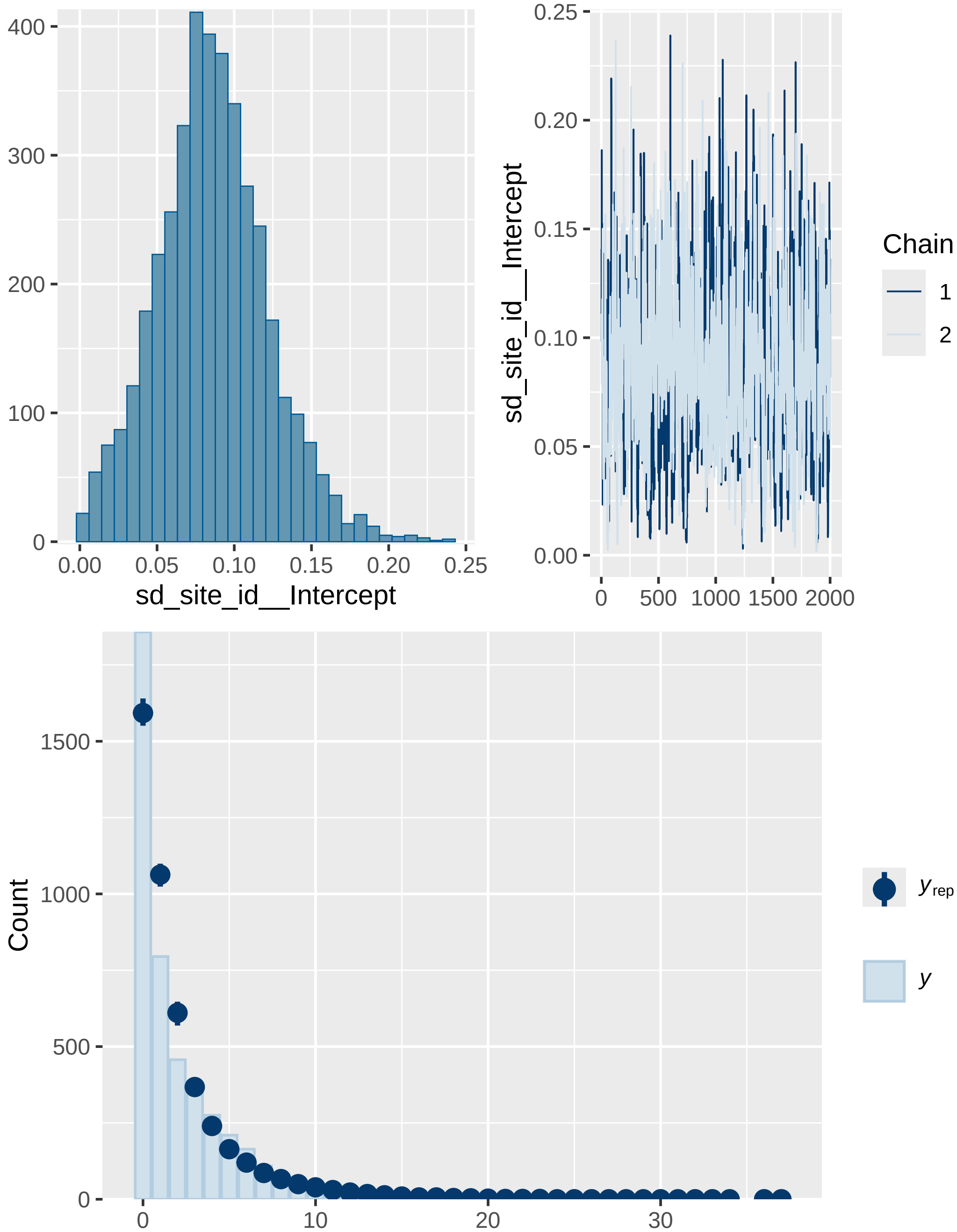
```
    Estimate   Est.Error      Q2.5    Q97.5
R2 0.7392941 0.007295095 0.7243178 0.752978


 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + in
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.871     0.021    0.831    0.913 1.001     1124     2045

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.086     0.036    0.017    0.162 1.004      473      604

Regression Coefficients:
                               Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                         0.158     0.037    0.083    0.229 1.001
breast_development_p_wave_2_sc    0.130     0.039    0.055    0.207 1.001
age_wave_2_sc                    -0.108     0.040   -0.185   -0.029 1.002
ethnraceblack_nh                 -0.068     0.063   -0.194    0.058 1.000
ethnracehispanic                 -0.017     0.056   -0.127    0.094 1.002
ethnraceother                    -0.013     0.062   -0.136    0.112 1.002
inr_wave_0_sc                     0.062     0.042   -0.021    0.143 1.000
adhd_traits_wave_0_sc             1.218     0.035    1.151    1.286 1.000
stimulant_status_wave_3Y          0.607     0.069    0.468    0.740 1.003
                               Bulk_ESS Tail_ESS
Intercept                          1655     2118
breast_development_p_wave_2_sc     1639     2517
age_wave_2_sc                      1140     2018
ethnraceblack_nh                   1398     2258
ethnracehispanic                   1093     2113
ethnraceother                      1478     2384
inr_wave_0_sc                      1366     2232
adhd_traits_wave_0_sc              1023     1712
stimulant_status_wave_3Y            969     1728
```

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

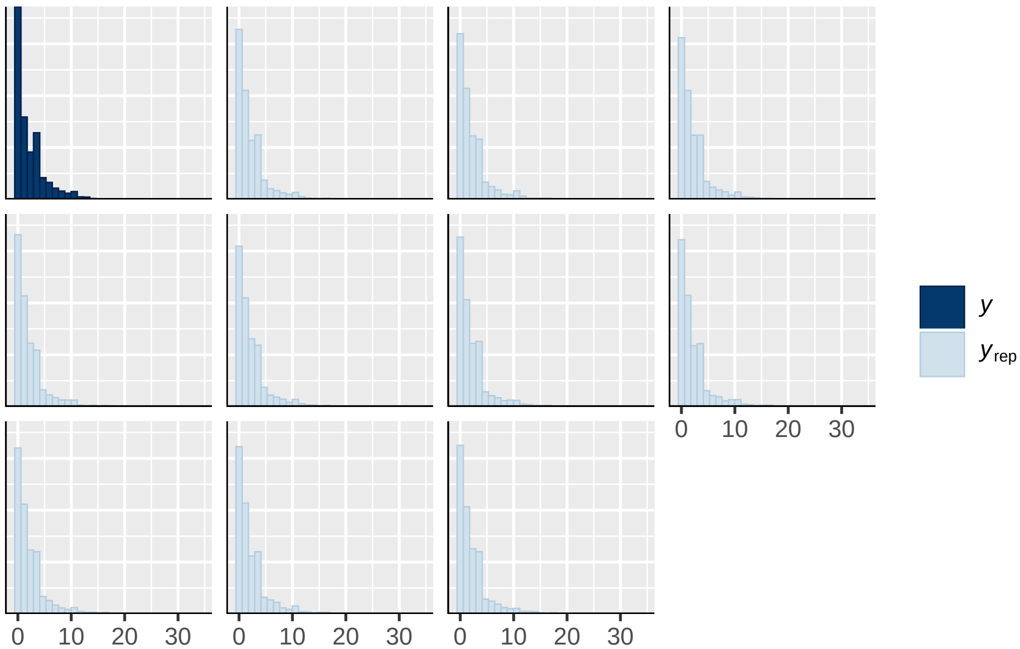
Model 1B longitudinal stimulant diagnostics

```
                prior     class                              coef     group resp
         normal(0, 1)         b                                                 
         normal(0, 1)         b          adhd_traits_wave_0_sc                  
         normal(0, 1)         b                  age_wave_2_sc                  
         normal(0, 1)         b breast_development_p_wave_2_sc                  
         normal(0, 1)         b               ethnraceblack_nh                  
         normal(0, 1)         b               ethnracehispanic                  
         normal(0, 1)         b                  ethnraceother                  
         normal(0, 1)         b                  inr_wave_0_sc                  
         normal(0, 1)         b       stimulant_status_wave_3Y                  
 student_t(3, 0, 2.7) Intercept                                                 
 student_t(3, 0, 2.7)        sd                                                 
 student_t(3, 0, 2.7)        sd                                 family_id       
 student_t(3, 0, 2.7)        sd                       Intercept family_id       
 student_t(3, 0, 2.7)        sd                                   site_id       
 student_t(3, 0, 2.7)        sd                       Intercept   site_id       
 dpar nlpar lb ub tag       source
                              user
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                           default
             0             default
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
```

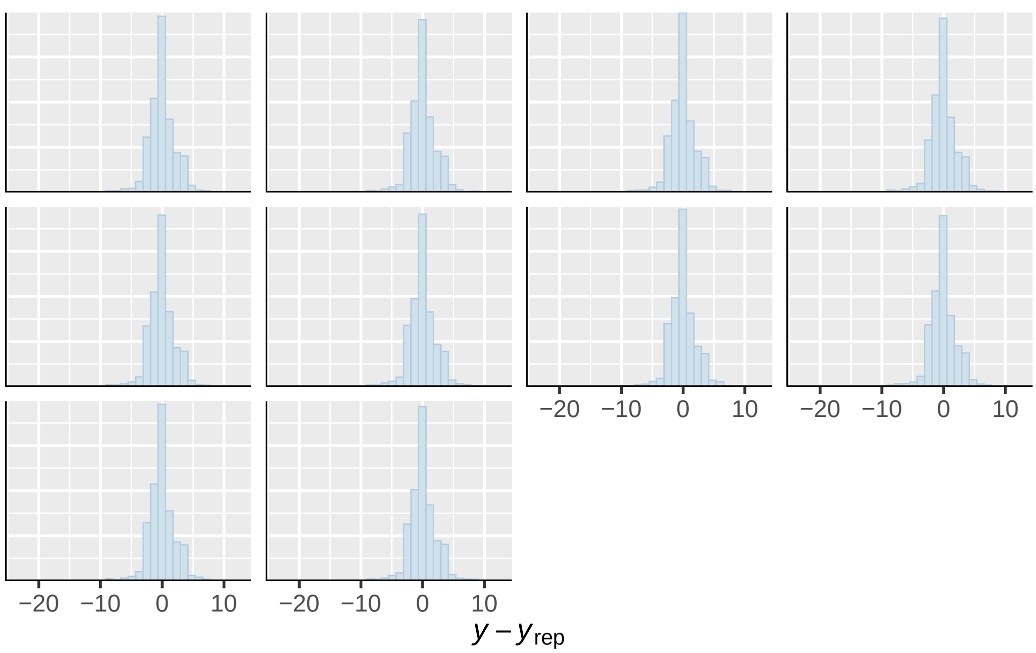

b_Intercept
b_breast_development_p_wave_2_sc
b_age_wave_2_sc
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnraceother
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
b_stimulant_status_wave_3Y
sd_family_id__Intercept
Chain
1
2



`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

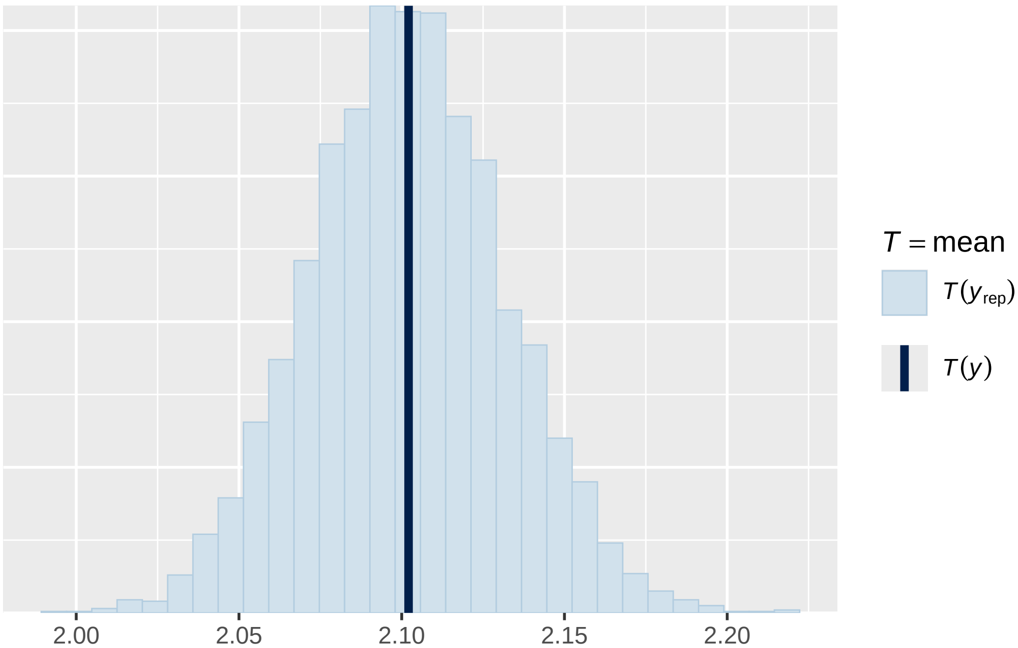



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

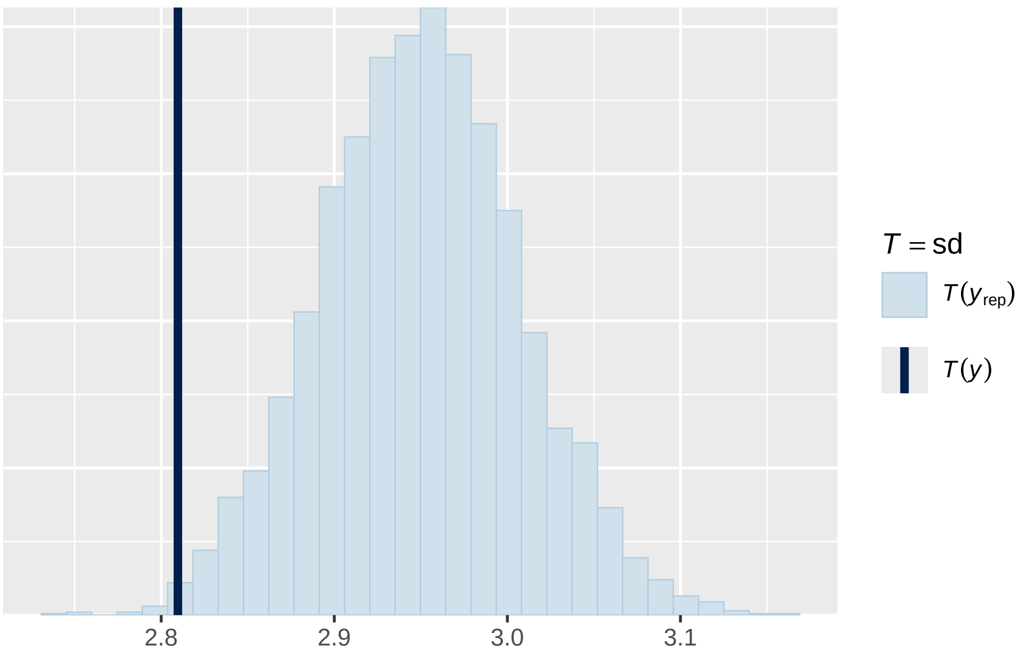



```
Using all posterior draws for ppc type 'stat' by default.
```

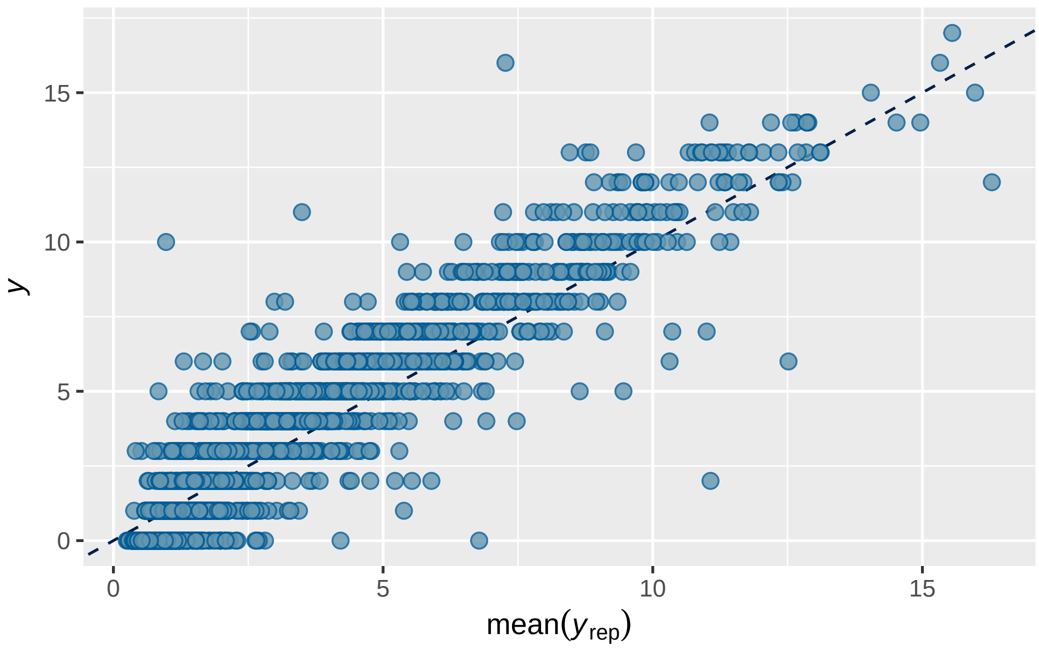
```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5B longitudinal stimulant output

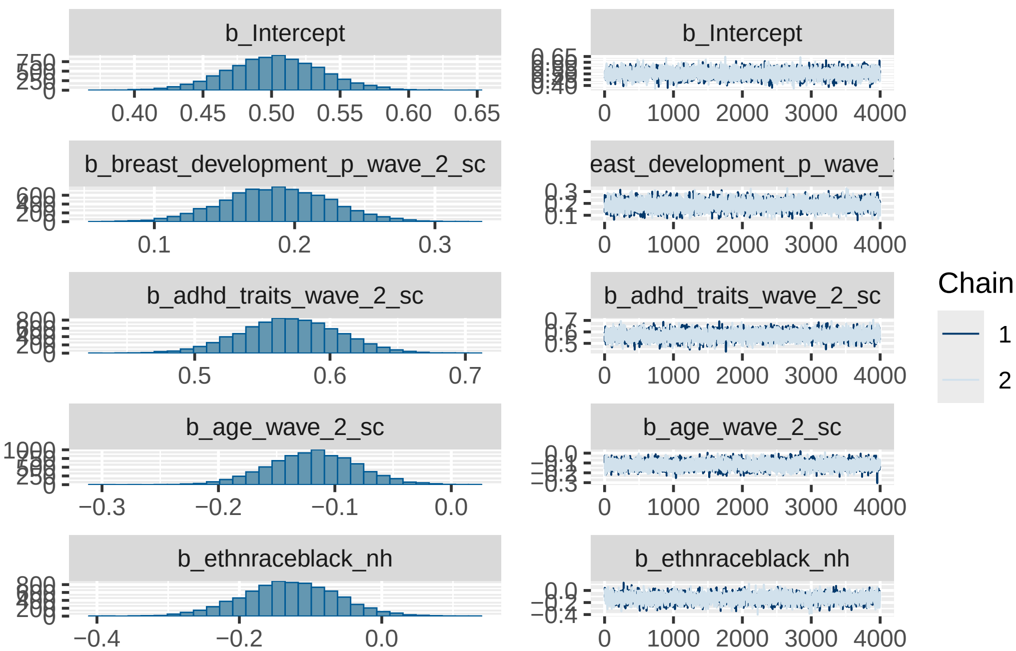
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8239053 0.005205048 0.8133844 0.8337296


 Family: poisson
  Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age_
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.986     0.020    0.946    1.026 1.001     2381     4669

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.065     0.037    0.005    0.145 1.009      611     1517
```

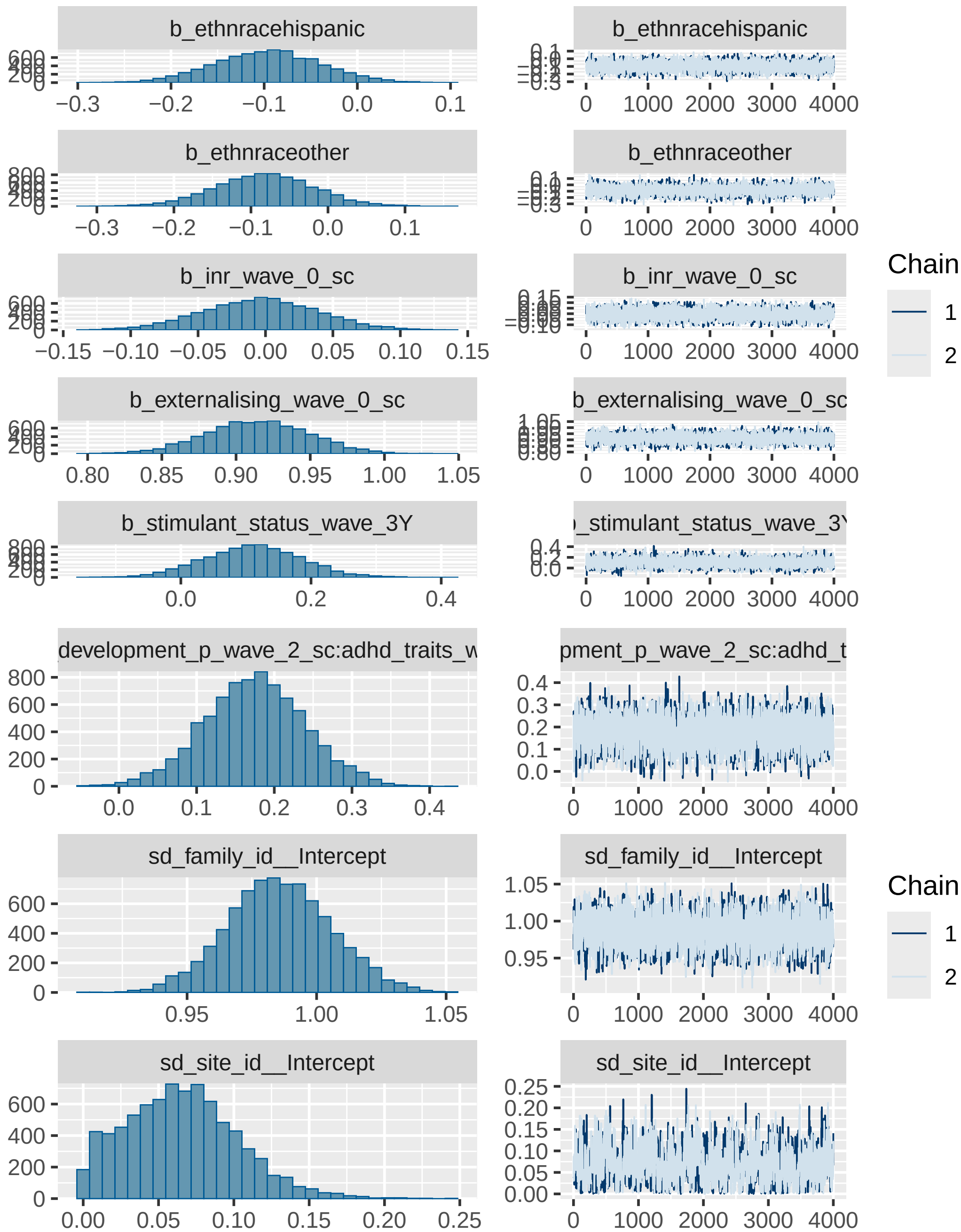
```
Regression Coefficients:
                                                  Estimate Est.Error
Intercept                                            0.503     0.035
breast_development_p_wave_2_sc                       0.188     0.038
adhd_traits_wave_2_sc                                0.570     0.036
age_wave_2_sc                                       -0.119     0.038
ethnraceblack_nh                                    -0.129     0.066
ethnracehispanic                                    -0.096     0.057
ethnraceother                                       -0.082     0.064
inr_wave_0_sc                                       -0.004     0.042
externalising_wave_0_sc                              0.914     0.035
stimulant_status_wave_3Y                             0.111     0.074
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc  0.175     0.065
                                                  l-95% CI u-95% CI  Rhat
Intercept                                            0.435    0.571 1.001
breast_development_p_wave_2_sc                       0.114    0.264 1.001
adhd_traits_wave_2_sc                                0.498    0.641 1.001
age_wave_2_sc                                       -0.193   -0.045 1.000
ethnraceblack_nh                                    -0.260    0.000 1.003
ethnracehispanic                                    -0.206    0.015 1.000
ethnraceother                                       -0.209    0.044 1.001
inr_wave_0_sc                                       -0.086    0.079 1.001
externalising_wave_0_sc                              0.846    0.982 1.001
stimulant_status_wave_3Y                            -0.033    0.257 1.001
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc  0.044    0.304 1.000
                                                  Bulk_ESS Tail_ESS
Intercept                                             4138     4789
breast_development_p_wave_2_sc                        4101     5676
adhd_traits_wave_2_sc                                 2777     4361
age_wave_2_sc                                         3692     5187
ethnraceblack_nh                                      2311     4174
ethnracehispanic                                      3236     4672
ethnraceother                                         2893     4509
inr_wave_0_sc                                         3251     5037
externalising_wave_0_sc                               2506     3980
stimulant_status_wave_3Y                              2319     3887
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc   2940     4482

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

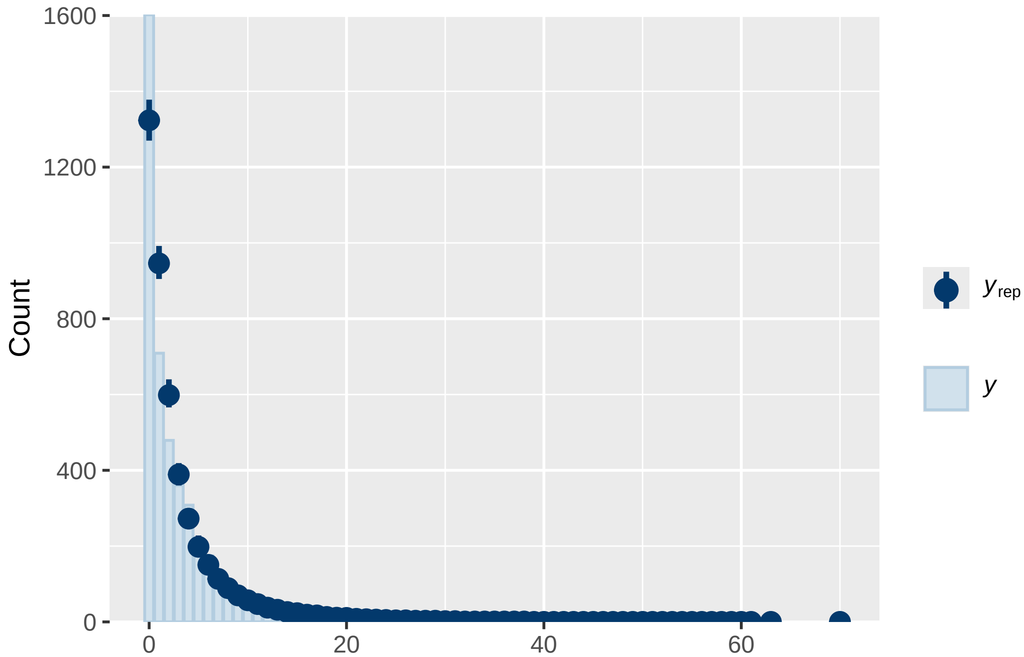
Model 5B longitudinal stimulant diagnostics

```
               prior     class
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
        normal(0, 1)         b
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
                                                  coef     group resp dpar nlpar

                               adhd_traits_wave_2_sc
                                       age_wave_2_sc
                      breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
                                    ethnraceblack_nh
                                    ethnracehispanic
                                       ethnraceother
                             externalising_wave_0_sc
                                       inr_wave_0_sc
                              stimulant_status_wave_3Y


                                                       family_id
                                            Intercept family_id
                                                         site_id
                                            Intercept   site_id
lb ub tag       source
                  user
          (vectorized)
          (vectorized)
```

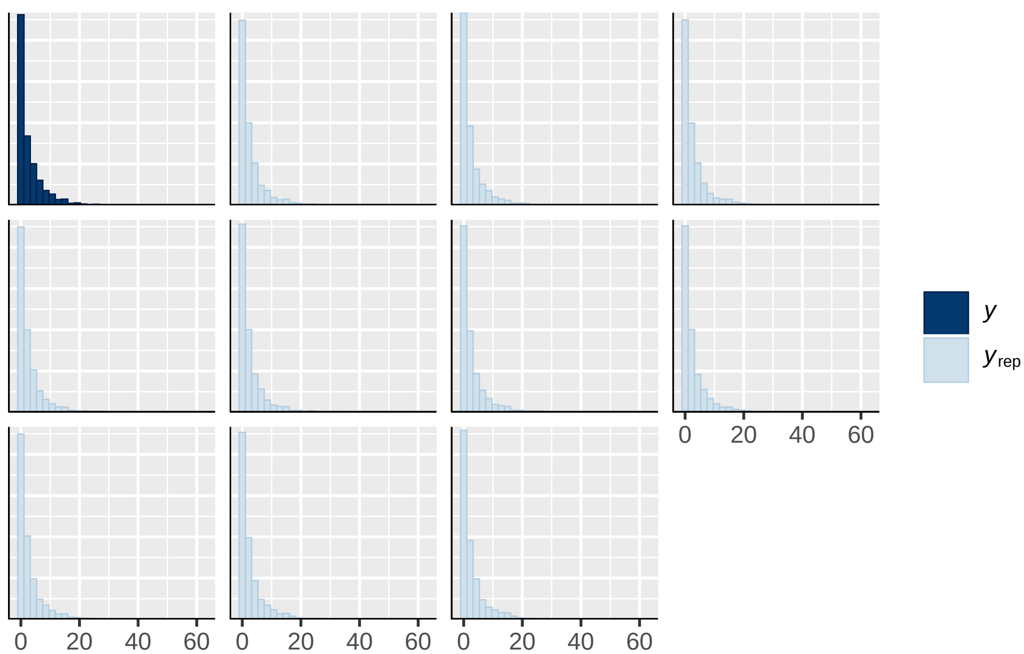
```
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
                default
0               default
0          (vectorized)
0          (vectorized)
0          (vectorized)
0          (vectorized)
```



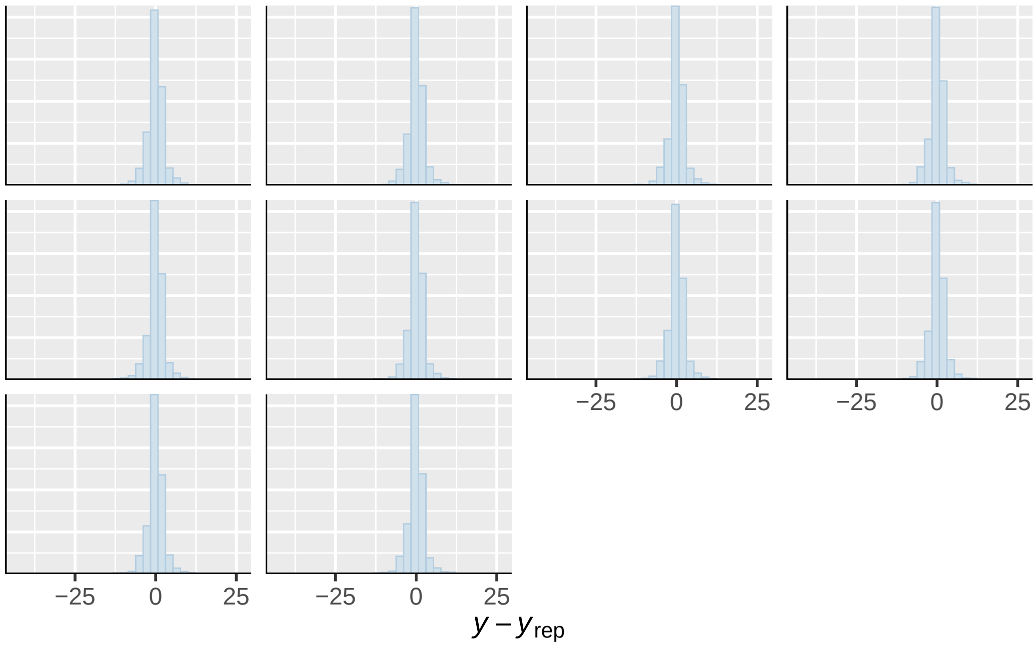

b_ethnracehispanic
b_ethnraceother
b_inr_wave_0_sc
b_externalising_wave_0_sc
b_stimulant_status_wave_3Y
development_p_wave_2_sc:adhd_traits_w
pment_p_wave_2_sc:adhd_t
sd_family_id__Intercept
sd_site_id__Intercept
Chain
1
2



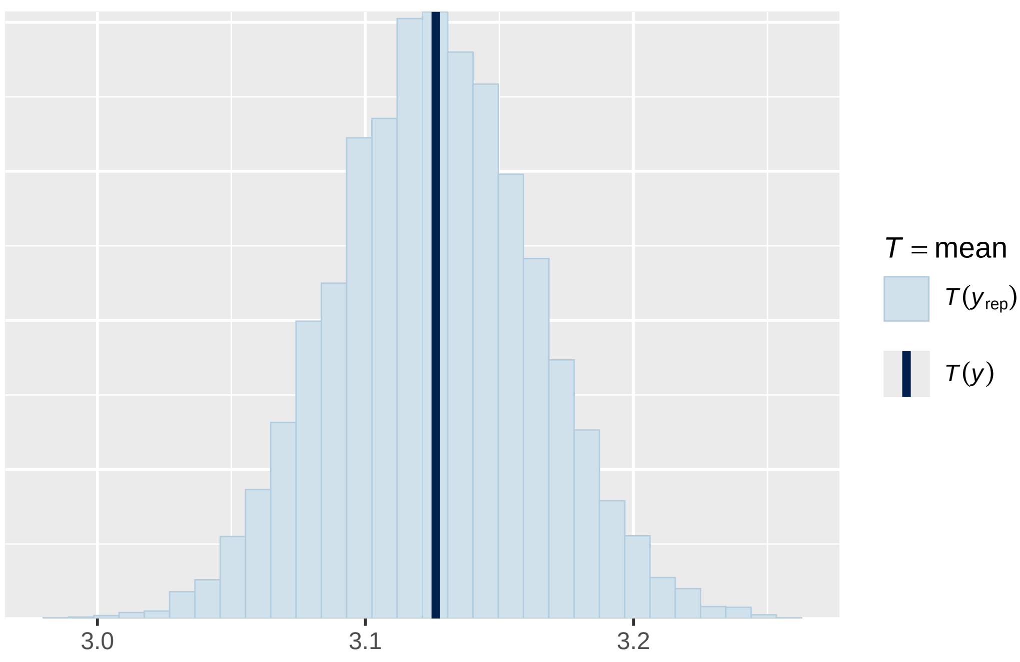

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



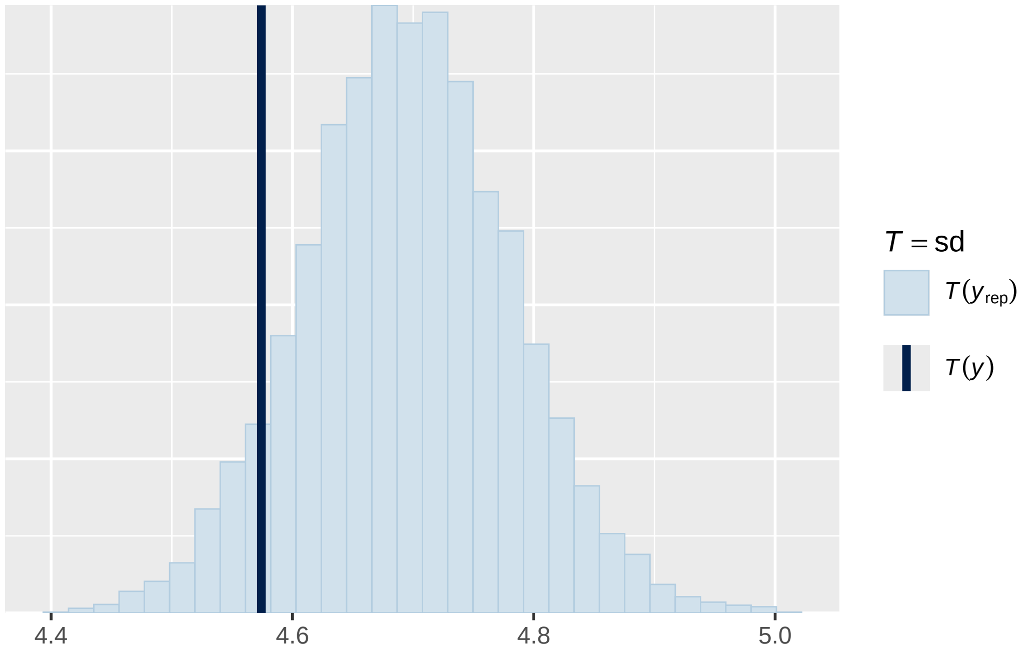

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



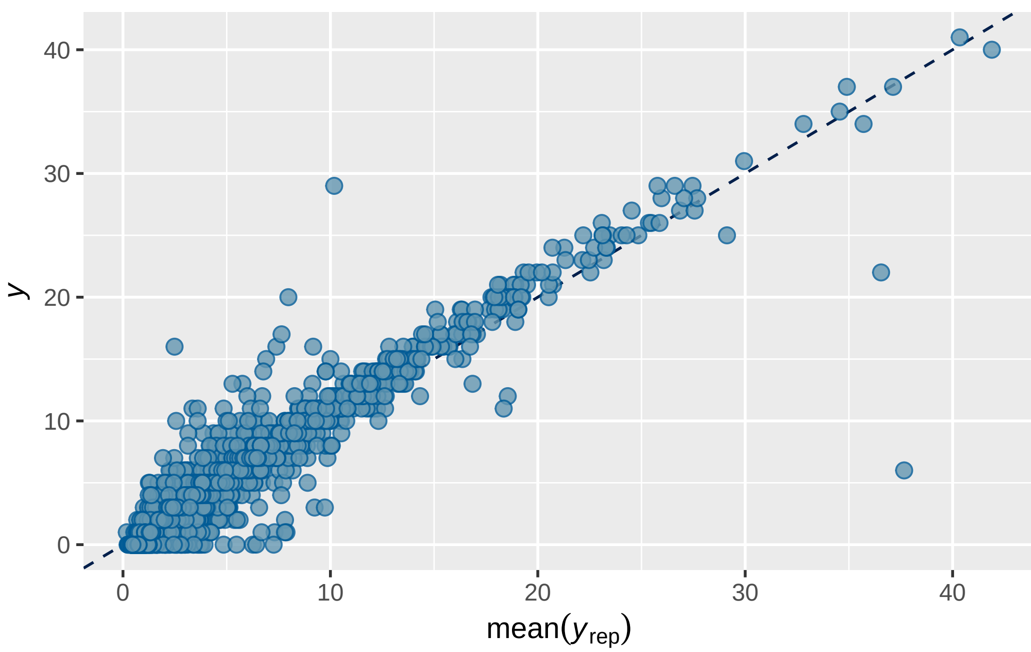

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



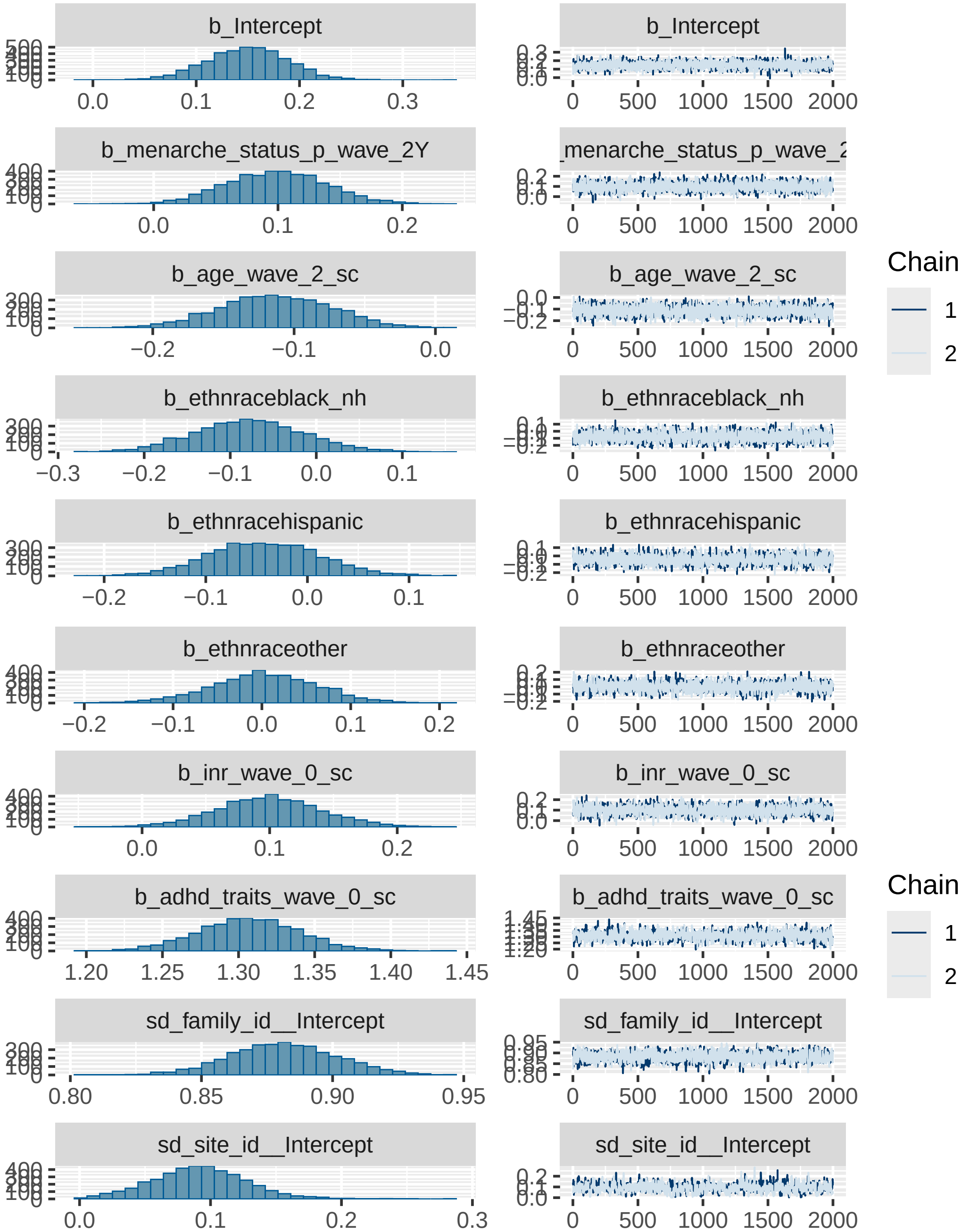
Model 1 prior 1 output

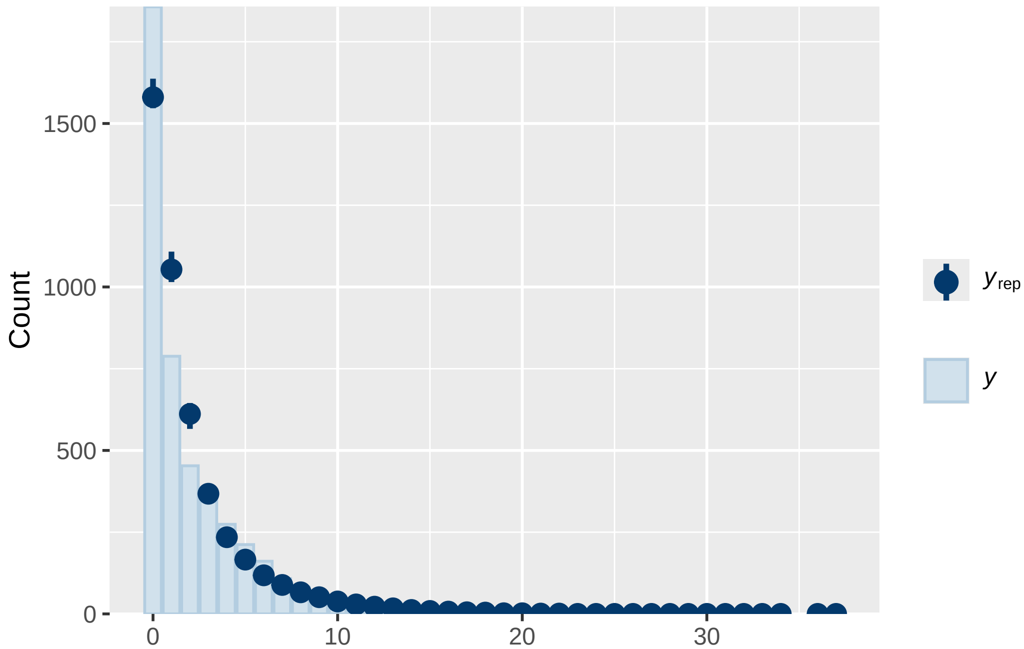
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7348938 0.007531416 0.7199932 0.7493823


 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave_
   Data: analytic_df_wide (Number of observations: 4496)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3915)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.881     0.020    0.841    0.922 1.007      937     1457

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.092     0.036    0.020    0.165 1.007      504      518

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                    0.150     0.039    0.072    0.224 1.003     1400
menarche_status_p_wave_2Y    0.099     0.040    0.022    0.178 1.001     1665
age_wave_2_sc               -0.113     0.040   -0.191   -0.038 1.001     1723
ethnraceblack_nh            -0.077     0.063   -0.197    0.049 1.001     1514
ethnracehispanic            -0.042     0.054   -0.146    0.066 1.001     1381
ethnraceother                0.003     0.060   -0.116    0.121 1.001     1324
inr_wave_0_sc                0.100     0.041    0.018    0.182 1.000     1489
adhd_traits_wave_0_sc        1.307     0.033    1.242    1.374 1.001     1073
                          Tail_ESS
Intercept                     2033
menarche_status_p_wave_2Y     2492
age_wave_2_sc                 2637
ethnraceblack_nh              2412
ethnracehispanic              2077
ethnraceother                 1677
inr_wave_0_sc                 2112
adhd_traits_wave_0_sc         1886

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
```

and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

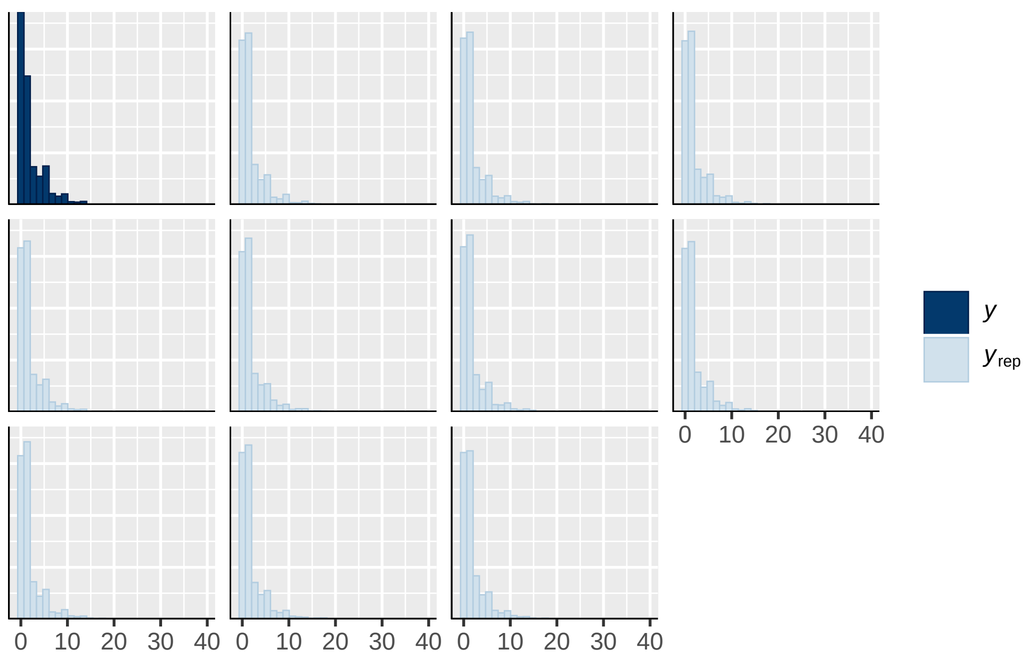
Model 1 longitudinal prior 1 diagnostics

```
               prior     class                        coef     group resp dpar
      normal(0, 0.25)         b                                               
      normal(0, 0.25)         b     adhd_traits_wave_0_sc                     
      normal(0, 0.25)         b              age_wave_2_sc                    
      normal(0, 0.25)         b          ethnraceblack_nh                     
      normal(0, 0.25)         b          ethnracehispanic                     
      normal(0, 0.25)         b              ethnraceother                    
      normal(0, 0.25)         b              inr_wave_0_sc                    
      normal(0, 0.25)         b menarche_status_p_wave_2Y                     
 student_t(3, 0, 2.7) Intercept                                               
 student_t(3, 0, 2.7)        sd                                               
 student_t(3, 0, 2.7)        sd                             family_id          
 student_t(3, 0, 2.7)        sd                  Intercept family_id          
 student_t(3, 0, 2.7)        sd                               site_id         
 student_t(3, 0, 2.7)        sd                  Intercept   site_id          
 nlpar lb ub tag       source
                         user
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                 (vectorized)
                      default
        0             default
        0        (vectorized)
        0        (vectorized)
        0        (vectorized)
        0        (vectorized)
```

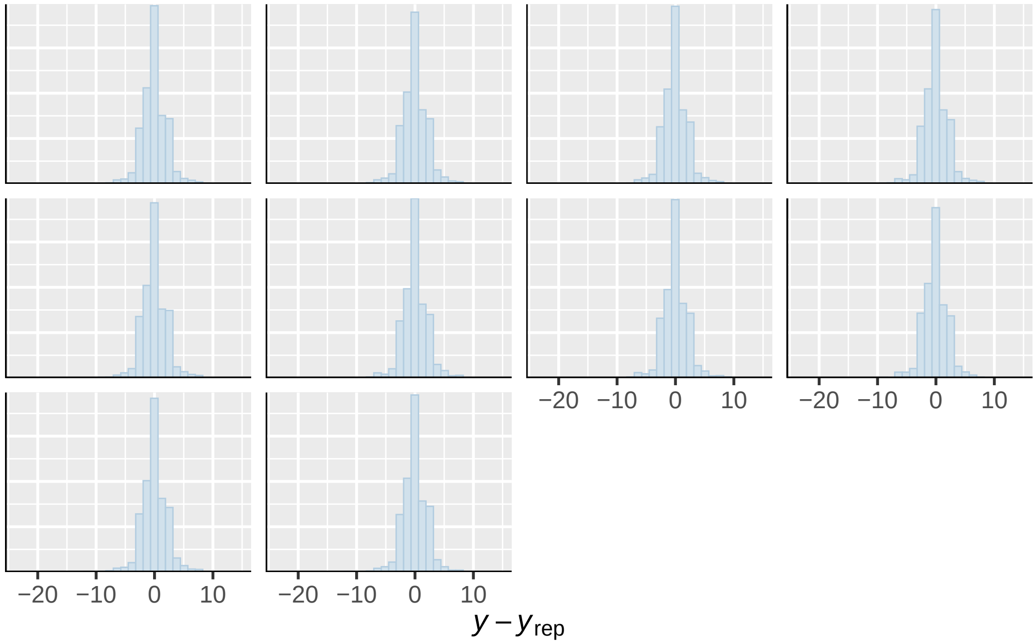

b_Intercept
b_menarche_status_p_wave_2Y
b_age_wave_2_sc
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnraceother
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
sd_family_id__Intercept
sd_site_id__Intercept
Chain
1
2



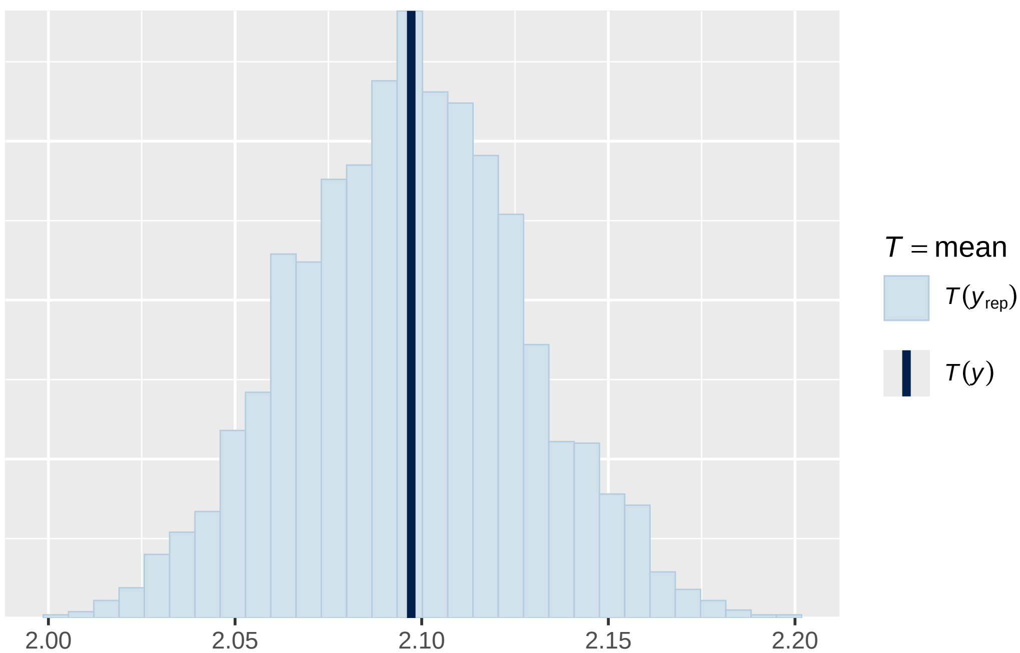

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



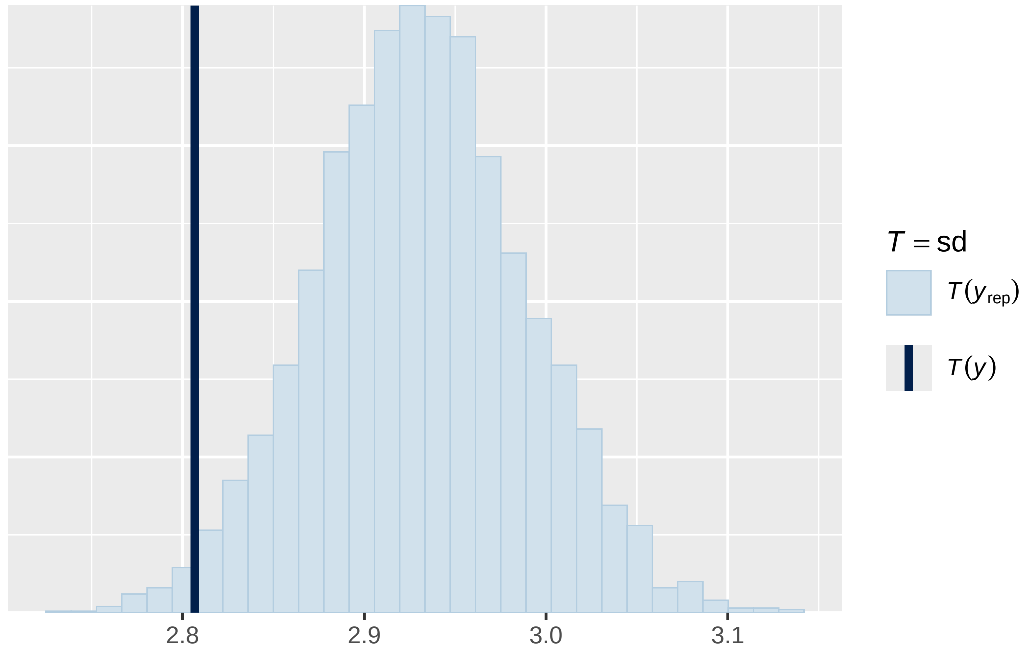

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



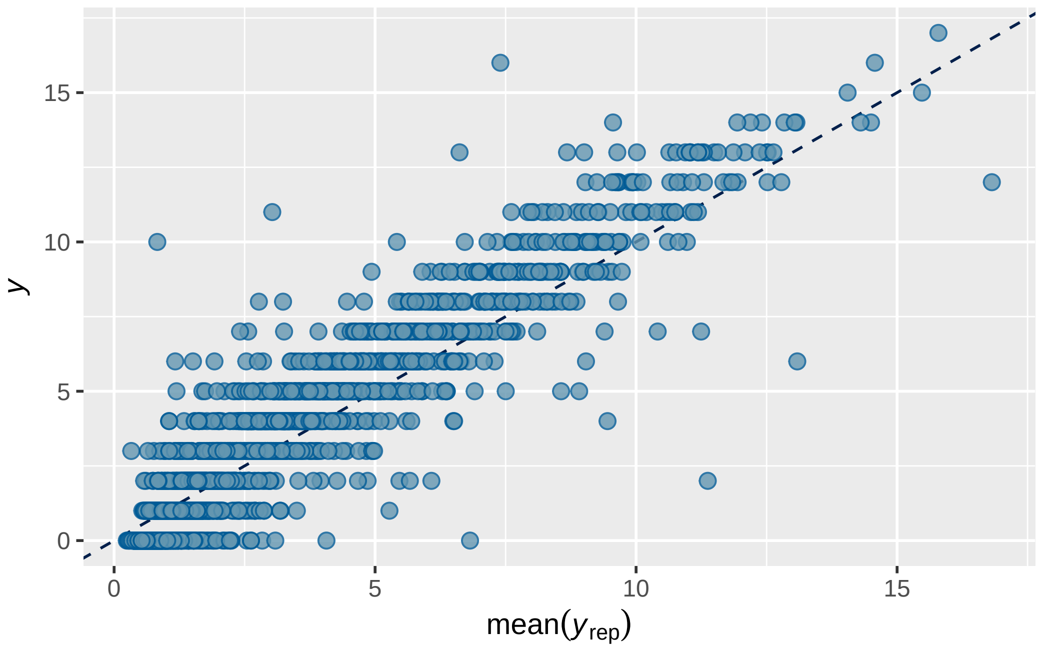

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



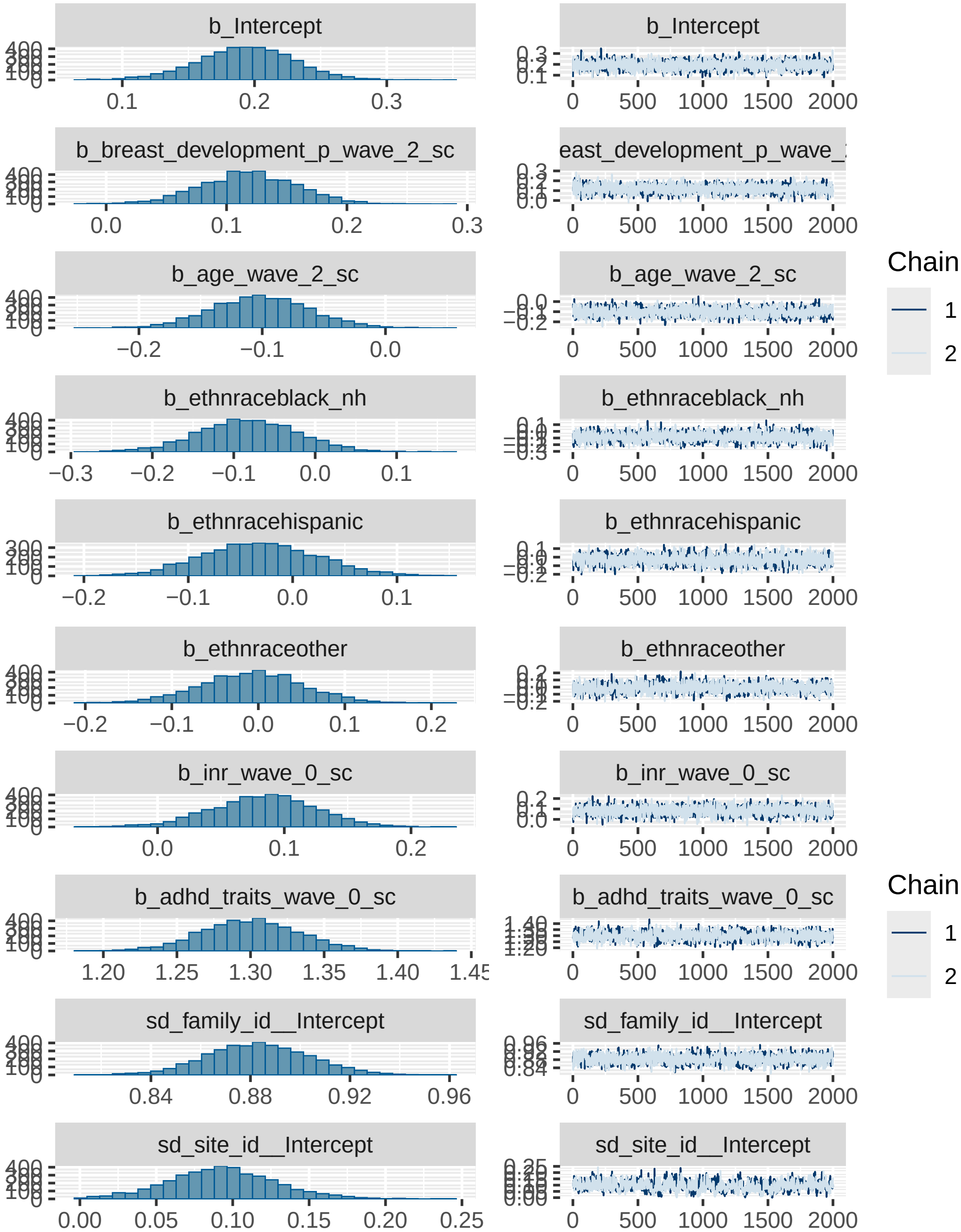
Model 1B longitudinal prior 1 output

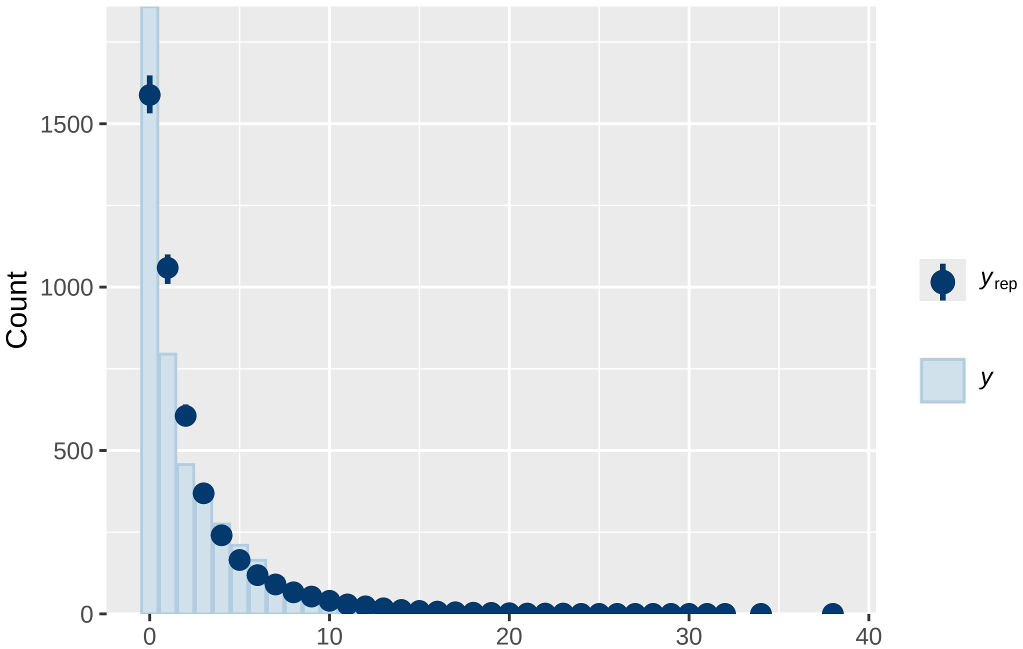
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7345825 0.007551707 0.7193221 0.7491023


 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + in
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.883     0.021    0.844    0.924 1.001     1351     2068

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.093     0.035    0.024    0.166 1.004      427      447

Regression Coefficients:
                               Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                         0.193     0.036    0.118    0.264 1.001
breast_development_p_wave_2_sc    0.118     0.040    0.039    0.194 1.002
age_wave_2_sc                    -0.101     0.039   -0.176   -0.024 1.002
ethnraceblack_nh                 -0.082     0.062   -0.205    0.039 1.000
ethnracehispanic                 -0.031     0.055   -0.135    0.081 1.001
ethnraceother                    -0.007     0.059   -0.125    0.107 1.001
inr_wave_0_sc                     0.084     0.041    0.001    0.162 1.001
adhd_traits_wave_0_sc             1.300     0.033    1.233    1.366 1.001
                               Bulk_ESS Tail_ESS
Intercept                          1961     2153
breast_development_p_wave_2_sc     1659     2617
age_wave_2_sc                      1453     2377
ethnraceblack_nh                   1491     2487
ethnracehispanic                   1159     1633
ethnraceother                      1565     2433
inr_wave_0_sc                      1462     2222
adhd_traits_wave_0_sc              1081     2297

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
```

and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

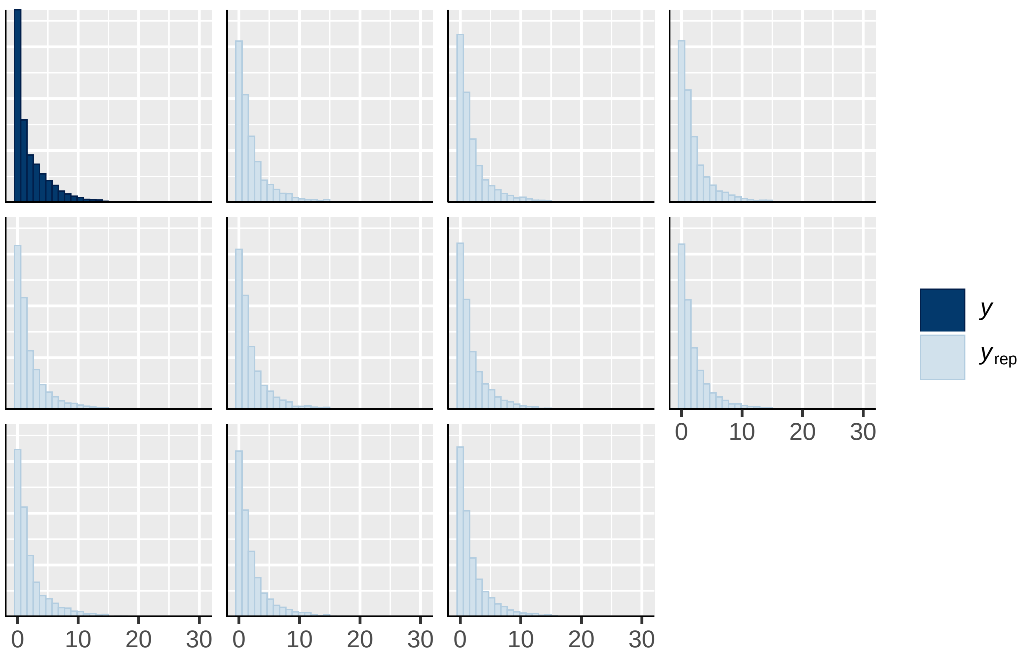
Model 1B longitudinal prior 1 diagnostics

```
                prior     class                          coef     group resp
      normal(0, 0.25)         b
      normal(0, 0.25)         b          adhd_traits_wave_0_sc
      normal(0, 0.25)         b                  age_wave_2_sc
      normal(0, 0.25)         b breast_development_p_wave_2_sc
      normal(0, 0.25)         b               ethnraceblack_nh
      normal(0, 0.25)         b              ethnracehispanic
      normal(0, 0.25)         b                  ethnraceother
      normal(0, 0.25)         b                  inr_wave_0_sc
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                                family_id
 student_t(3, 0, 2.7)        sd                      Intercept family_id
 student_t(3, 0, 2.7)        sd                                  site_id
 student_t(3, 0, 2.7)        sd                      Intercept   site_id
 dpar nlpar lb ub tag       source
                              user
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                           default
             0             default
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
```

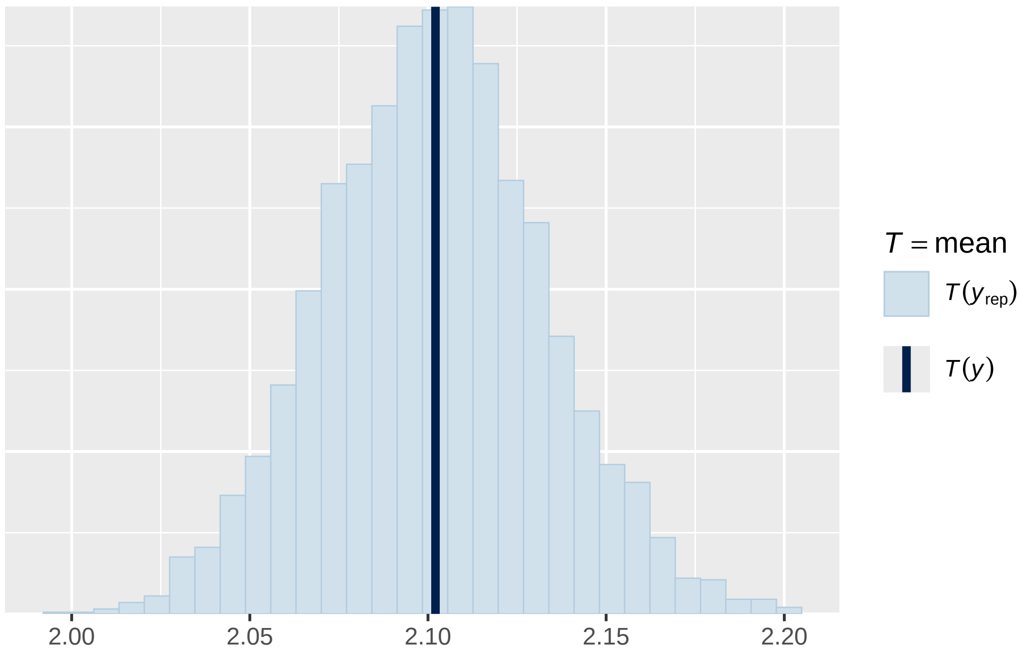

b_Intercept
b_Intercept
b_breast_development_p_wave_2_sc
east_development_p_wave_
b_age_wave_2_sc
b_age_wave_2_sc
b_ethnraceblack_nh
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnracehispanic
b_ethnraceother
b_ethnraceother
b_inr_wave_0_sc
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
b_adhd_traits_wave_0_sc
sd_family_id__Intercept
sd_family_id__Intercept
sd_site_id__Intercept
sd_site_id__Intercept
Chain
1
2
Chain
1
2



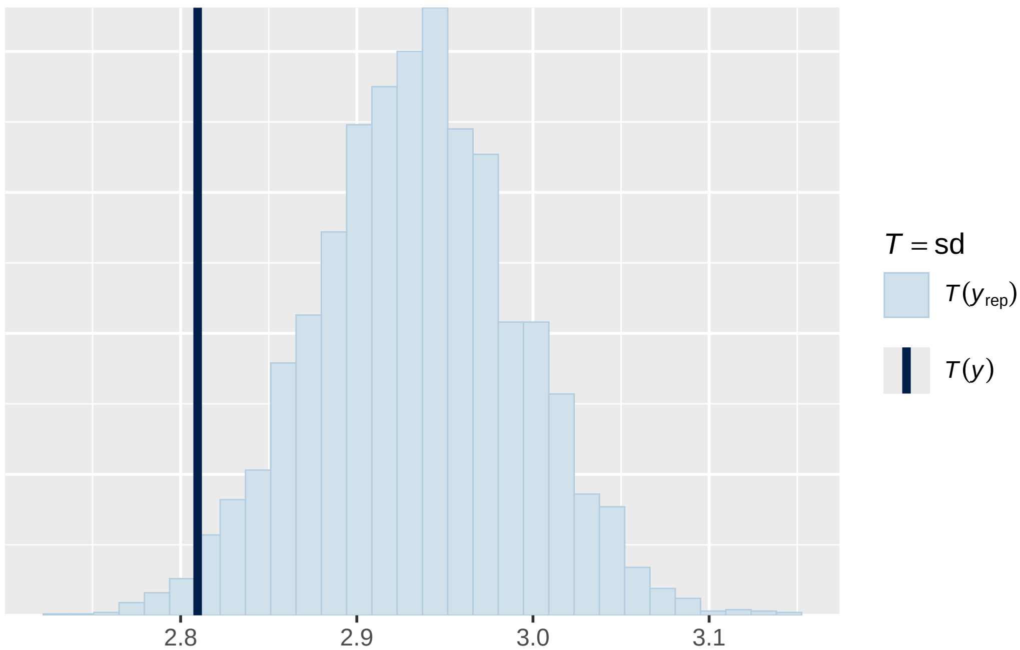

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



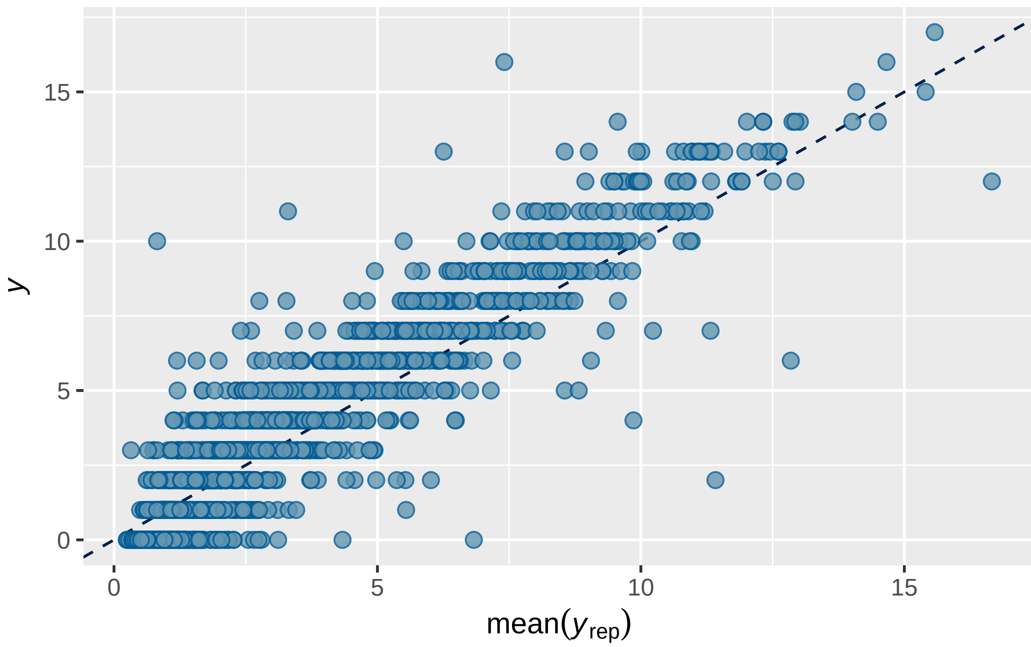

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5B longitudinal prior 1 output

```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8229693 0.005223904 0.8125308 0.8329675


 Family: poisson
  Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age_
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.983     0.020    0.946    1.023 1.002     1810     3364

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.067     0.037    0.005    0.147 1.002      549     1105

Regression Coefficients:
```

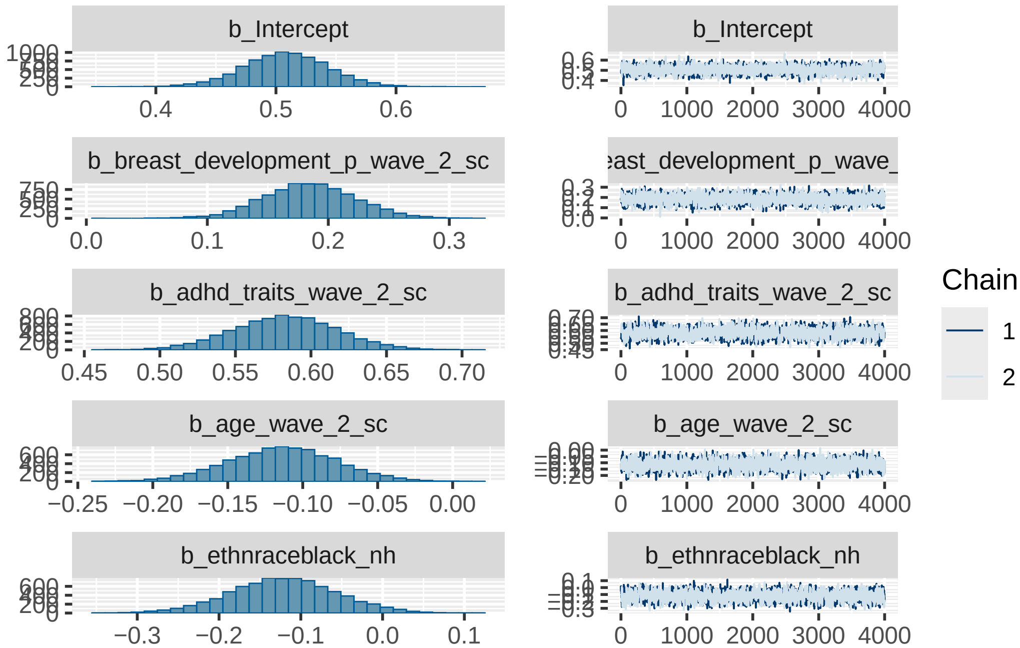
```
                                                    Estimate Est.Error
Intercept                                              0.508     0.035
breast_development_p_wave_2_sc                         0.185     0.038
adhd_traits_wave_2_sc                                  0.583     0.034
age_wave_2_sc                                         -0.114     0.036
ethnraceblack_nh                                      -0.121     0.065
ethnracehispanic                                      -0.090     0.056
ethnraceother                                         -0.072     0.062
inr_wave_0_sc                                          0.000     0.042
externalising_wave_0_sc                                0.903     0.034
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc   0.159     0.063
                                                    l-95% CI u-95% CI  Rhat
Intercept                                              0.437    0.577 1.001
breast_development_p_wave_2_sc                         0.111    0.260 1.001
adhd_traits_wave_2_sc                                  0.515    0.651 1.002
age_wave_2_sc                                         -0.186   -0.043 1.001
ethnraceblack_nh                                      -0.251    0.008 1.000
ethnracehispanic                                      -0.199    0.020 1.001
ethnraceother                                         -0.192    0.049 1.002
inr_wave_0_sc                                         -0.081    0.085 1.001
externalising_wave_0_sc                                0.838    0.968 1.000
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc   0.033    0.282 1.001
                                                    Bulk_ESS Tail_ESS
Intercept                                               1540     2546
breast_development_p_wave_2_sc                          2252     3825
adhd_traits_wave_2_sc                                   1485     2841
age_wave_2_sc                                           2118     4282
ethnraceblack_nh                                        1606     2644
ethnracehispanic                                        1581     2946
ethnraceother                                           1532     2744
inr_wave_0_sc                                           1856     3055
externalising_wave_0_sc                                 1264     2791
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc    1515     3545

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

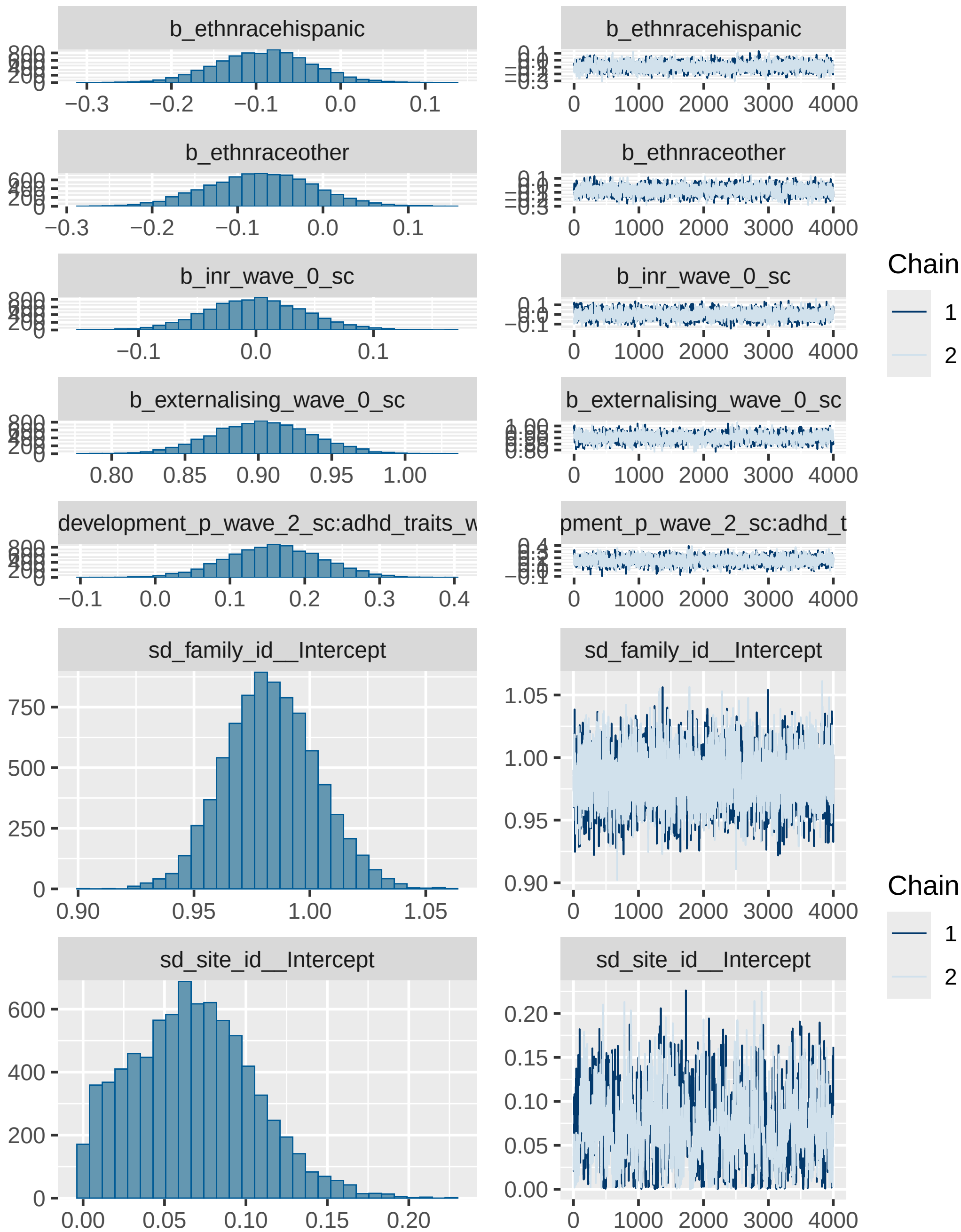
Model 5B longitudinal prior 1 diagnostics

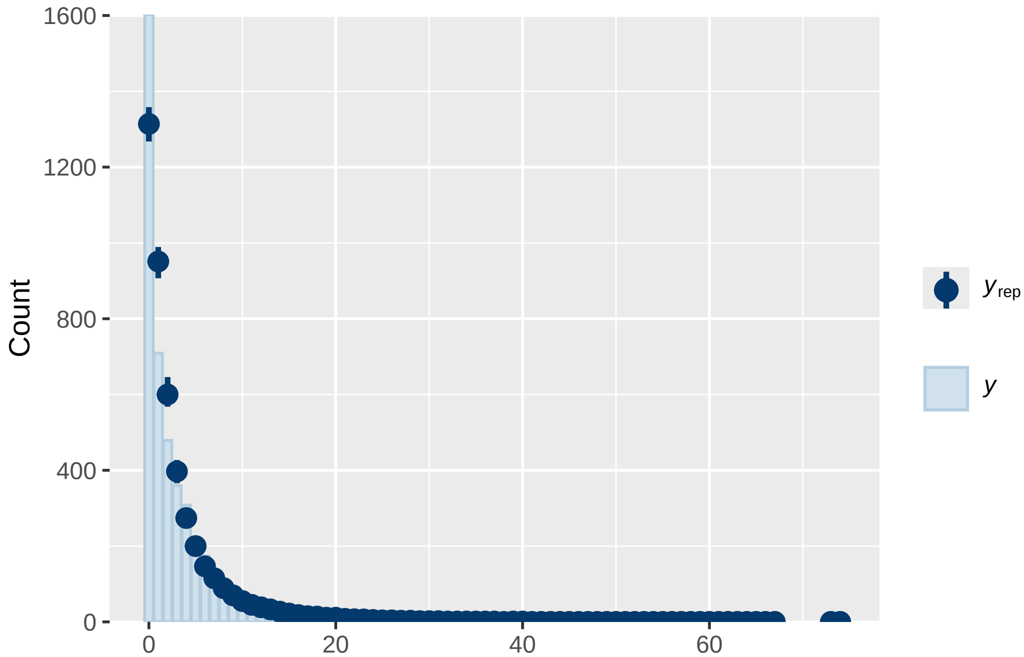
```
           prior     class
      normal(0, 0.25)         b
```

```
     normal(0, 0.25)         b
     normal(0, 0.25)         b
     normal(0, 0.25)         b
     normal(0, 0.25)         b
     normal(0, 0.25)         b
     normal(0, 0.25)         b
     normal(0, 0.25)         b
     normal(0, 0.25)         b
     normal(0, 0.25)         b
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
student_t(3, 0, 2.9)        sd
                                                   coef     group resp dpar nlpar

                                 adhd_traits_wave_2_sc
                                         age_wave_2_sc
                      breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
                                     ethnraceblack_nh
                                     ethnracehispanic
                                         ethnraceother
                              externalising_wave_0_sc
                                         inr_wave_0_sc


                                                          family_id
                                              Intercept family_id
                                                            site_id
                                              Intercept   site_id
lb ub tag       source
                  user
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
```

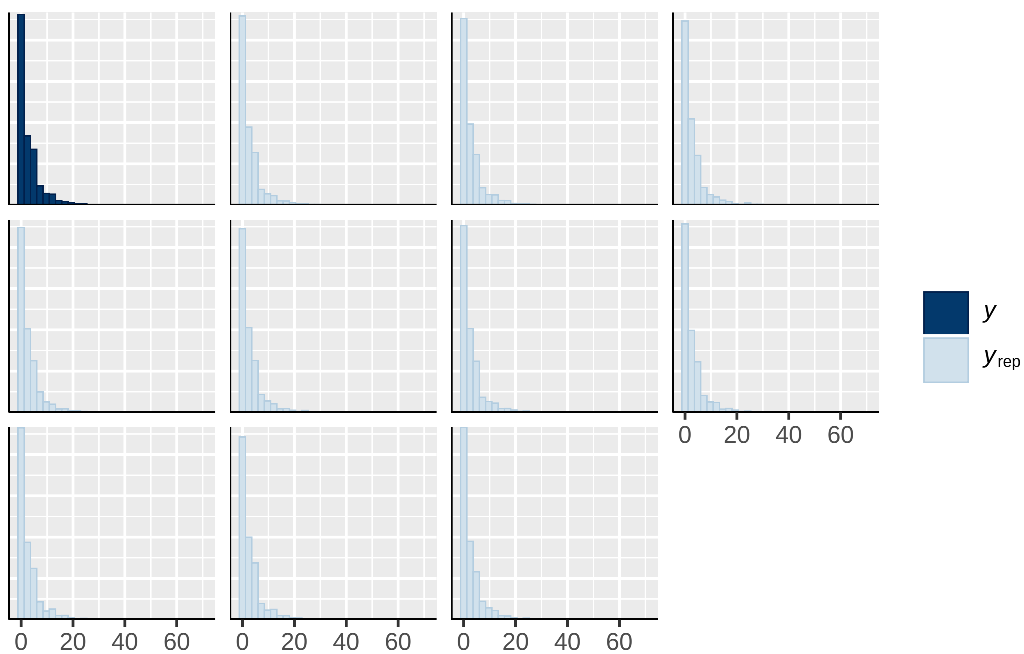
```
                  default
0                 default
0            (vectorized)
0            (vectorized)
0            (vectorized)
0            (vectorized)
```



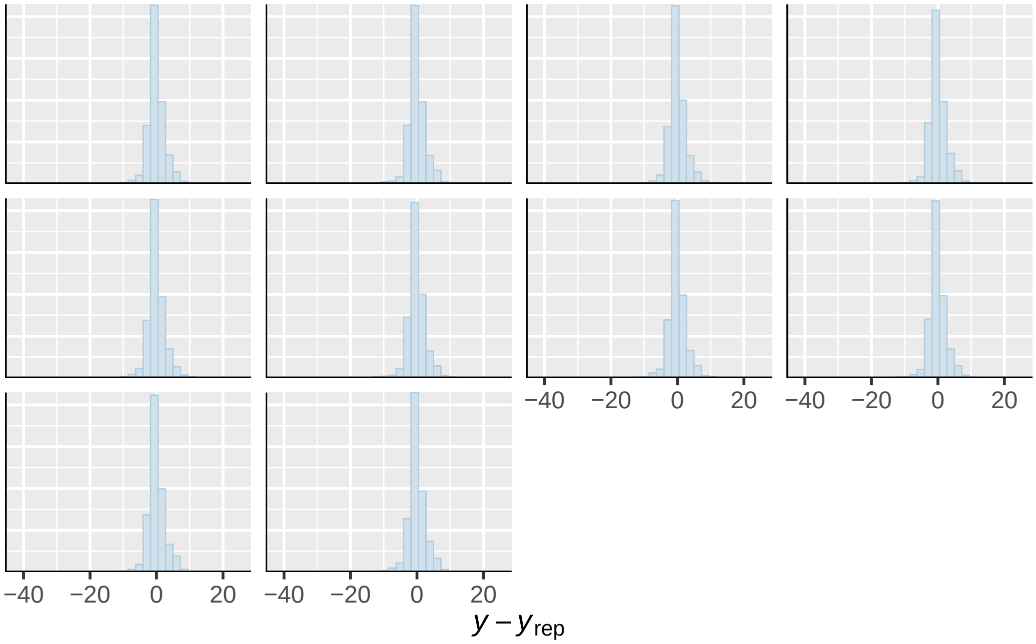

b_ethnracehispanic
b_ethnraceother
b_inr_wave_0_sc
b_externalising_wave_0_sc
development_p_wave_2_sc:adhd_traits_w
pment_p_wave_2_sc:adhd_t
sd_family_id__Intercept
sd_site_id__Intercept
Chain
1
2
Chain
1
2



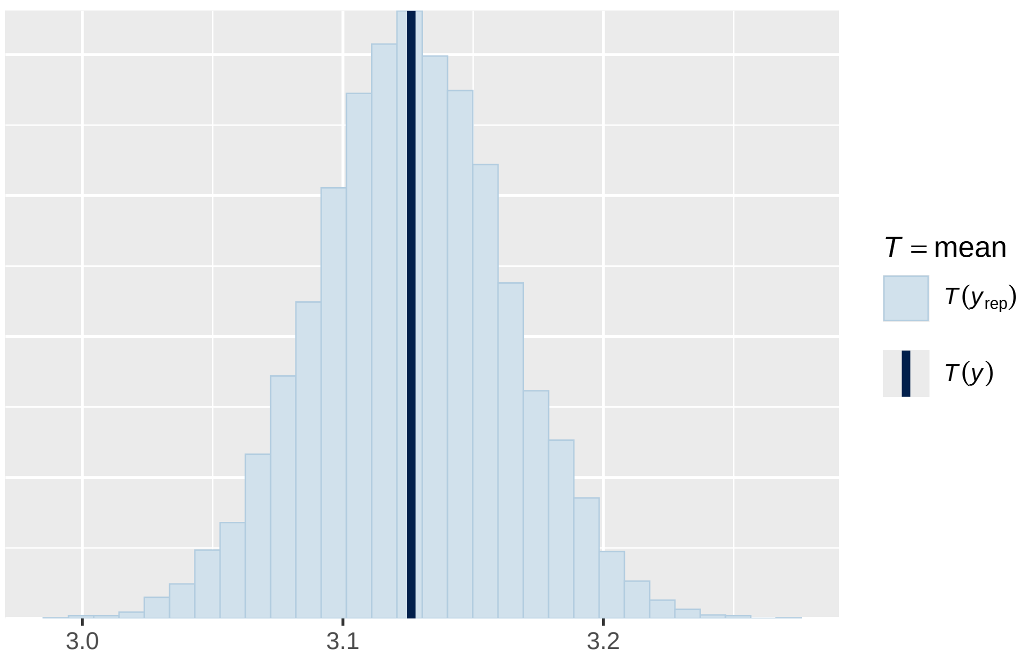

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



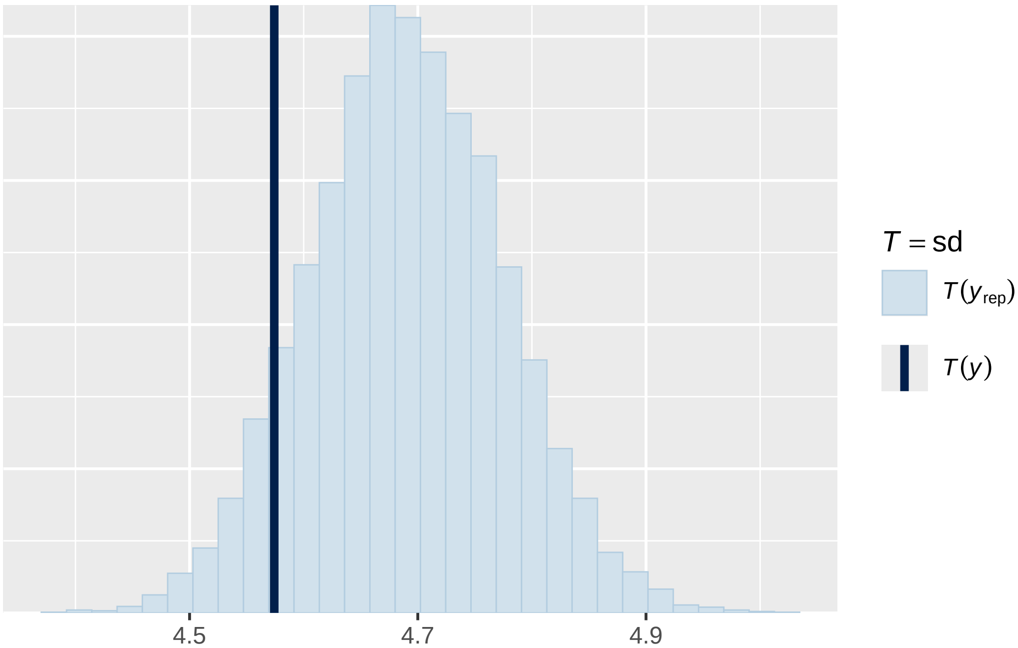

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



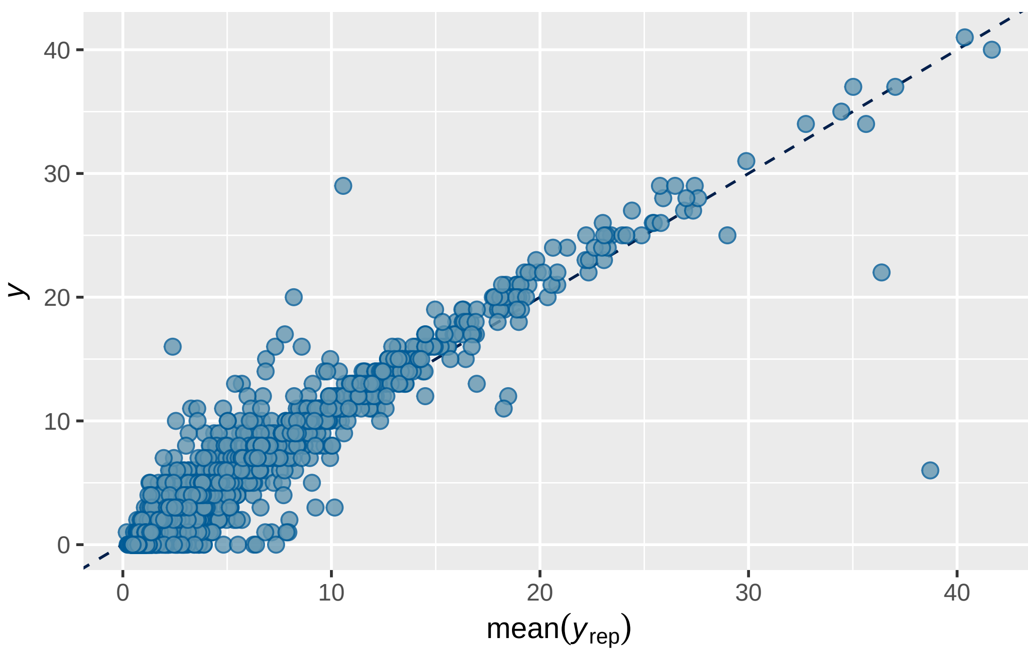

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



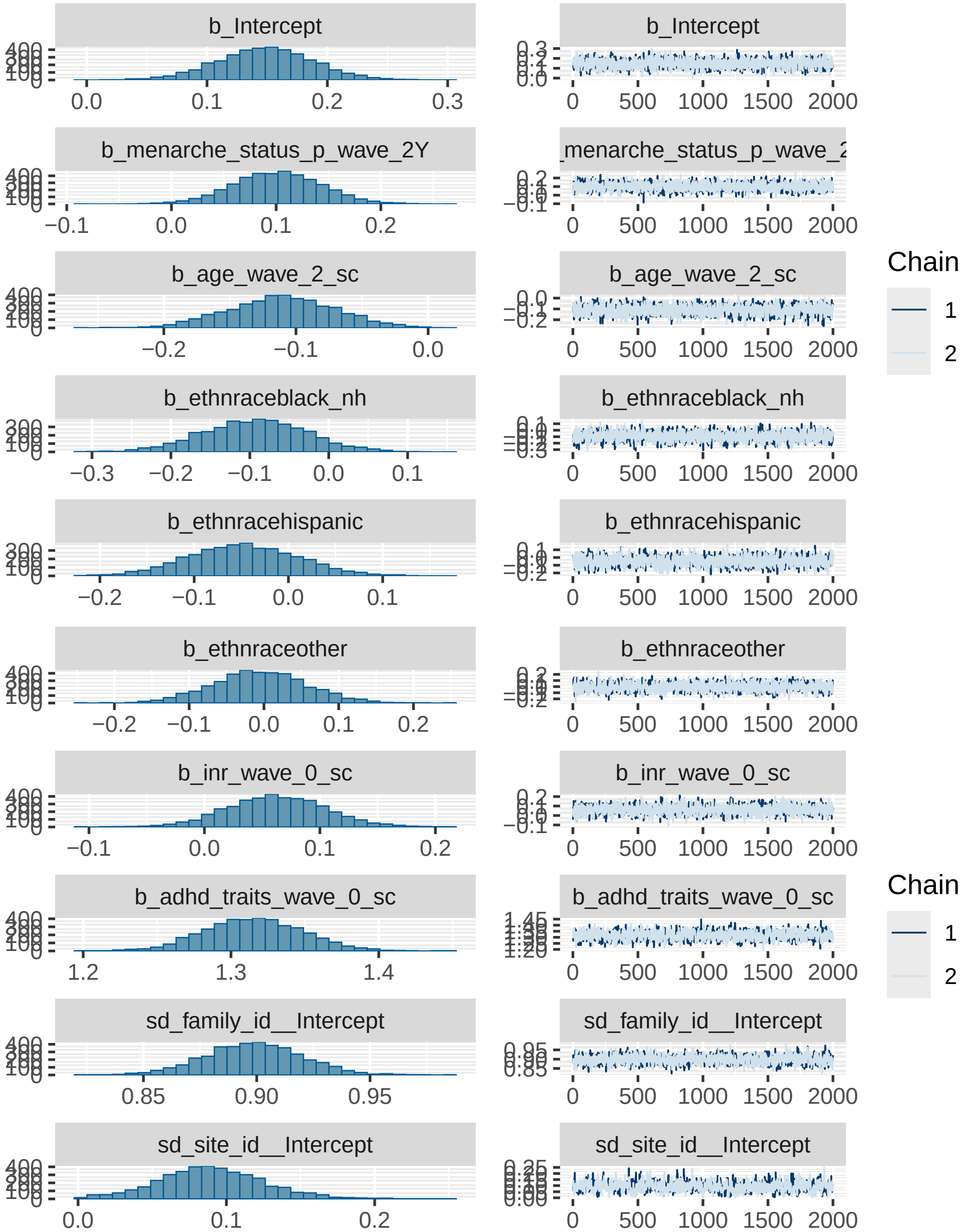
Model 1 longitudinal prior 2 output

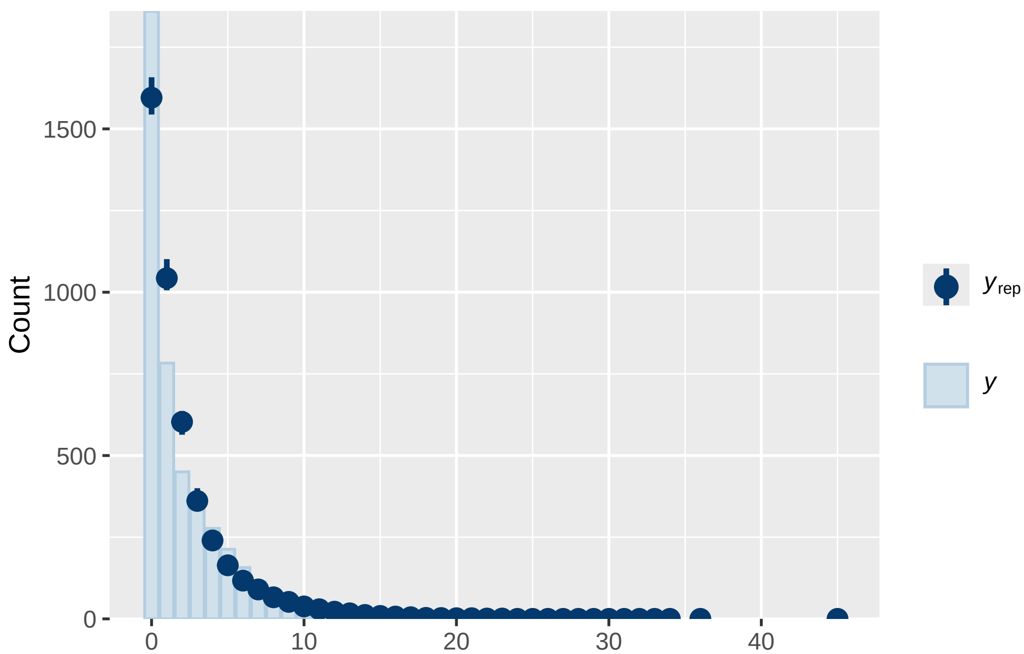
```
    Estimate   Est.Error      Q2.5    Q97.5
R2 0.7393702 0.007767695 0.7238316 0.754201


 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave_
   Data: analytic_df_wide (Number of observations: 4494)
  Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~family_id (Number of levels: 3913)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.899     0.021    0.857    0.940 1.000     1230     2009

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.090     0.036    0.021    0.163 1.006      556      535

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
Intercept                    0.148     0.040    0.070    0.227 1.001     1838
menarche_status_p_wave_2Y    0.101     0.042    0.019    0.182 1.001     1819
age_wave_2_sc               -0.111     0.041   -0.189   -0.032 1.003     1935
ethnraceblack_nh            -0.093     0.066   -0.224    0.039 1.000     1639
ethnracehispanic            -0.047     0.058   -0.158    0.068 1.000     1431
ethnraceother               -0.006     0.062   -0.128    0.118 1.000     1613
inr_wave_0_sc                0.062     0.043   -0.020    0.149 1.000     1648
adhd_traits_wave_0_sc        1.315     0.033    1.253    1.381 1.002     1270
                          Tail_ESS
Intercept                     2837
menarche_status_p_wave_2Y     2607
age_wave_2_sc                 2664
ethnraceblack_nh              2775
ethnracehispanic              2236
ethnraceother                 2739
inr_wave_0_sc                 2050
adhd_traits_wave_0_sc         2183

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
```

and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

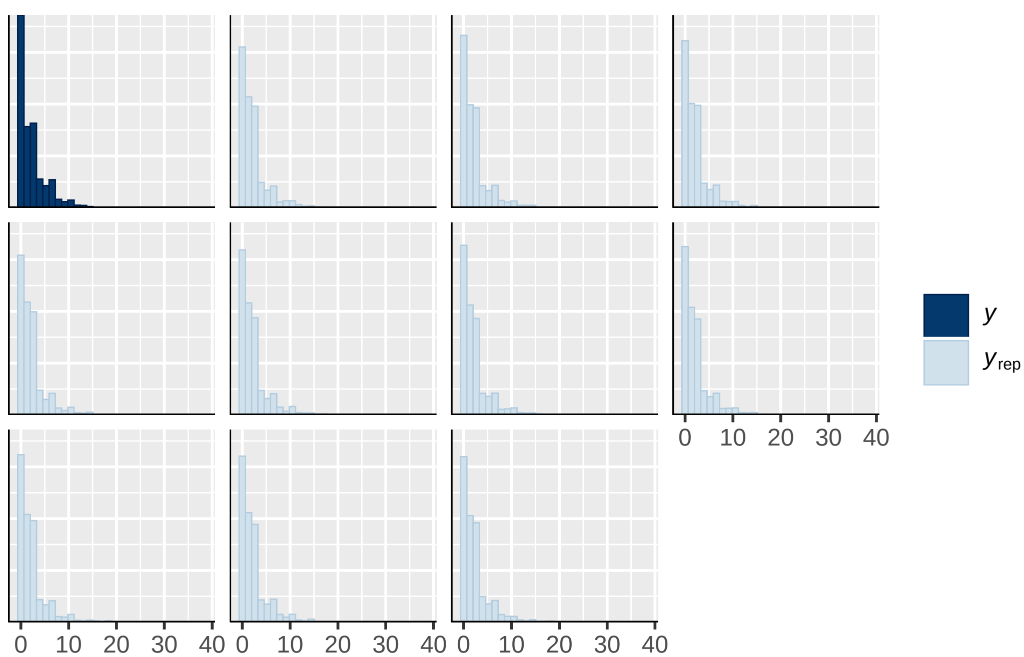
Model 1 longitudinal prior 2 diagnostics

```
               prior     class                      coef     group resp dpar
       normal(0, 10)         b                                              
       normal(0, 10)         b     adhd_traits_wave_0_sc                    
       normal(0, 10)         b             age_wave_2_sc                    
       normal(0, 10)         b          ethnraceblack_nh                    
       normal(0, 10)         b          ethnracehispanic                    
       normal(0, 10)         b             ethnraceother                    
       normal(0, 10)         b             inr_wave_0_sc                    
       normal(0, 10)         b menarche_status_p_wave_2Y                    
student_t(3, 0, 2.9) Intercept                                              
student_t(3, 0, 2.9)        sd                                              
student_t(3, 0, 2.9)        sd                           family_id          
student_t(3, 0, 2.9)        sd                 Intercept family_id          
student_t(3, 0, 2.9)        sd                             site_id          
student_t(3, 0, 2.9)        sd                 Intercept   site_id          
nlpar lb ub tag       source
                        user
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                (vectorized)
                     default
       0             default
       0        (vectorized)
       0        (vectorized)
       0        (vectorized)
       0        (vectorized)
```

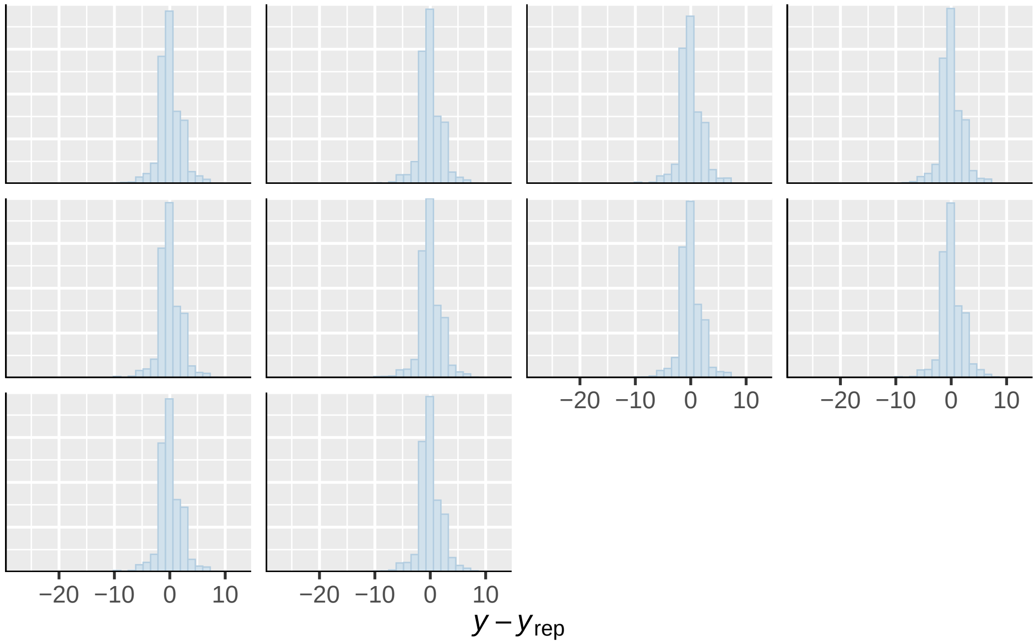

b_Intercept
b_menarche_status_p_wave_2Y
b_age_wave_2_sc
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnraceother
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
sd_family_id__Intercept
sd_site_id__Intercept
Chain
1
2



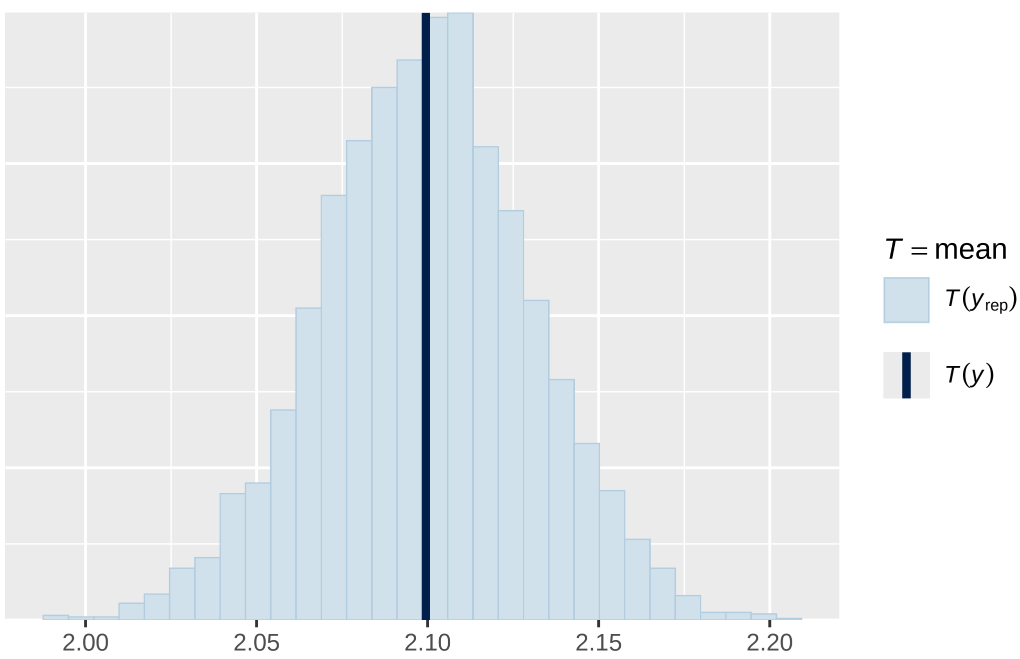

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



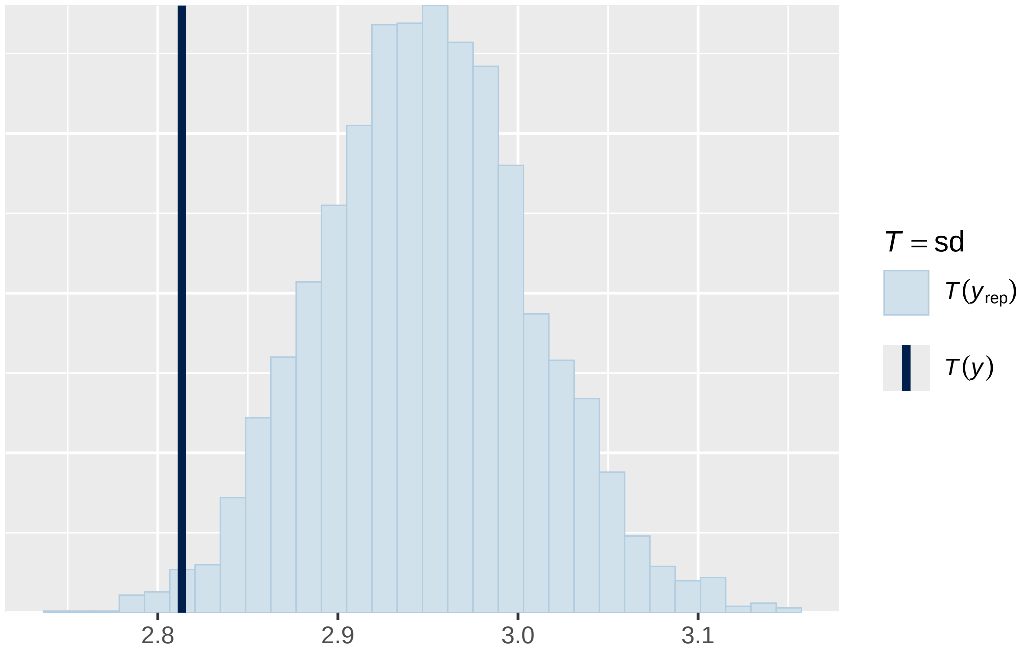

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



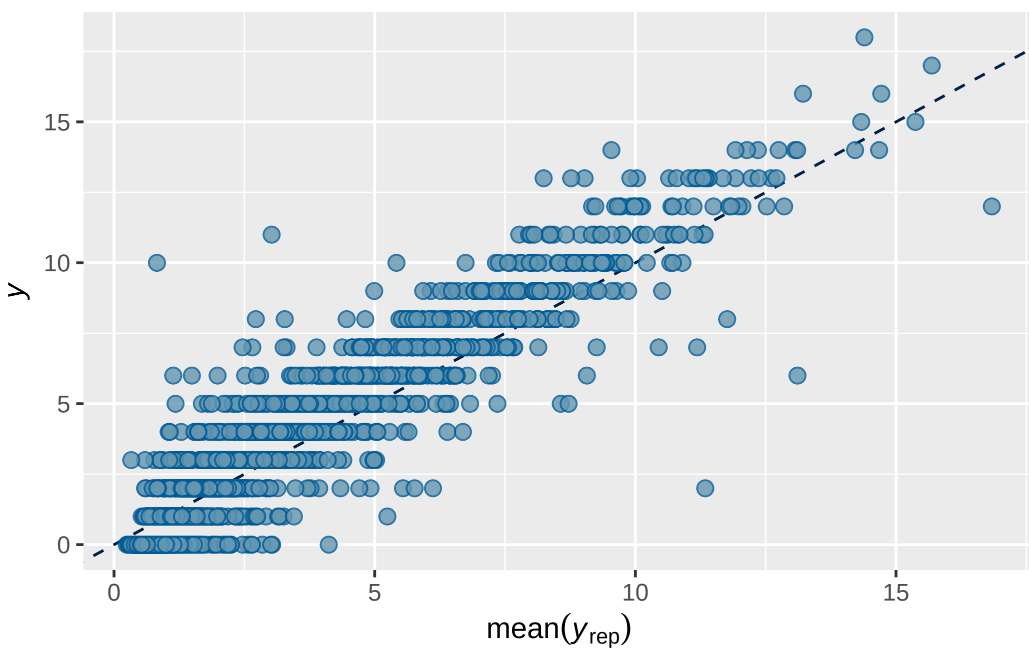

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



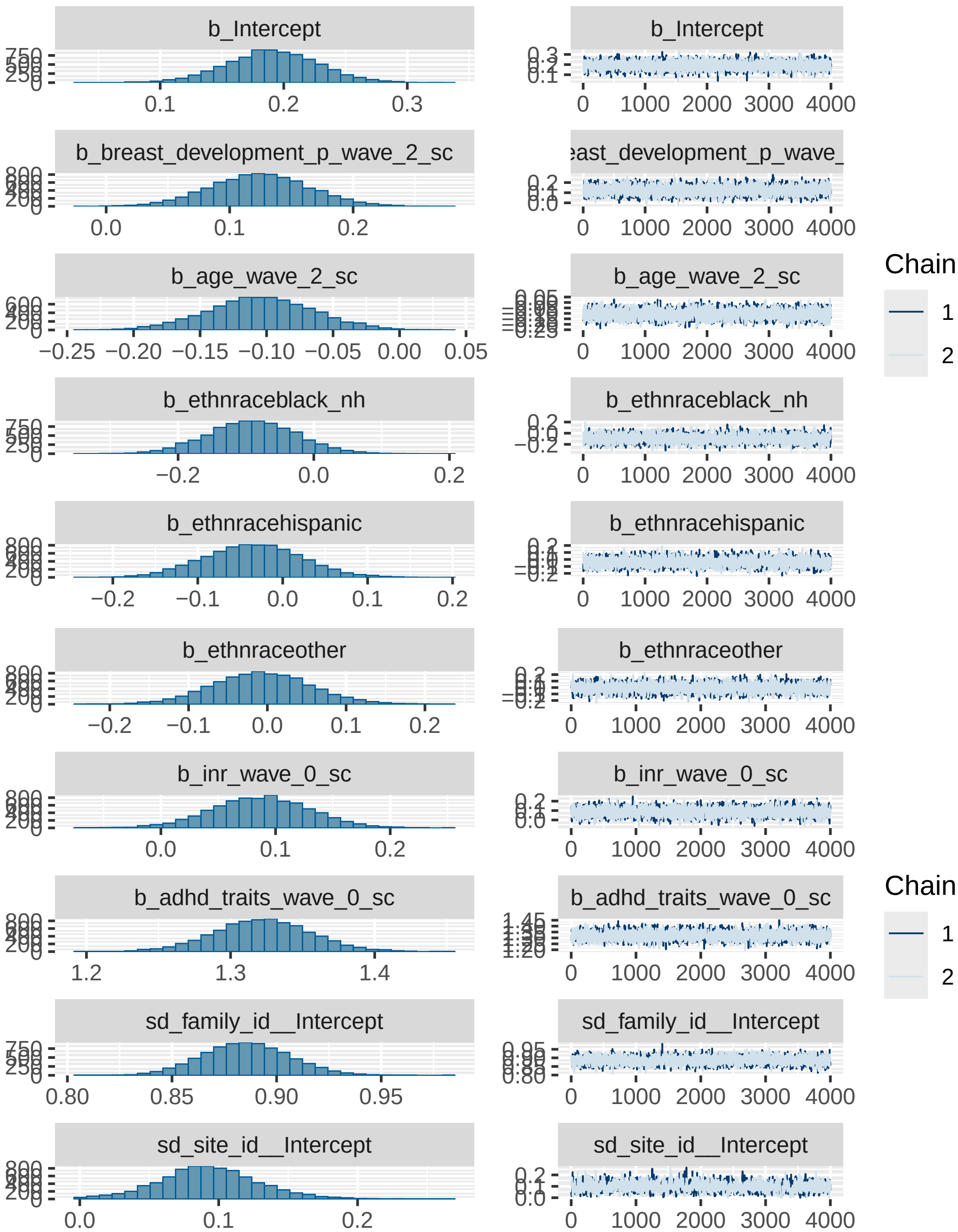
Model 1B longitudinal prior 2 output

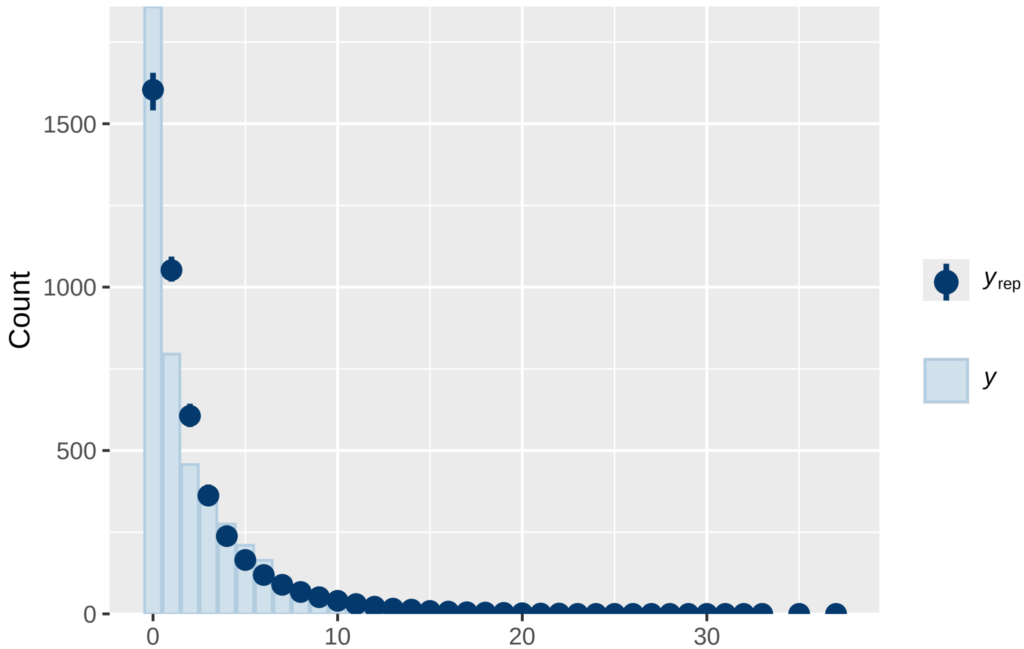
```
    Estimate   Est.Error      Q2.5    Q97.5
R2 0.7357083 0.007529665 0.7207626 0.750273


 Family: poisson
  Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + in
   Data: analytic_df_wide (Number of observations: 4515)
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.886     0.021    0.846    0.927 1.004     2360     4681

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.093     0.036    0.022    0.168 1.001     1068     1112

Regression Coefficients:
                               Estimate Est.Error l-95% CI u-95% CI  Rhat
Intercept                         0.191     0.037    0.116    0.264 1.001
breast_development_p_wave_2_sc    0.125     0.040    0.047    0.204 1.000
age_wave_2_sc                    -0.104     0.040   -0.182   -0.026 1.000
ethnraceblack_nh                 -0.091     0.065   -0.216    0.037 1.001
ethnracehispanic                 -0.035     0.057   -0.145    0.076 1.001
ethnraceother                    -0.007     0.062   -0.129    0.118 1.001
inr_wave_0_sc                     0.087     0.043    0.003    0.170 1.000
adhd_traits_wave_0_sc             1.322     0.034    1.256    1.388 1.001
                               Bulk_ESS Tail_ESS
Intercept                          3537     4081
breast_development_p_wave_2_sc     3892     4613
age_wave_2_sc                      3724     5234
ethnraceblack_nh                   2931     4747
ethnracehispanic                   3005     4569
ethnraceother                      3139     4952
inr_wave_0_sc                      3087     4724
adhd_traits_wave_0_sc              2630     4155

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
```

and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

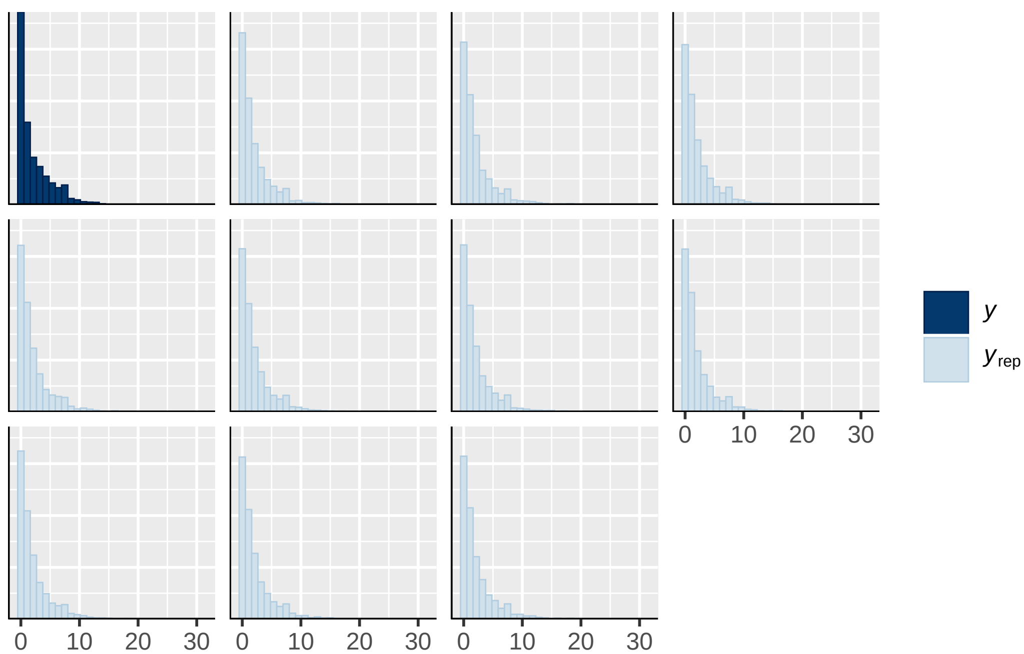
Model 1B longitudinal prior 2 diagnostics

```
                prior     class                           coef     group resp
        normal(0, 10)         b
        normal(0, 10)         b          adhd_traits_wave_0_sc
        normal(0, 10)         b                  age_wave_2_sc
        normal(0, 10)         b breast_development_p_wave_2_sc
        normal(0, 10)         b               ethnraceblack_nh
        normal(0, 10)         b               ethnracehispanic
        normal(0, 10)         b                  ethnraceother
        normal(0, 10)         b                  inr_wave_0_sc
 student_t(3, 0, 2.7) Intercept
 student_t(3, 0, 2.7)        sd
 student_t(3, 0, 2.7)        sd                                family_id
 student_t(3, 0, 2.7)        sd                      Intercept family_id
 student_t(3, 0, 2.7)        sd                                  site_id
 student_t(3, 0, 2.7)        sd                      Intercept   site_id
 dpar nlpar lb ub tag       source
                              user
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                      (vectorized)
                           default
             0             default
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
             0        (vectorized)
```

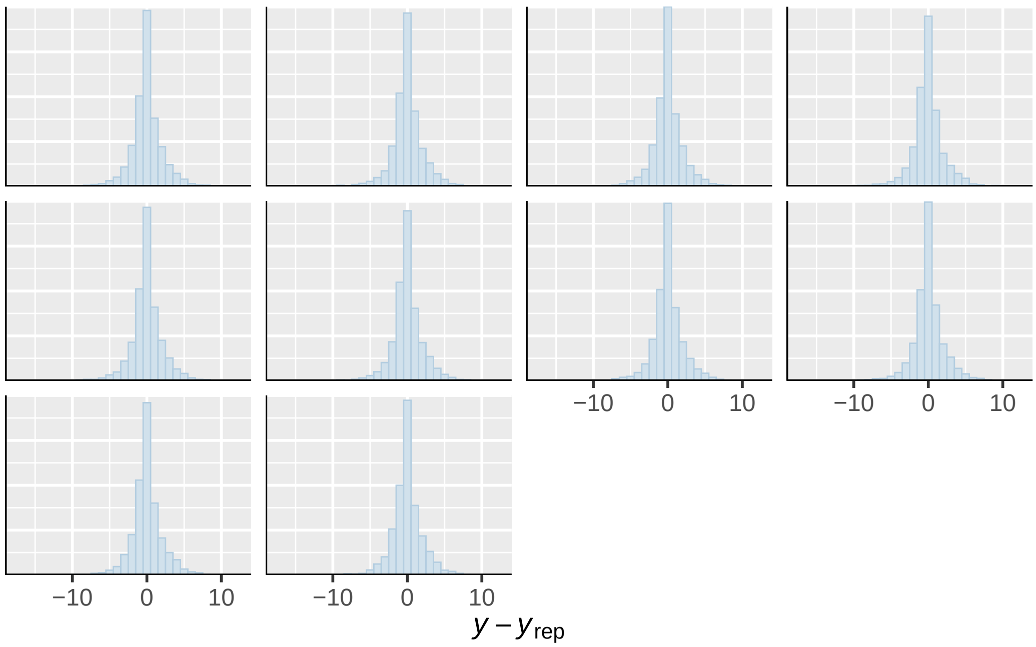

b_Intercept
0.1
0.2
0.3
b_Intercept
0
1000
2000
3000
4000
b_breast_development_p_wave_2_sc
0.0
0.1
0.2
b_age_wave_2_sc
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
0.05
b_age_wave_2_sc
b_ethnraceblack_nh
−0.2
0.0
0.2
b_ethnraceblack_nh
b_ethnracehispanic
−0.2
−0.1
0.0
0.1
0.2
b_ethnracehispanic
b_ethnraceother
b_ethnraceother
b_inr_wave_0_sc
b_inr_wave_0_sc
b_adhd_traits_wave_0_sc
1.2
1.3
1.4
b_adhd_traits_wave_0_sc
sd_family_id__Intercept
0.80
0.85
0.90
0.95
sd_family_id__Intercept
sd_site_id__Intercept
sd_site_id__Intercept
Chain
1
2



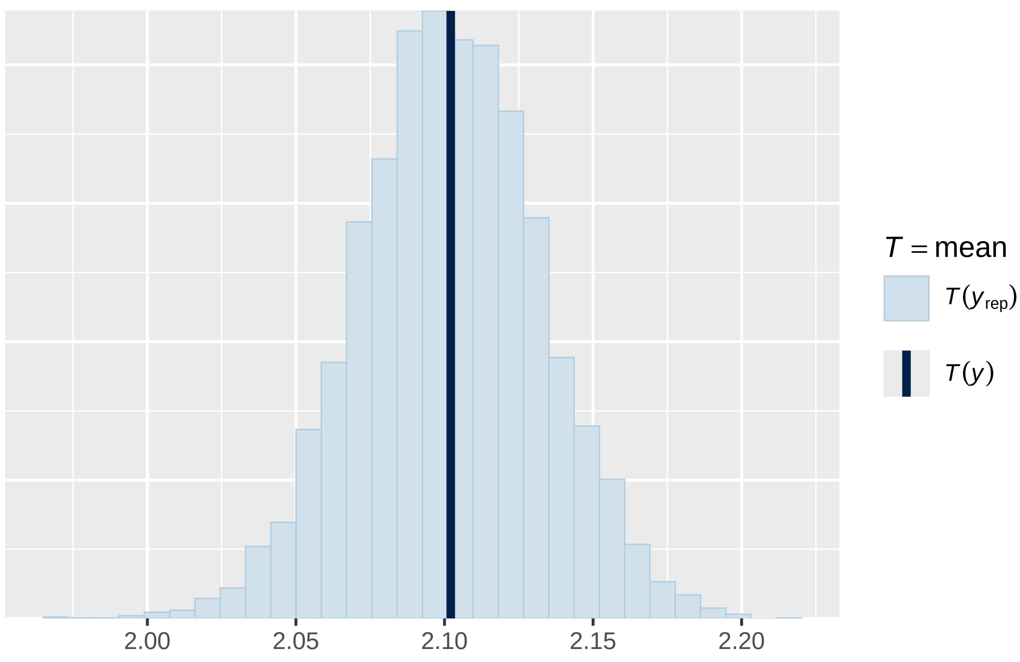

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



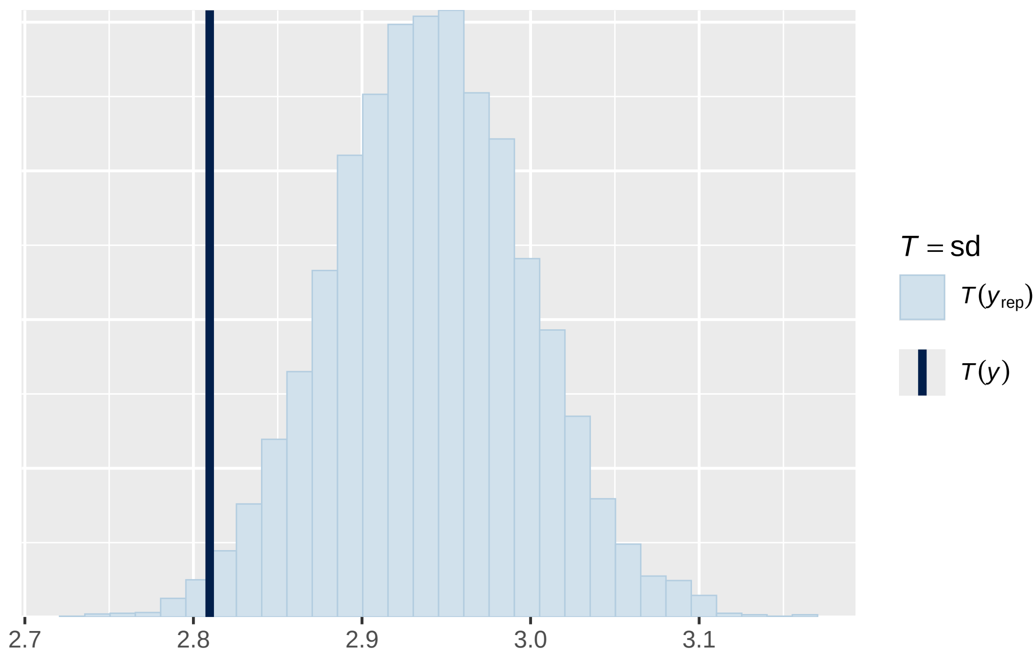

```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



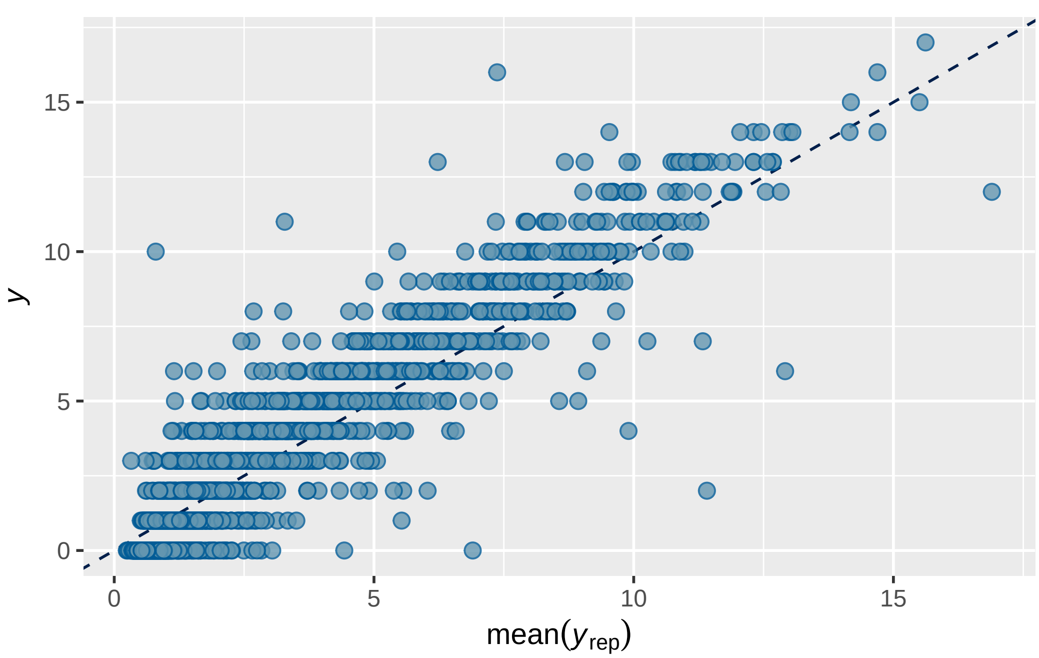

```
Using all posterior draws for ppc type 'stat' by default.
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'scatter_avg' by default.
```



Model 5B longitudinal prior 2 output

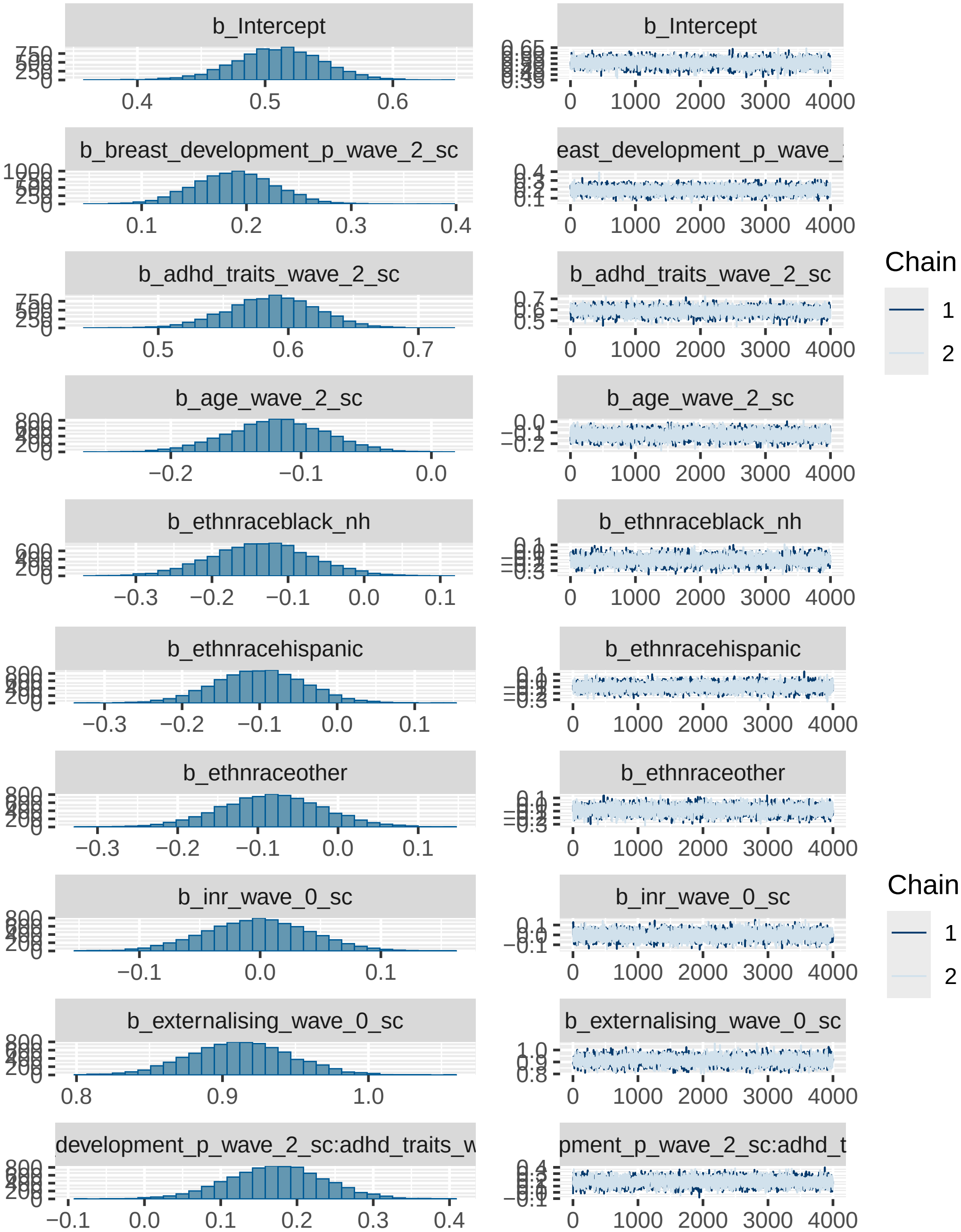
```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8232708 0.005211247 0.8127758 0.8330009


 Family: poisson 
  Links: mu = log 
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age_
   Data: analytic_df_wide (Number of observations: 4515) 
  Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~family_id (Number of levels: 3928) 
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.985     0.019    0.947    1.023 1.000     2396     4408

~site_id (Number of levels: 22) 
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.066     0.038    0.004    0.150 1.003      747     1960

Regression Coefficients:
                                                     Estimate Est.Error
Intercept                                               0.510     0.035
breast_development_p_wave_2_sc                          0.189     0.038
adhd_traits_wave_2_sc                                   0.588     0.034
age_wave_2_sc                                          -0.119     0.037
ethnraceblack_nh                                       -0.134     0.066
ethnracehispanic                                       -0.099     0.058
ethnraceother                                          -0.082     0.064
inr_wave_0_sc                                          -0.002     0.043
externalising_wave_0_sc                                 0.912     0.035
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc    0.173     0.064
                                                     l-95% CI u-95% CI  Rhat
Intercept                                               0.440    0.577 1.000
breast_development_p_wave_2_sc                          0.115    0.264 1.001
adhd_traits_wave_2_sc                                   0.520    0.653 1.001
age_wave_2_sc                                          -0.193   -0.046 1.000
ethnraceblack_nh                                       -0.261   -0.006 1.000
ethnracehispanic                                       -0.212    0.015 1.000
ethnraceother                                          -0.209    0.044 1.000
inr_wave_0_sc                                          -0.087    0.082 1.001
```

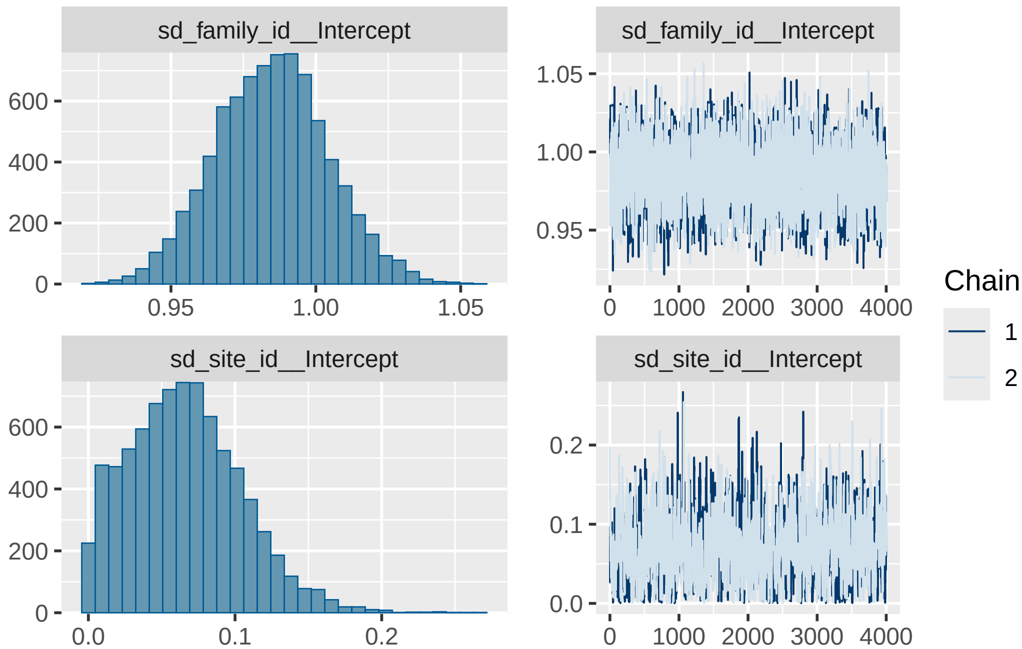
```
externalising_wave_0_sc                                0.845    0.980 1.000
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc    0.046    0.298 1.001
                                                     Bulk_ESS Tail_ESS
Intercept                                                3957     5113
breast_development_p_wave_2_sc                           4506     5827
adhd_traits_wave_2_sc                                    3055     4147
age_wave_2_sc                                            4175     5517
ethnraceblack_nh                                         2842     4278
ethnracehispanic                                         3424     5003
ethnraceother                                            3134     4414
inr_wave_0_sc                                            3475     5000
externalising_wave_0_sc                                  2660     4781
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc     3070     4827

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

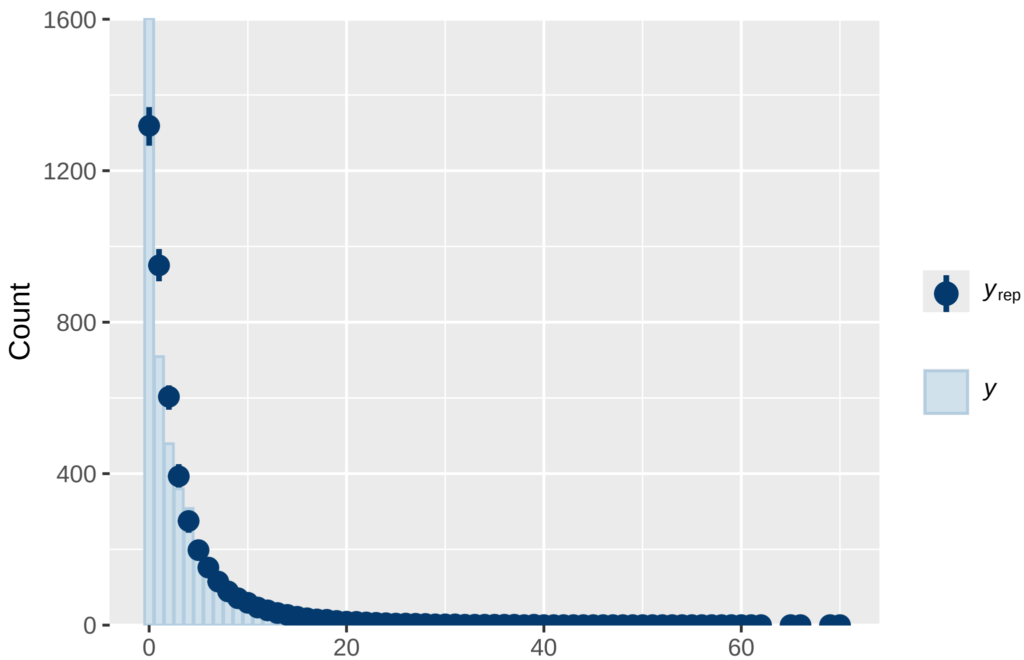
Model 5B longitudinal prior 2 diagnostics

```
               prior     class
       normal(0, 10)         b
       normal(0, 10)         b
       normal(0, 10)         b
       normal(0, 10)         b
       normal(0, 10)         b
       normal(0, 10)         b
       normal(0, 10)         b
       normal(0, 10)         b
       normal(0, 10)         b
       normal(0, 10)         b
 student_t(3, 0, 2.9) Intercept
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
 student_t(3, 0, 2.9)        sd
                                                    coef     group resp dpar nlpar

                                   adhd_traits_wave_2_sc
                                           age_wave_2_sc
                          breast_development_p_wave_2_sc
```

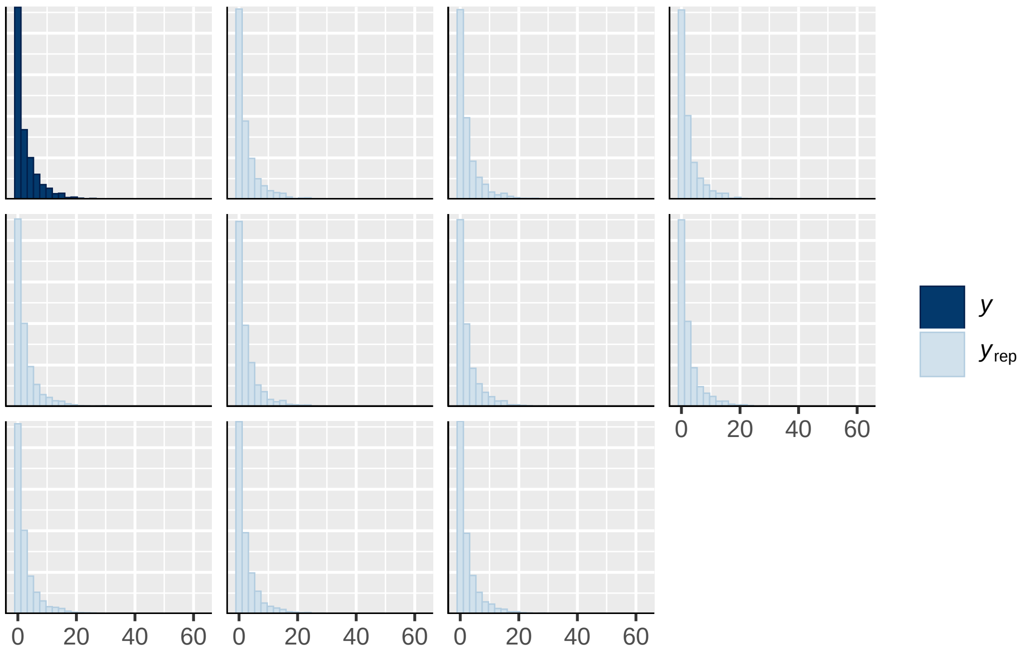
```
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
                                   ethnraceblack_nh
                                  ethnracehispanic
                                     ethnraceother
                            externalising_wave_0_sc
                                     inr_wave_0_sc


                                                            family_id
                                                  Intercept family_id
                                                              site_id
                                                  Intercept   site_id
lb ub tag       source
                  user
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
          (vectorized)
               default
 0             default
 0        (vectorized)
 0        (vectorized)
 0        (vectorized)
 0        (vectorized)
```


b_Intercept
b_Intercept
b_breast_development_p_wave_2_sc
east_development_p_wave_
b_adhd_traits_wave_2_sc
b_adhd_traits_wave_2_sc
b_age_wave_2_sc
b_age_wave_2_sc
b_ethnraceblack_nh
b_ethnraceblack_nh
b_ethnracehispanic
b_ethnracehispanic
b_ethnraceother
b_ethnraceother
b_inr_wave_0_sc
b_inr_wave_0_sc
b_externalising_wave_0_sc
b_externalising_wave_0_sc
development_p_wave_2_sc:adhd_traits_w
pment_p_wave_2_sc:adhd_t
Chain
1
2
Chain
1
2

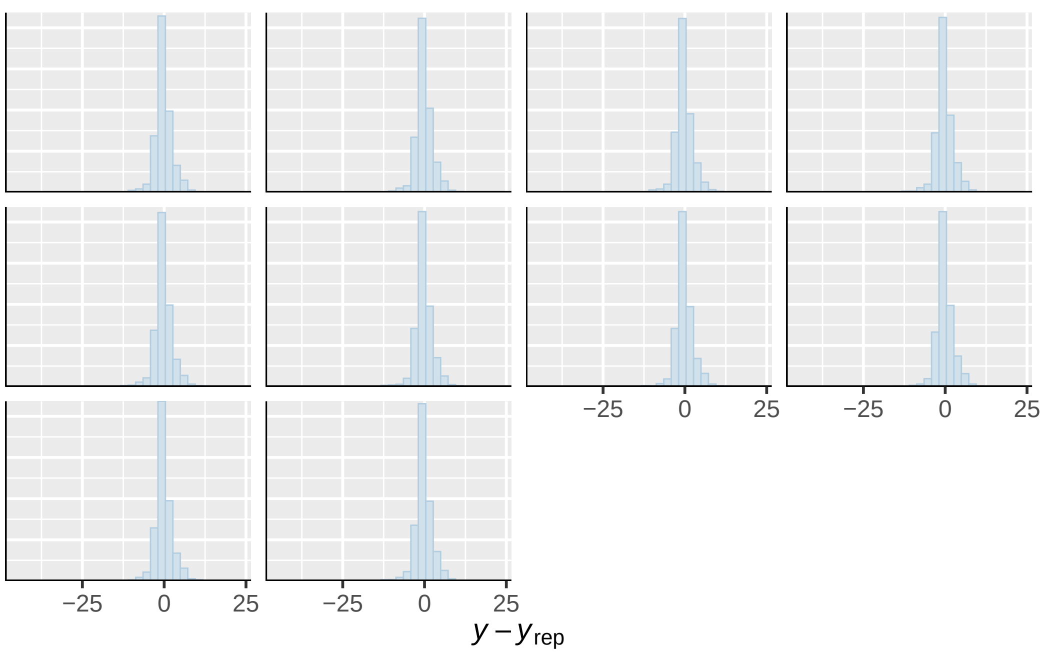





```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

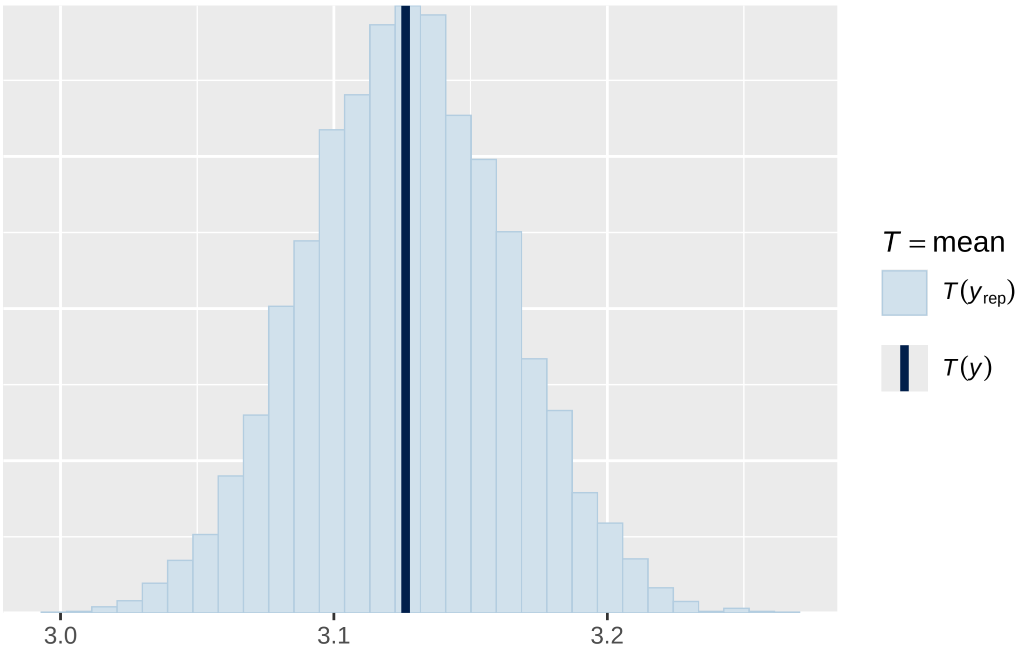



```
Using 10 posterior draws for ppc type 'error_hist' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

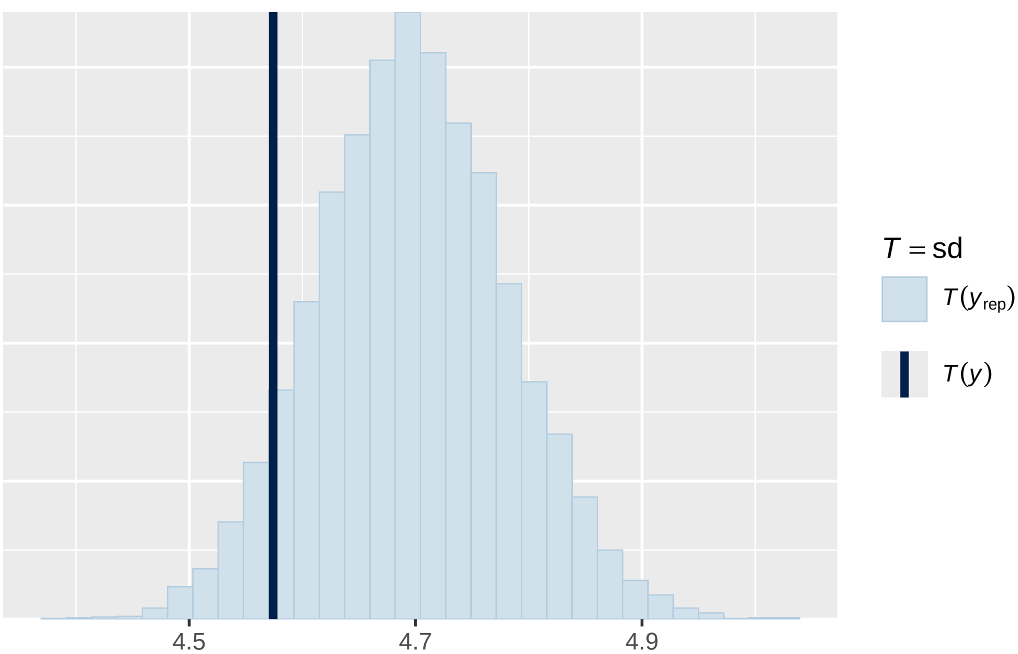



```
Using all posterior draws for ppc type 'stat' by default.
```

```
Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
Using all posterior draws for ppc type 'stat' by default.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
Using all posterior draws for ppc type 'scatter_avg' by default.
```